Model Structure (25.12):

This is a panmictic model (do_spatial == 0).
There are 2 sexes, 2 fishing fleets, 3 survey fleets, and 1 regions being modeled.
The model has 67 years, starting in 1960 and ending in 2026.
There are 30 ages, where recruitment occurs at age−2 and the plus group occurs at age−31.
There are 30 length bins, which start at 41cm and end at 99cm.

Growth Estimation:

This model uses the growth and weight estimates as updated in the 2021 SAFE (Echave et al., 2021).
There are 2 (pre/post 1996) growth blocks and a single weight block (use_new_bio == 0).
For fishery composition data aggregation catch−weighting, the historic sex−aggregated Weight parameters are used to convert to catch in numbers (sex_disagg_comp_wt == 0).

Aleutian Islands (AI) Data Use:

This model uses data from the Aleutian Islands (AI) −− no data is being removed (drop_AI == 0).

Longline Survey Index Construction:

There was a longline survey in 2026, so there is a terminal year LLS RPN value (LLS_active == 1).

The extrapolated values for the western AI longline survey stations WERE included in the LLS RPN index (drop_WAI_LLS == 0).

The longline survey used off−year extrapolations for the BS (even years) or AI (odd years) based on the year over year change in the GOA RPNs.
Starting in 2025, carryover values for the BS+AI (odd years) or GOA (even years) were used (drop_LLS_extrap == 0 & do_LLS_carry == 0).
The SPoRC model is being passed the index SEs, no conversions needed (do_cont_SE == 1).

Composition Data Structure and ISS:

Composition proportions by age/length are aggregated across regions after weighting by catch in numbers or RPNs.

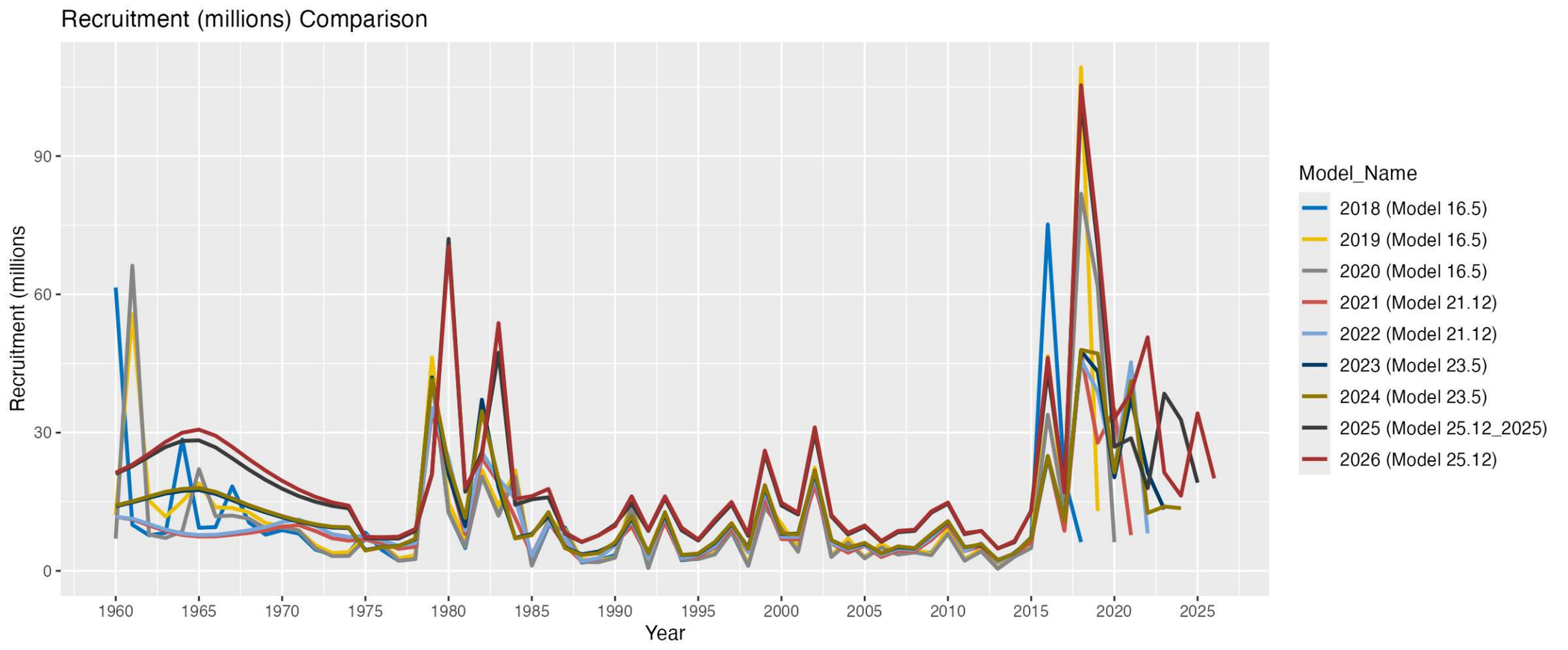
Age and length compositions are fit 'joint' by sex, but lengths are dropped for years when ages exist (JPN LLS lengths are kept for all years)
(do_spatial == 0 & sex_age_comps == 5).
The Input Sample Size (ISS) for composition data is fixed across years at 20
(do_spatial == 0 & do_scale_ISS == 0).

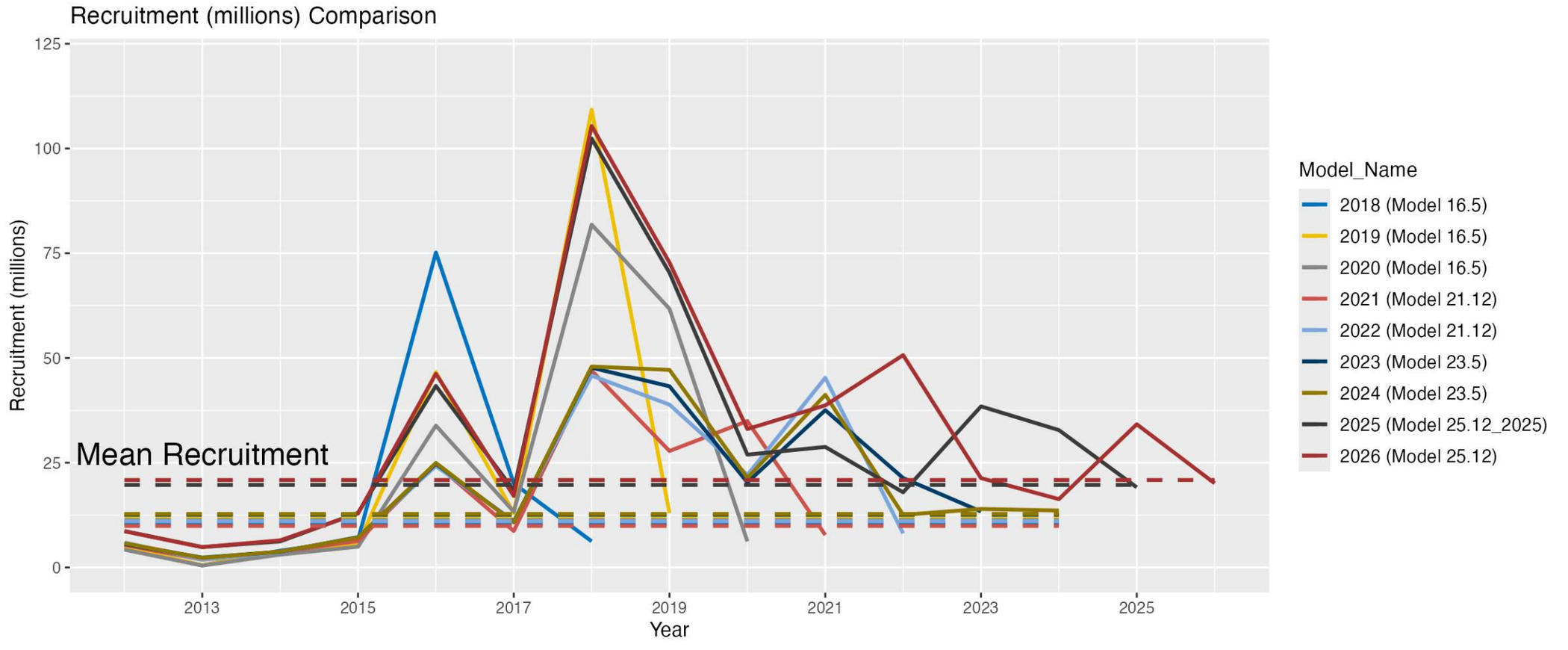
Length Composition Data Use:

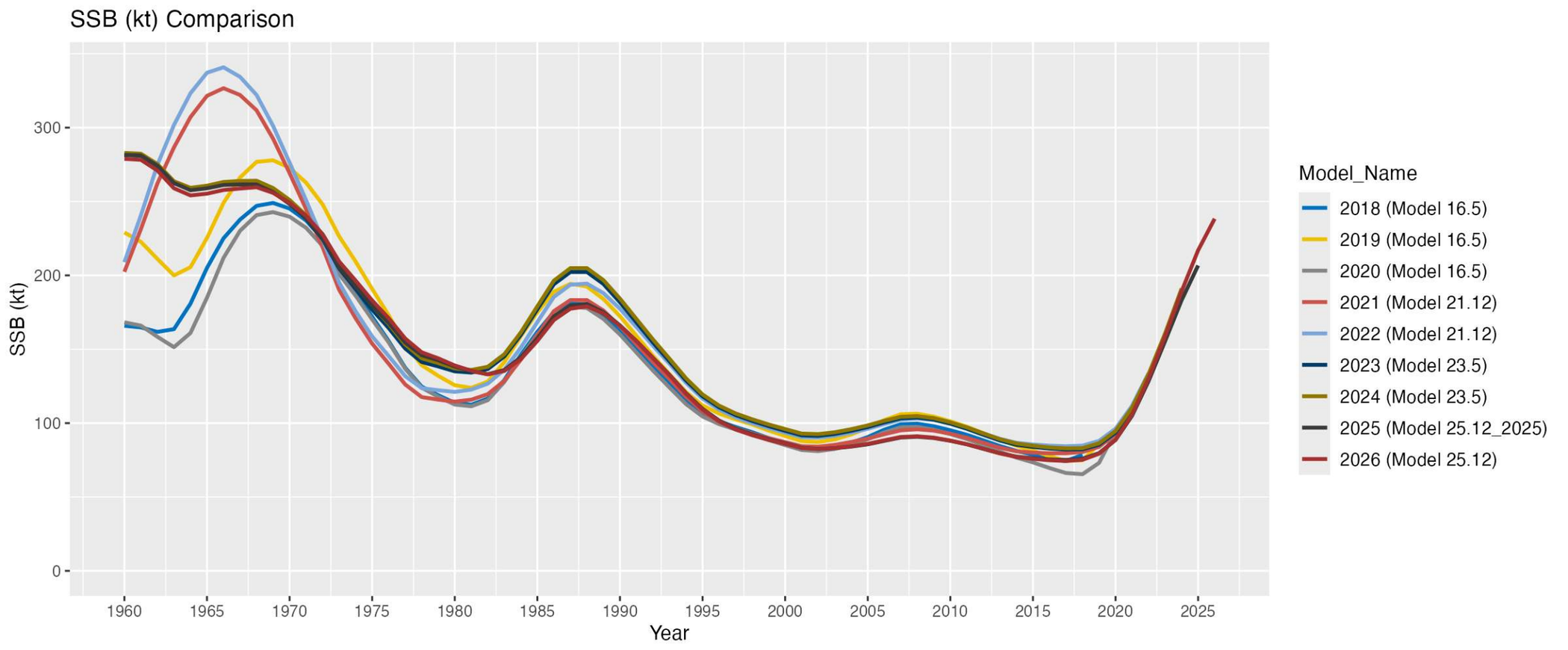
The longline survey length composition data is weighted/calculated using the raw length data and weighted by area RPNs consistent with the age composition data
(LLS_length_comp_source == 1).
Longline survey length data are fit in the model as determined by the sex_age_comps switch BUT terminal in−year length data are NOT fit
(fit_LLS_len == 1 & fit_LLS_len_term == 0).
Fixed gear fishery length data are fit in the model as determined by the sex_age_comps switch (fit_LLF_len == 1).

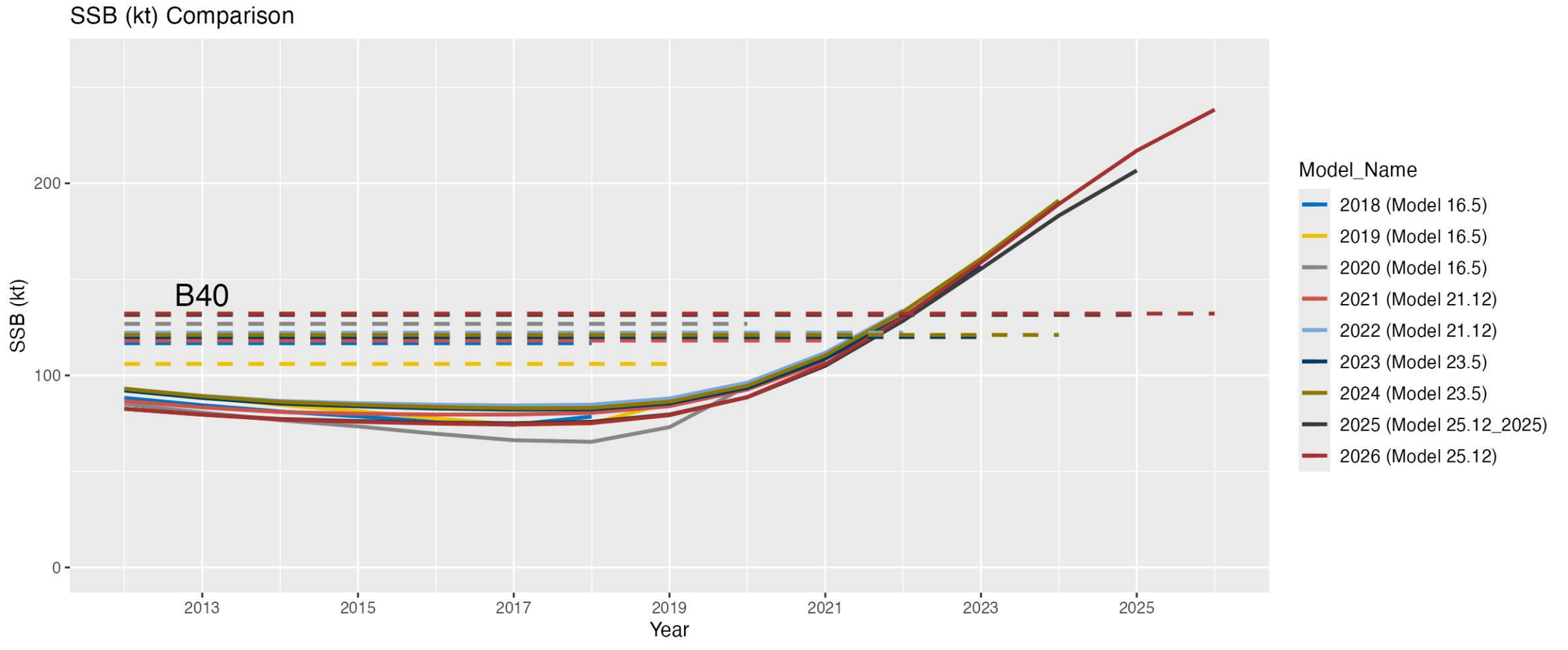
General Data Fitting Notes:

Fishery whale depredation estimates were fixed/static at the previous year's values (whale_dep_static == 1).
The GOA trawl survey data was NOT fit in this model (use_trwl_srvy == 0).
The fishery CPUE index WAS fit in this model (terminal year 2022) using the STANDARDIZED index values (Cheng et al., 2023; ICES JMS)
(use_CPUE == 1 & use_CPUE_stand == 1).
Non−fishery catch (e.g., scientific removals) WAS included in the model as fixed gear catch, where all sources of removals were available for terminal year
(add_non_fish_catch == 1 & non_fsh_catch_updated == 1).


Recruitment (millions) Comparison
Recruitment (millions
0
30
60
90
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 25.12)


Recruitment (millions) Comparison
Recruitment (millions)
0
25
50
75
100
125
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 25.12)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 25.12)


SSB (kt) Comparison
B40
SSB (kt)
200
100
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 25.12)

colour — First Iter — Last Iter

Fixed Gear Fishery

Trawl Fishery

Fishery Ages

Fishery Lengths

Mean Composition Bin

Mean Com

14
12
10
8
6
4

70
65
60
55

1990
2000
2010
2020

Year

colour — First Iter — Last Iter

Japan LL Survey

NOAA LL Survey

Survey Ages

Survey Lengths

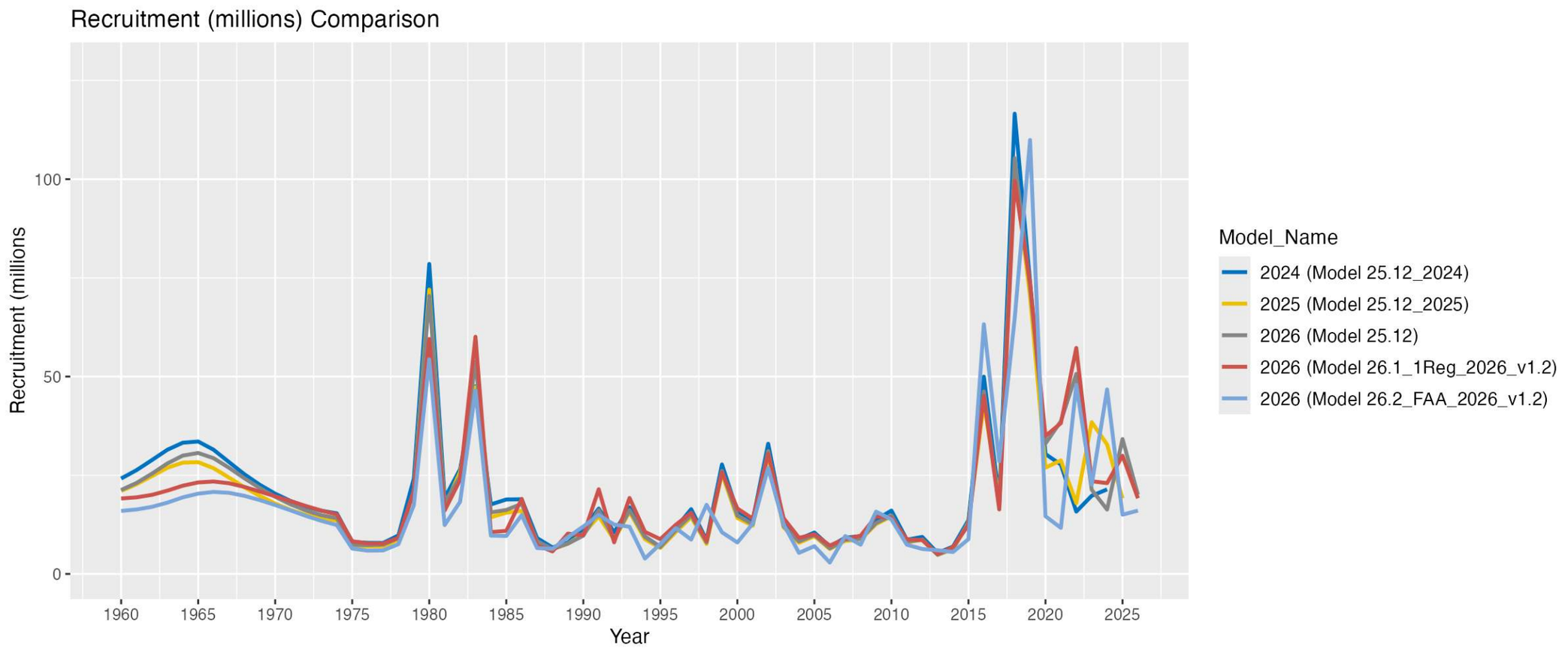
Mean Composition Bin

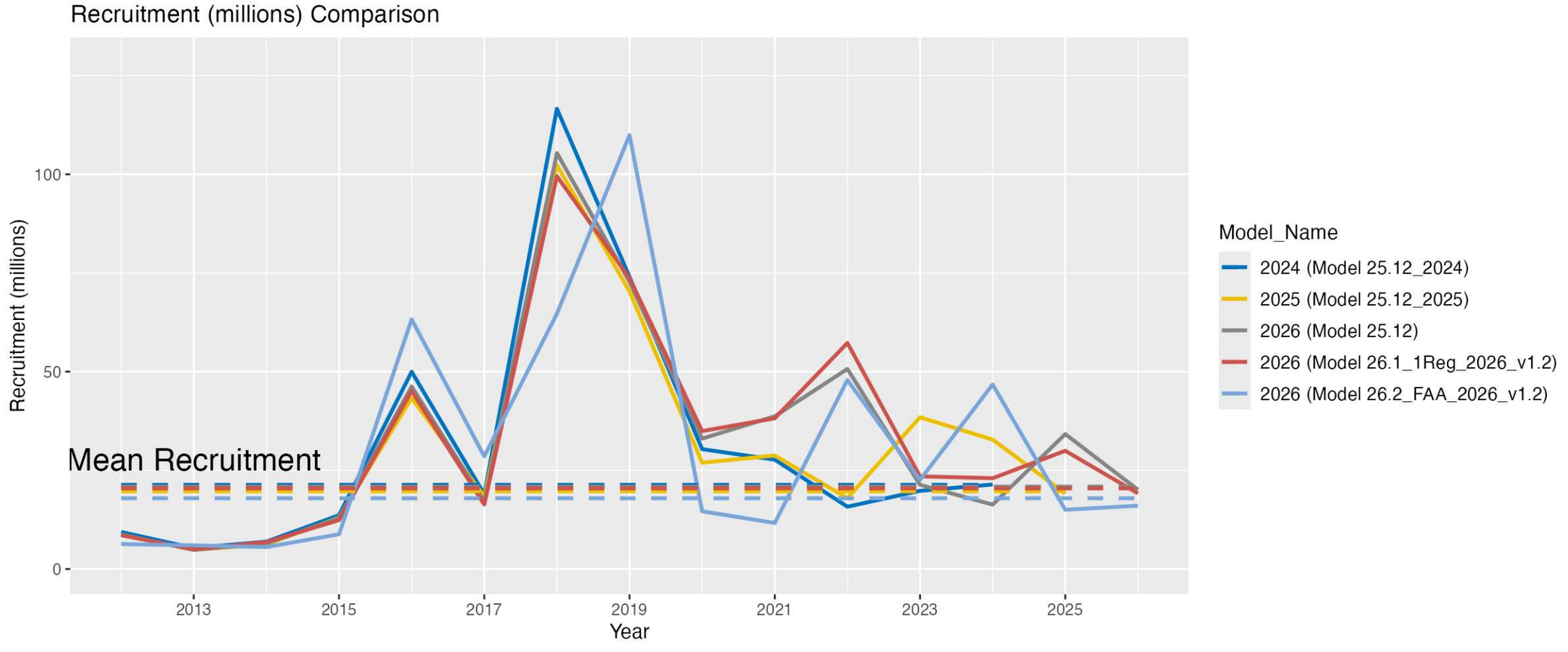
12
10
8
6
4

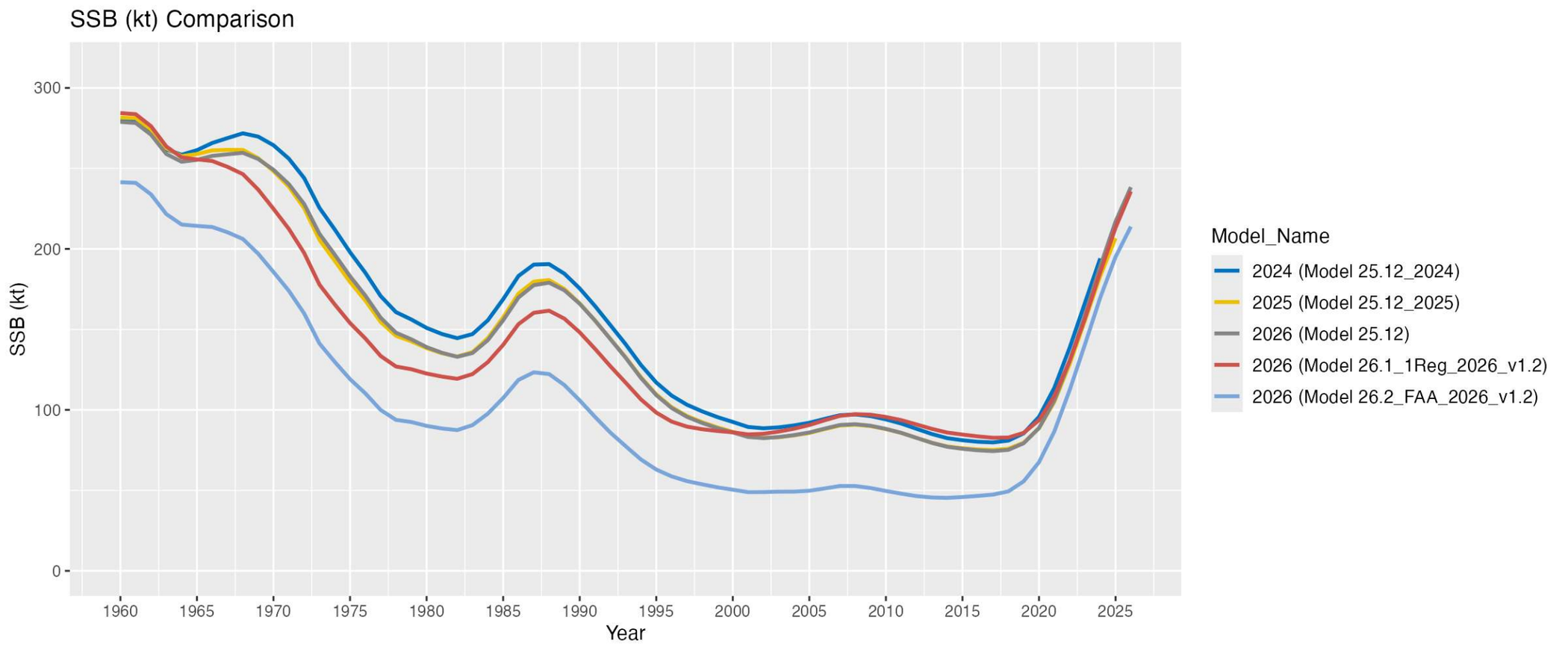
70.0
67.5
65.0
62.5

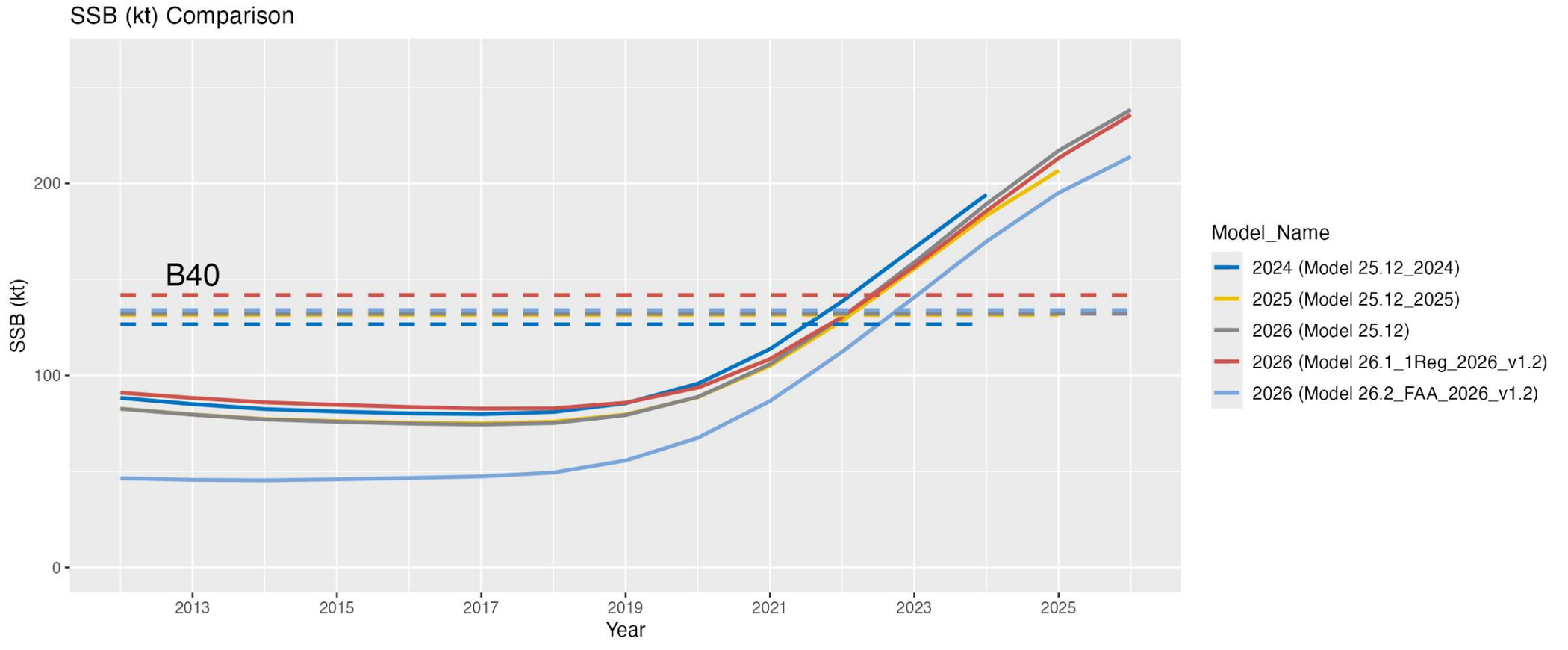
1980
1990
2000
2010
2020

Year


Recruitment (millions) Comparison
Recruitment (millions)
100
50
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2026 (Model 25.12)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)


Recruitment (millions) Comparison
Recruitment (millions)
100
50
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2026 (Model 25.12)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2026 (Model 25.12)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)


SSB (kt) Comparison
B40
SSB (kt)
200
100
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2026 (Model 25.12)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)

Model 25.12 Population Population 1

Female

Male

AK

Spawning Weight-at-Age (kg)

0.100

0.075

0.050

0.025

0.000

0

10

20

30

Age

Model 25.12 Population Population 1

Female

Male

AK

Maturity at Age

1.00

0.75

0.50

0.25

0.00

0

10

20

30

Age

Model 25.12 Population Population 1

Female

Male

AK

Spawning Weight at Age

6

4

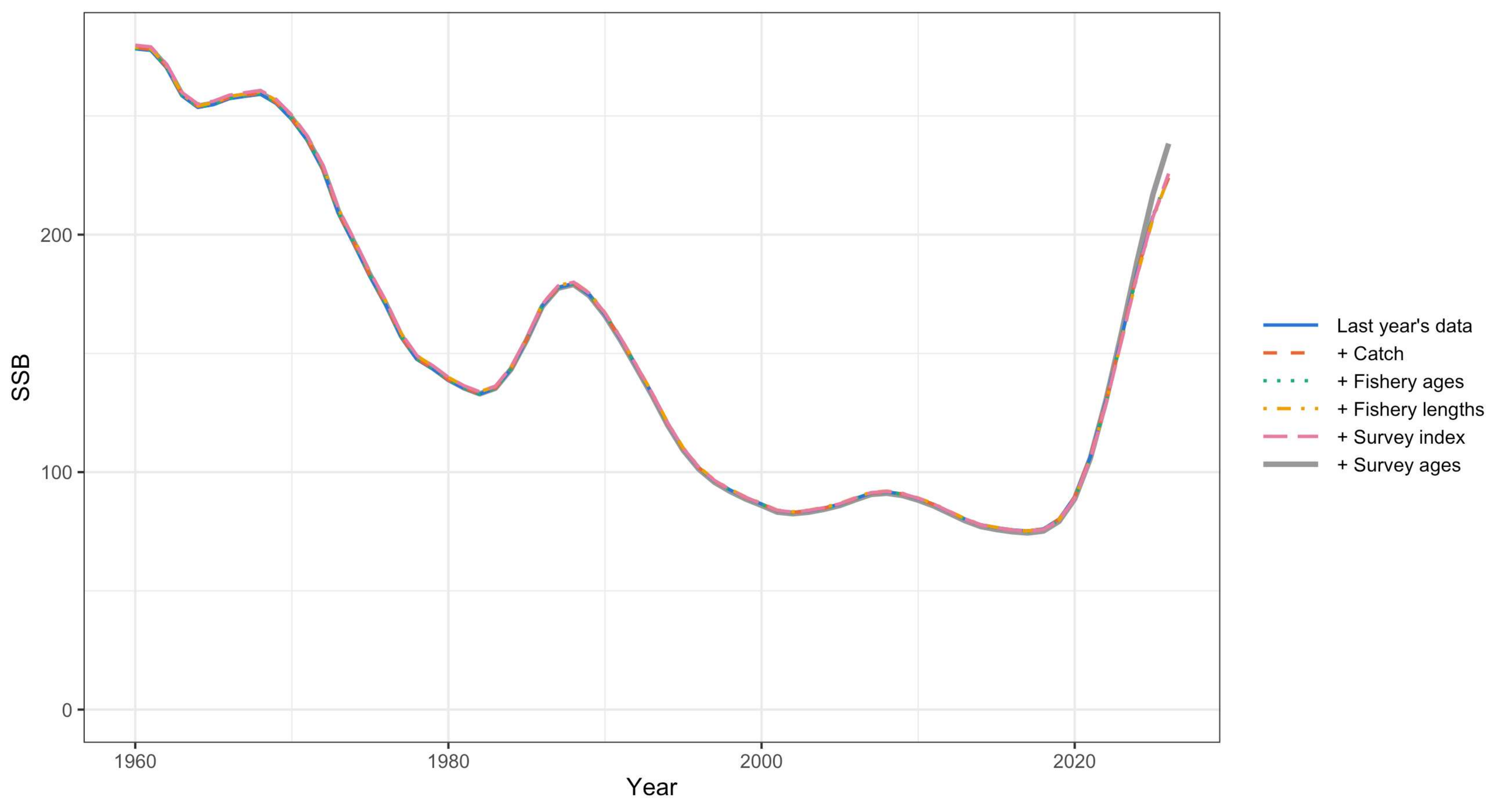
2

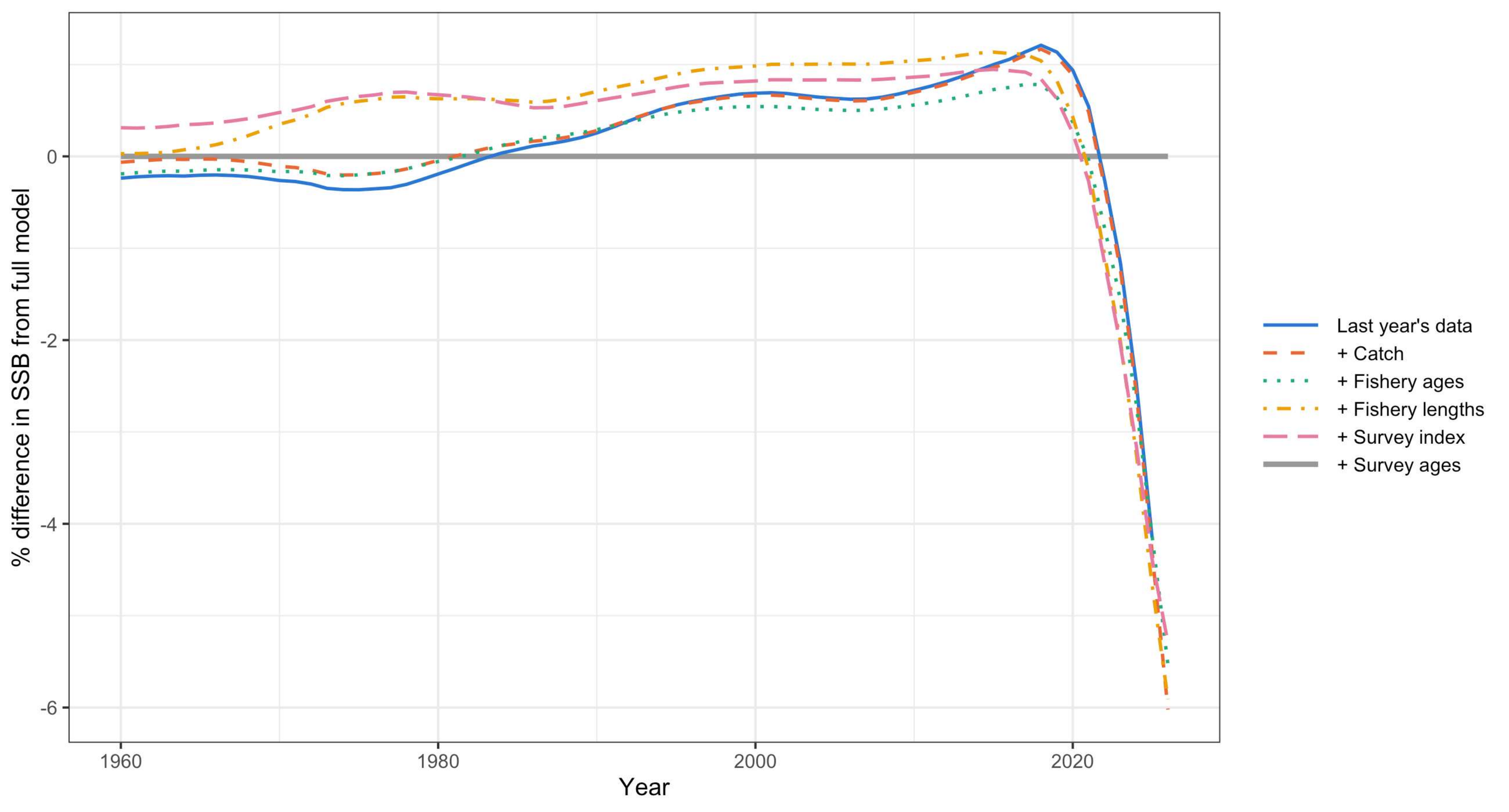
0

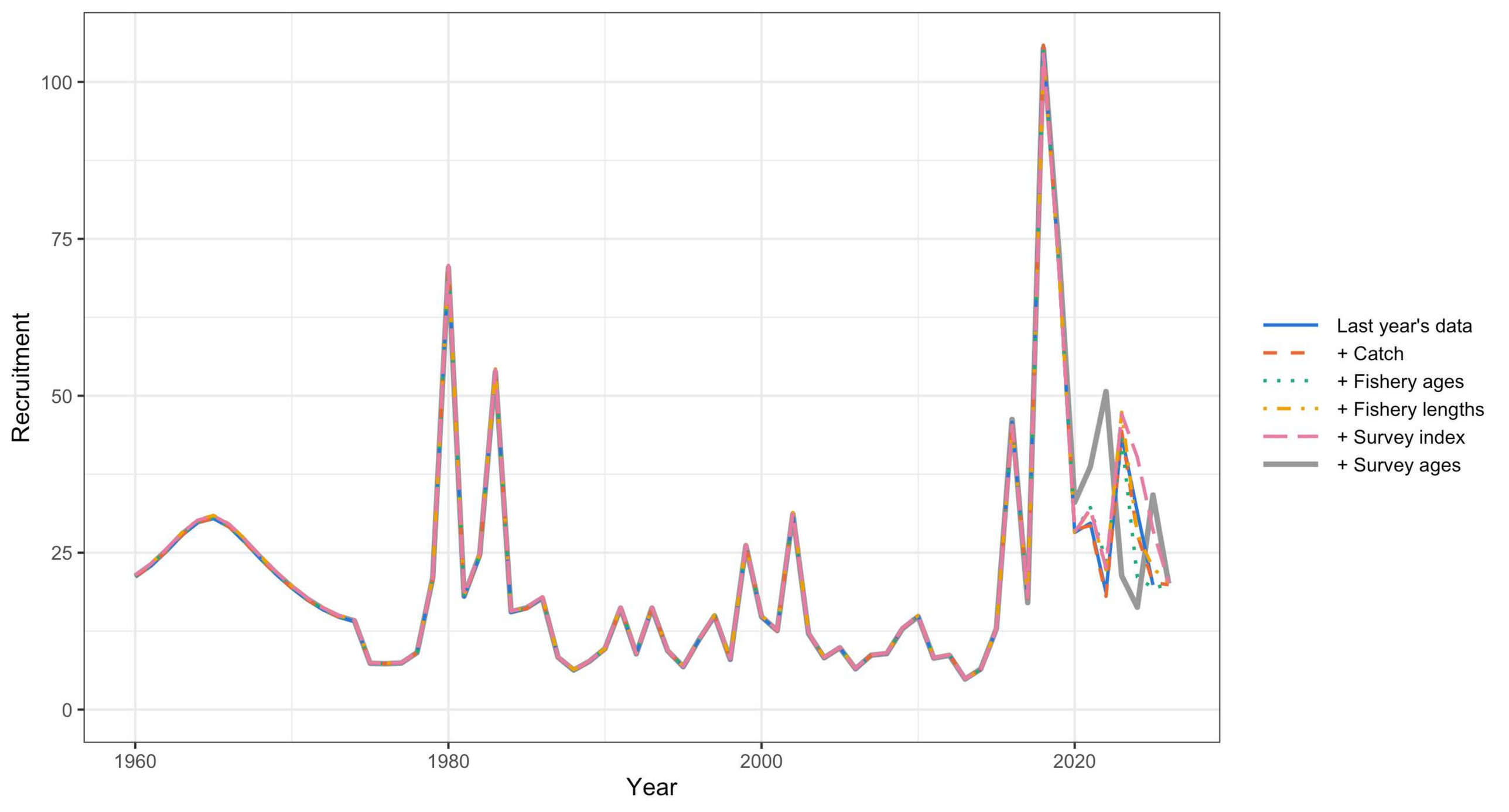
0 10 20 30

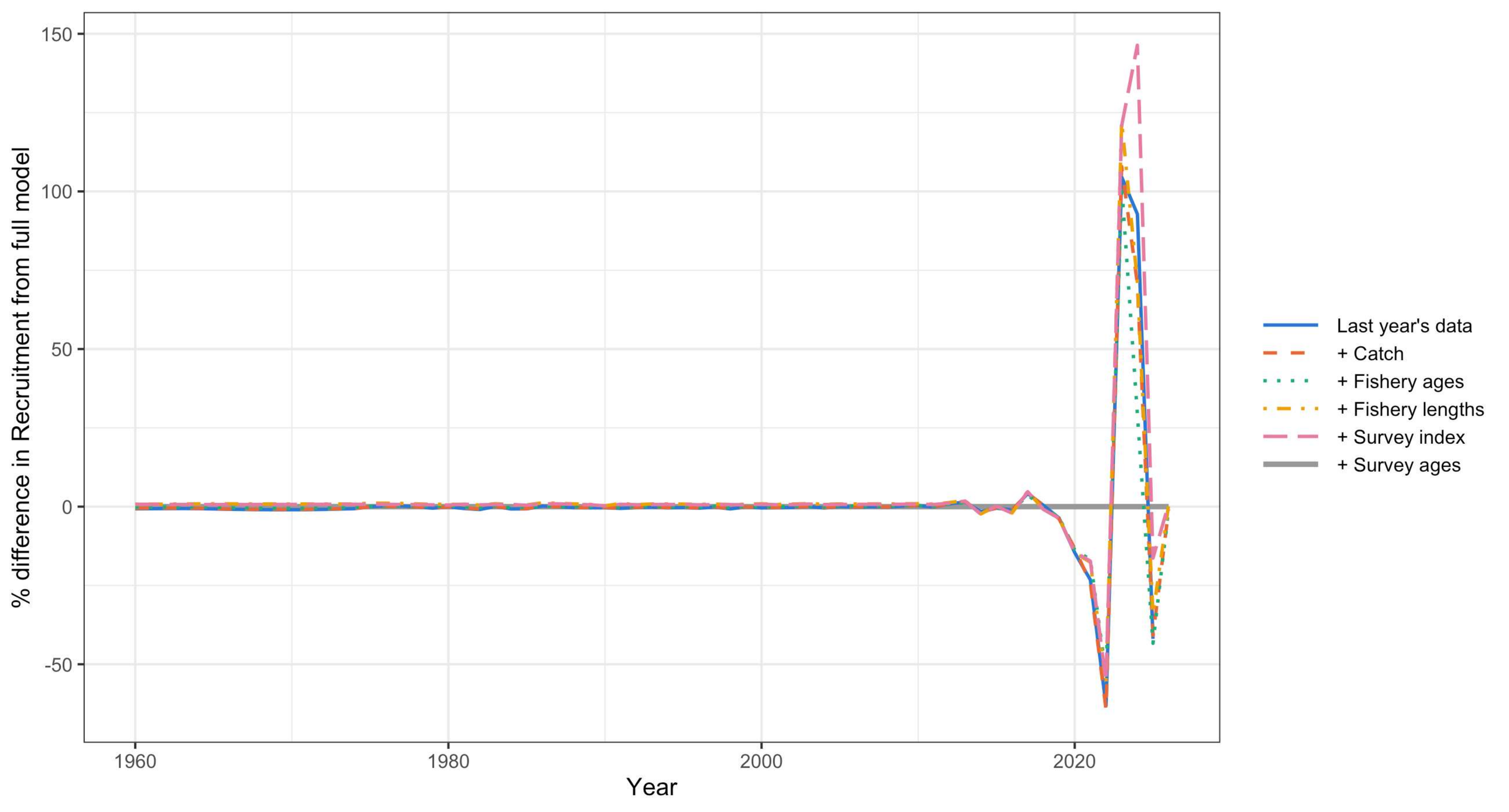
0 10 20 30

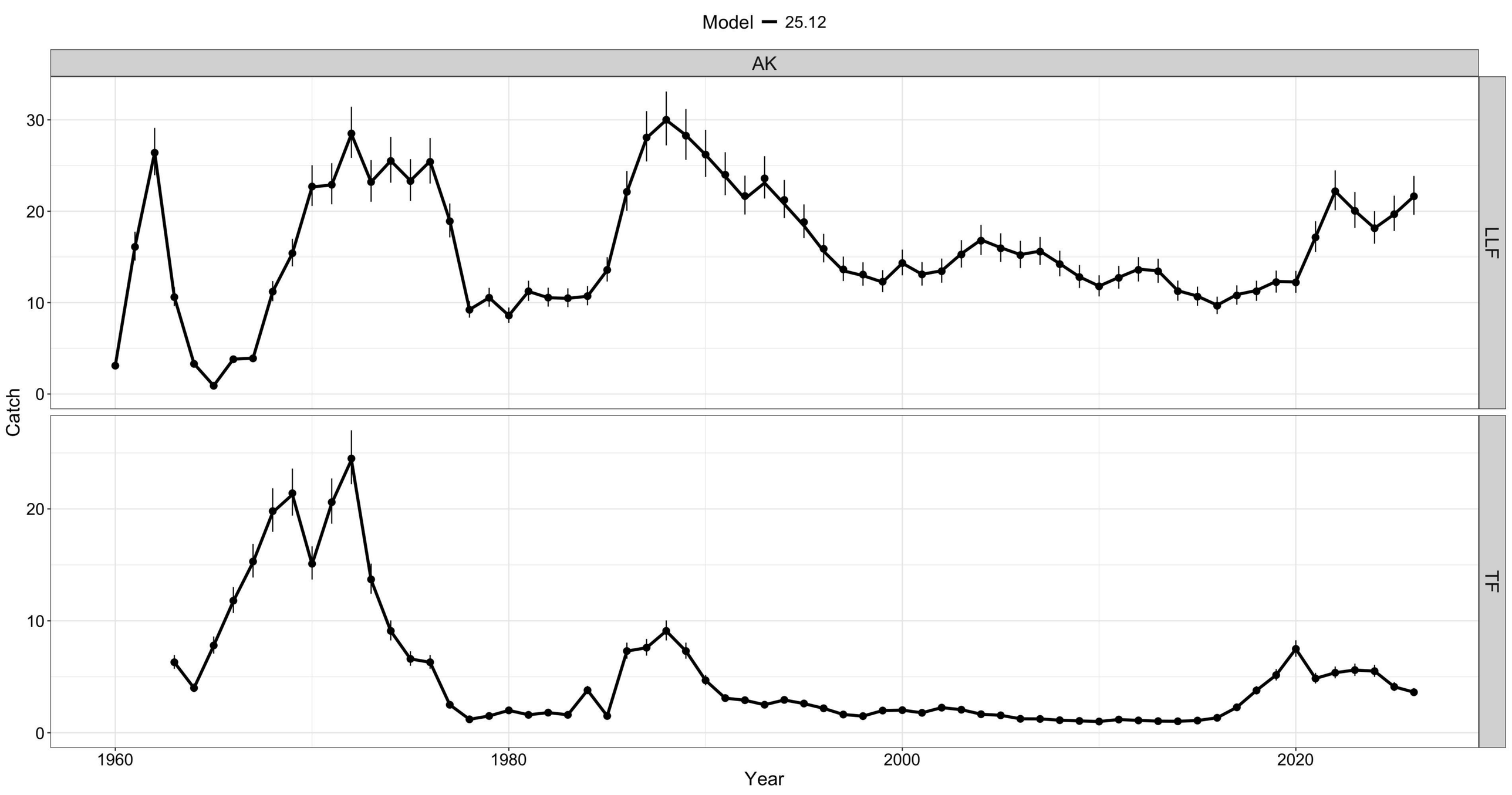
Age


SSB
200
100
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


% difference in SSB from full model
0
-2
-4
-6
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


Recruitment
100
75
50
25
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


150
100
50
0
-50
% difference in Recruitment from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


Model — 25.12
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

AK

JPN LLS Lengths

LLS Lengths

JPN LLS Index

LLS Index

JPN LLS Ages

LLS Ages

TF Lengths

LLF Lengths

LLF Index

TF Catch

LLF Catch

LLF Ages

25.12

1960

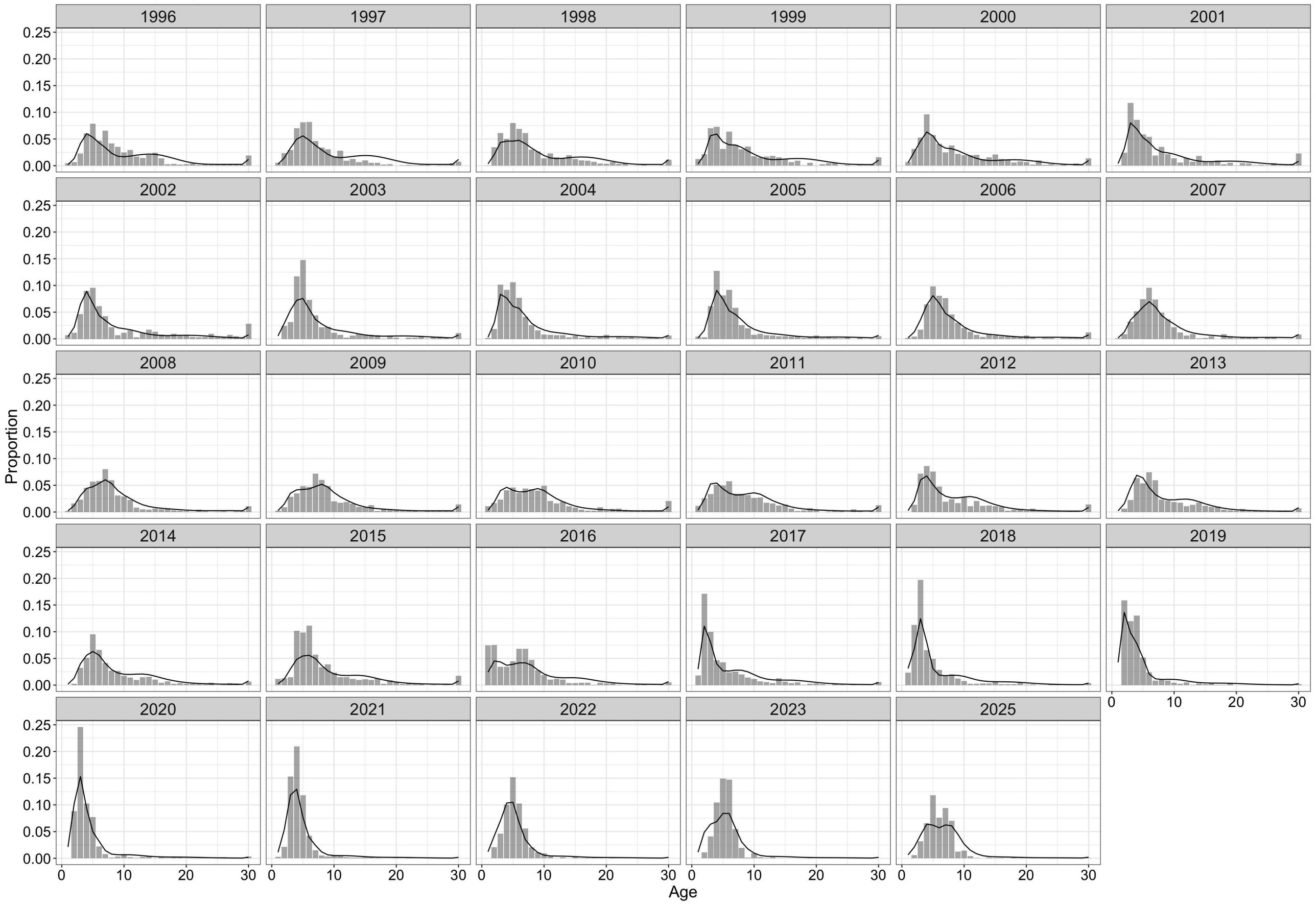
1980

2000

2020

Year

LL Survey Age Comps Female AK



LL Survey Age Comps Male AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

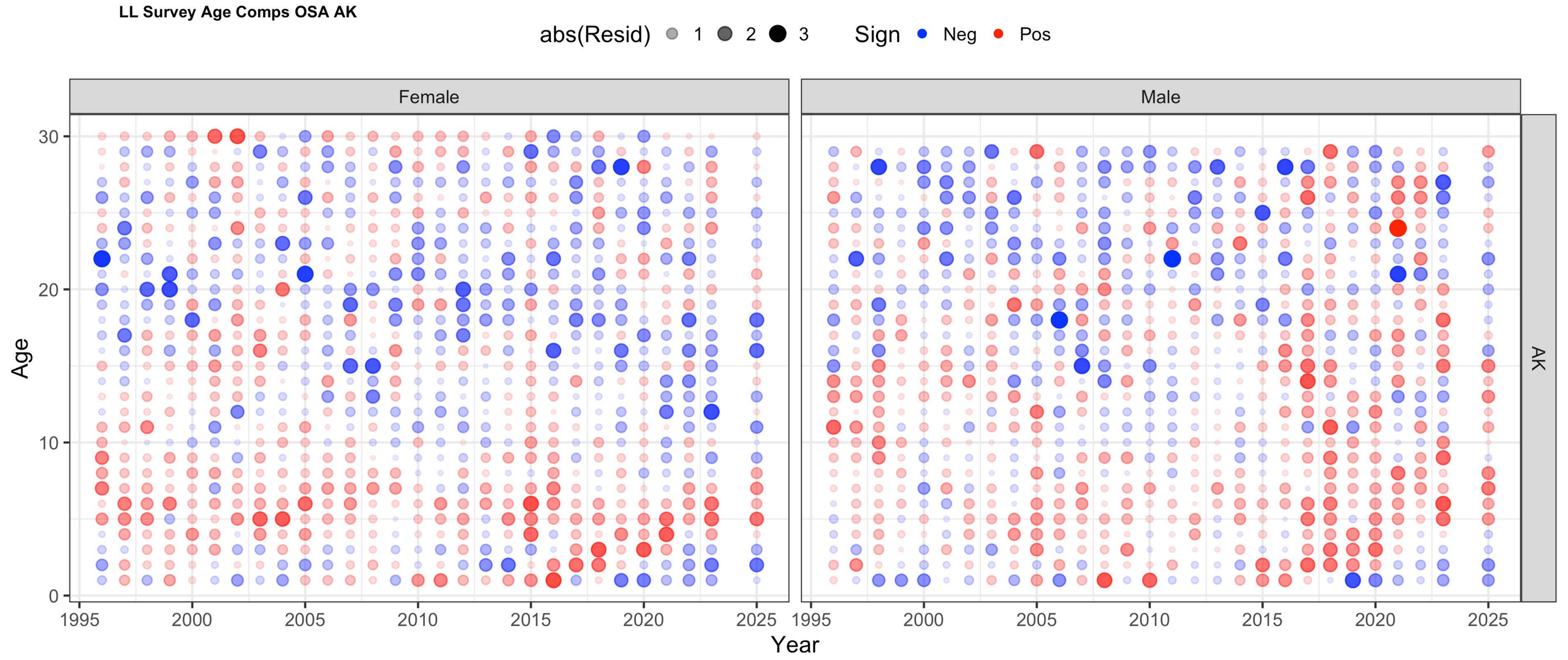
Proportion

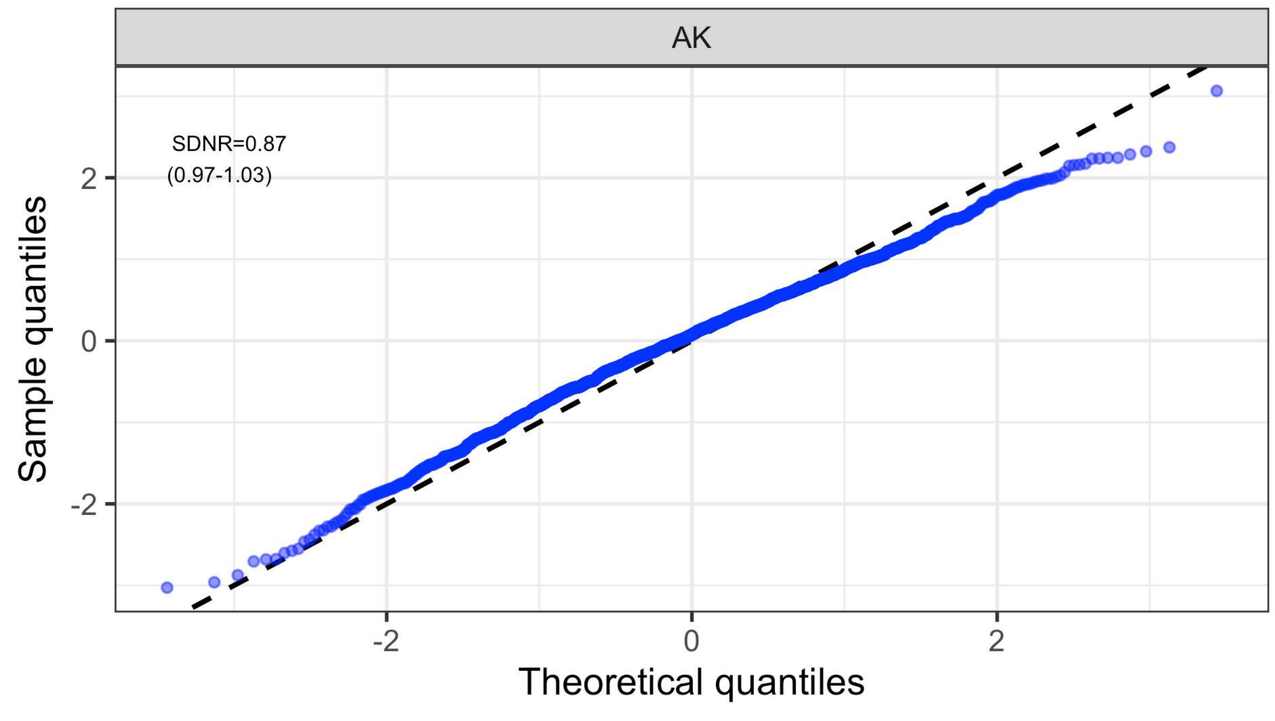
0.00 0.05 0.10 0.15

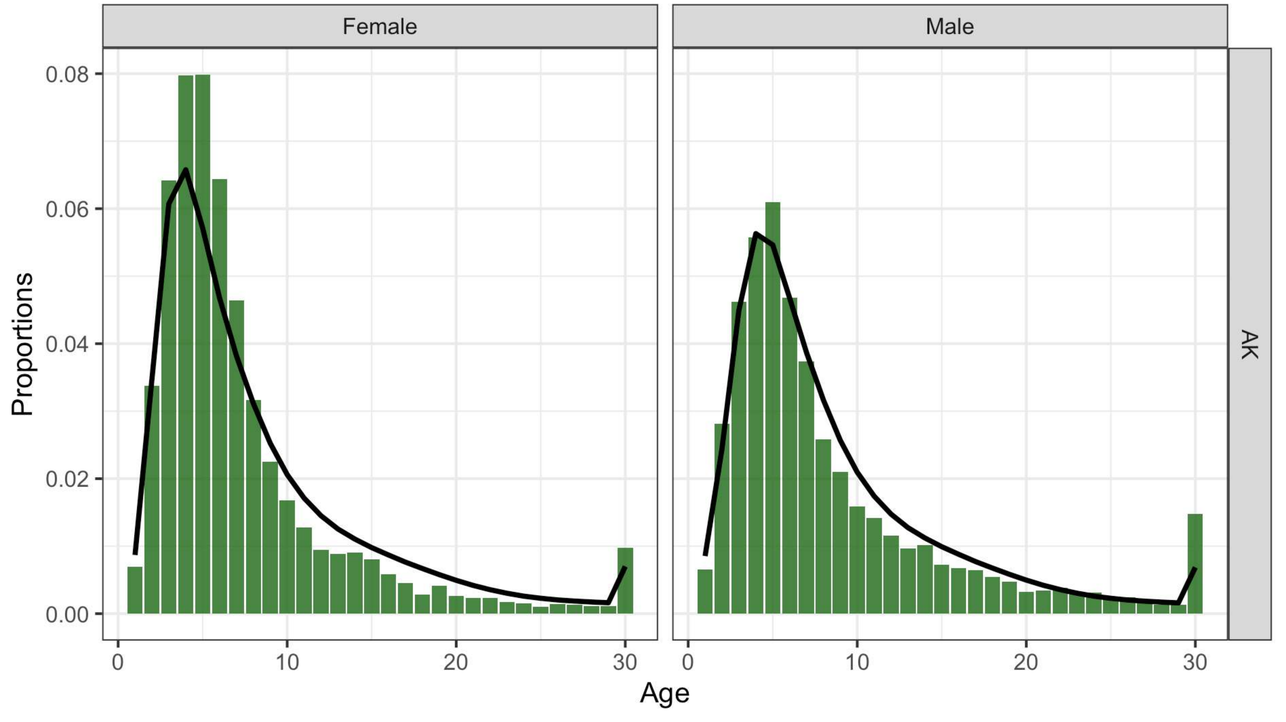
0 10 20 30

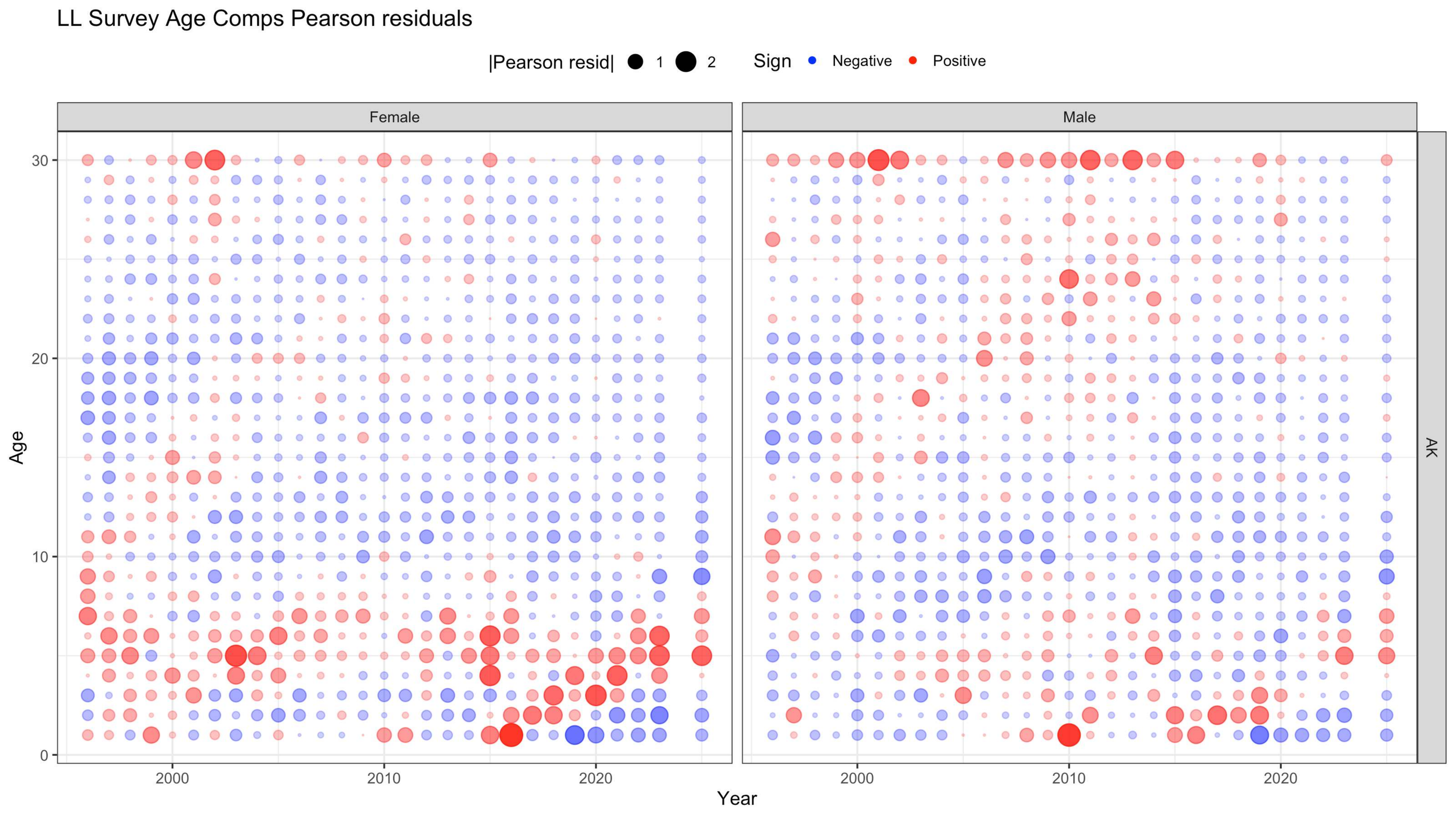
Age

LL Survey Age Comps OSA AK








LL Survey Age Comps Pearson residuals
|Pearson resid|
1
2
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

Model — 25.12

LLF Q1

LLF Q2

LLF Q3

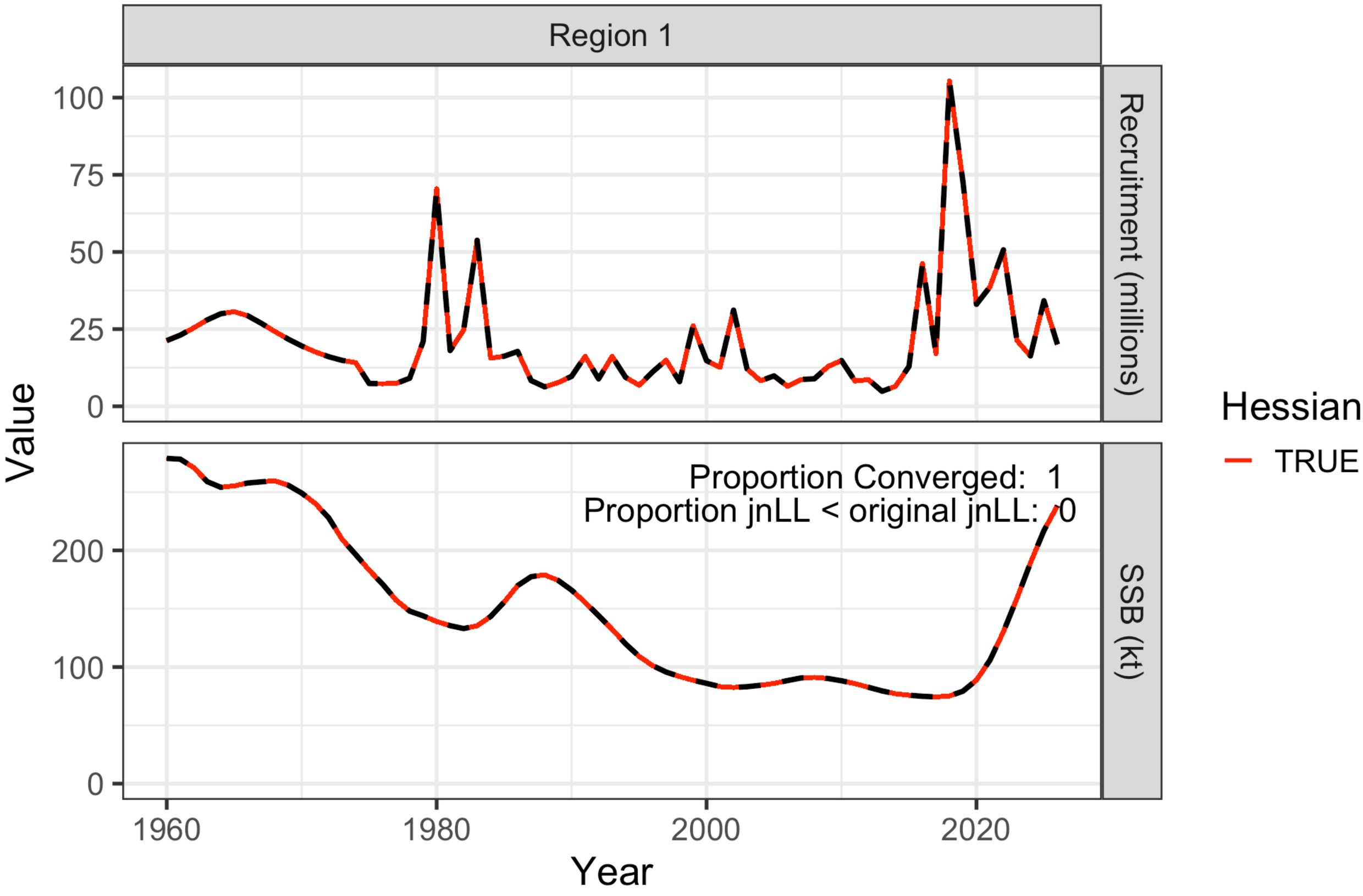
LLS

JPN LLS

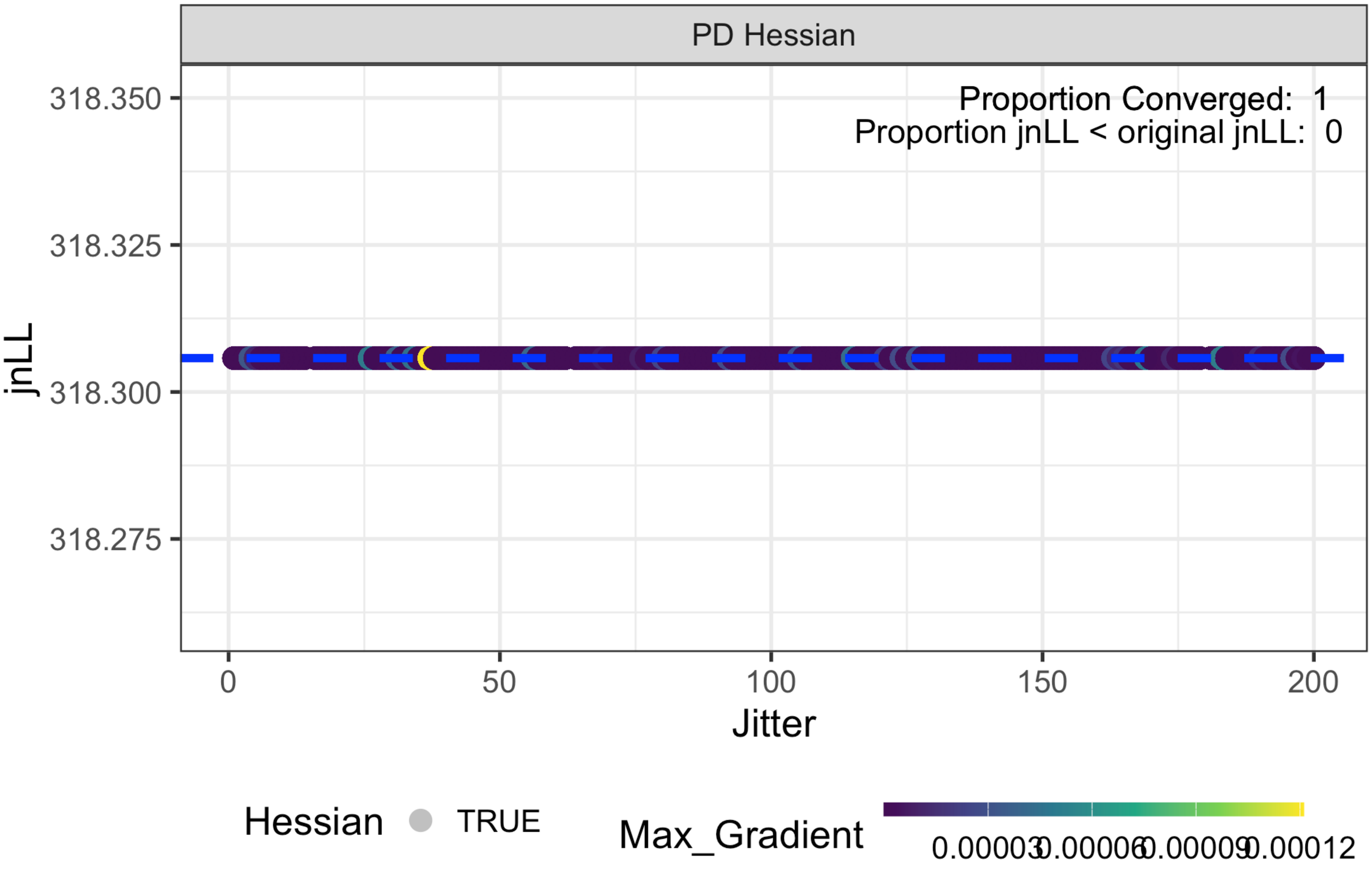
RPNs (1000s)

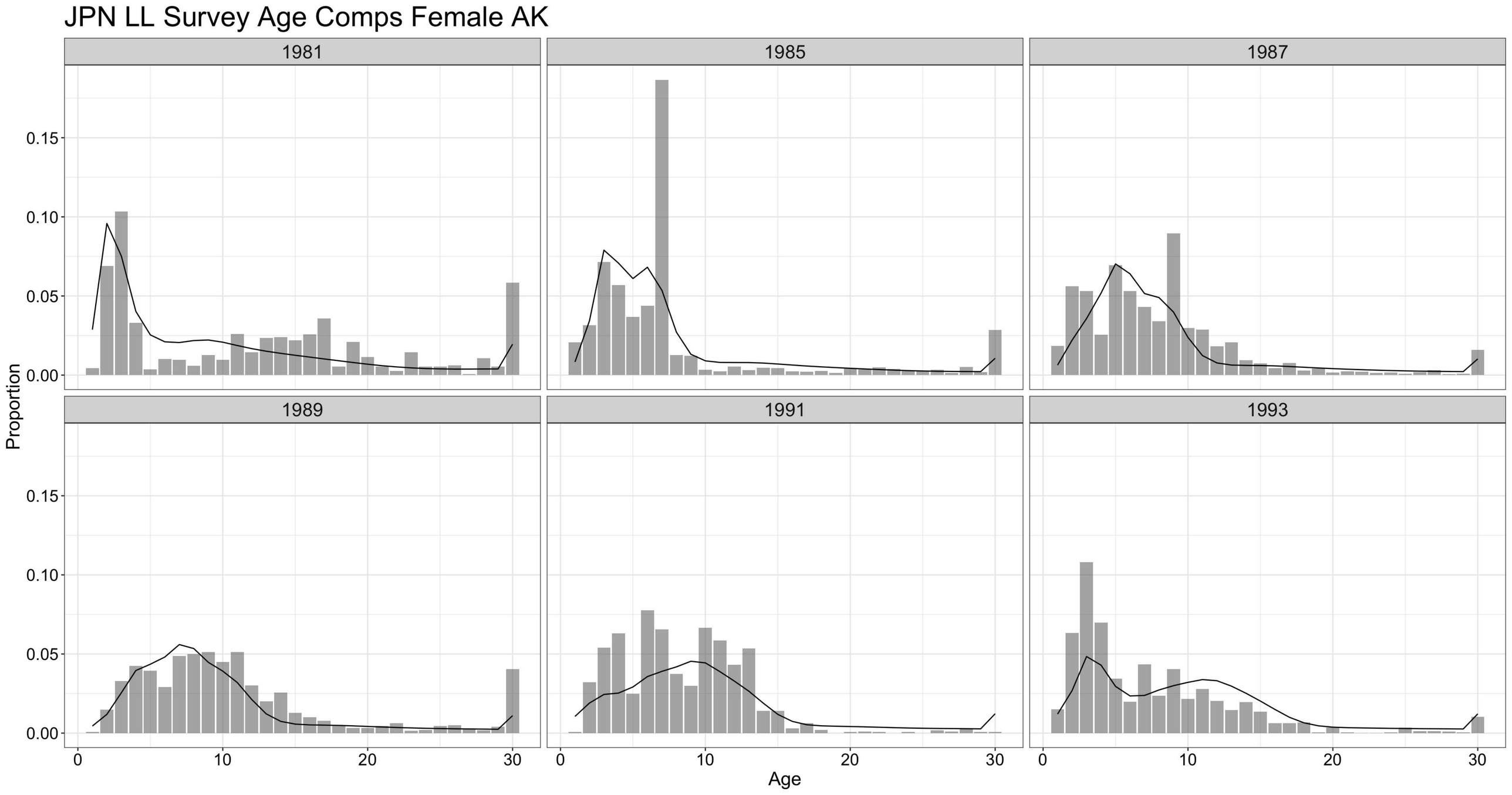
Year

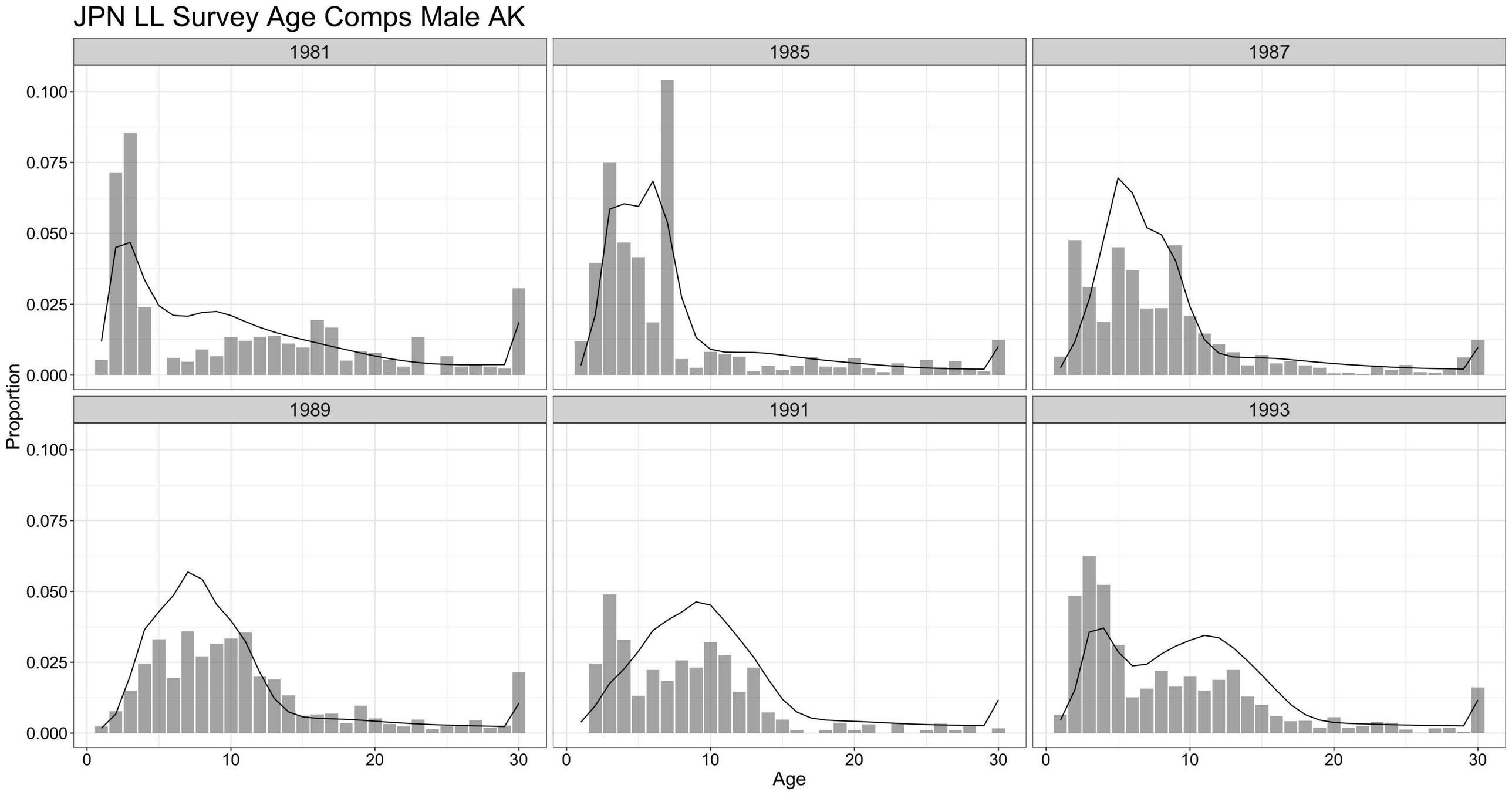
Jitter SSB and Recruitment



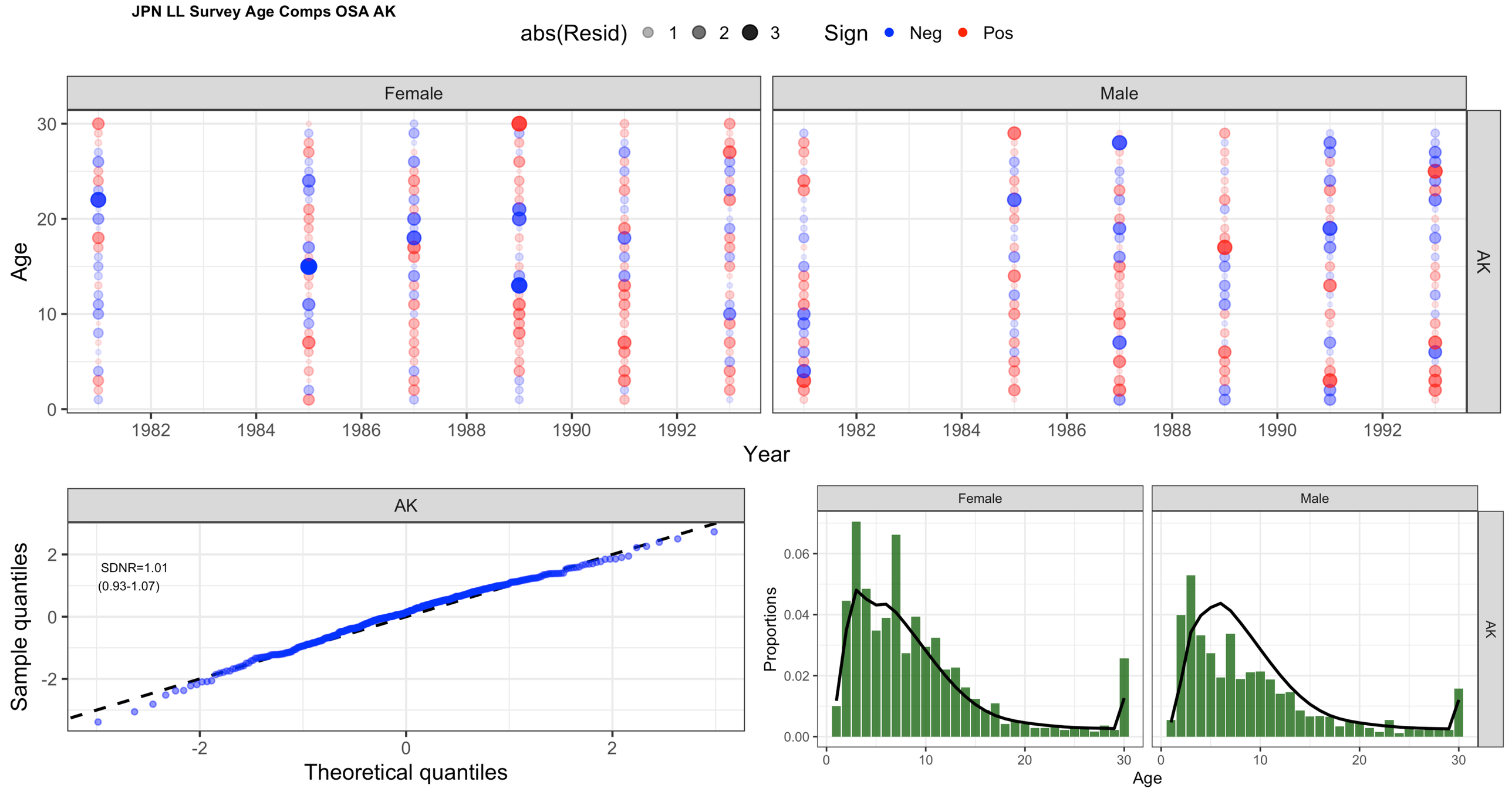
Jitter nLL and Convergence

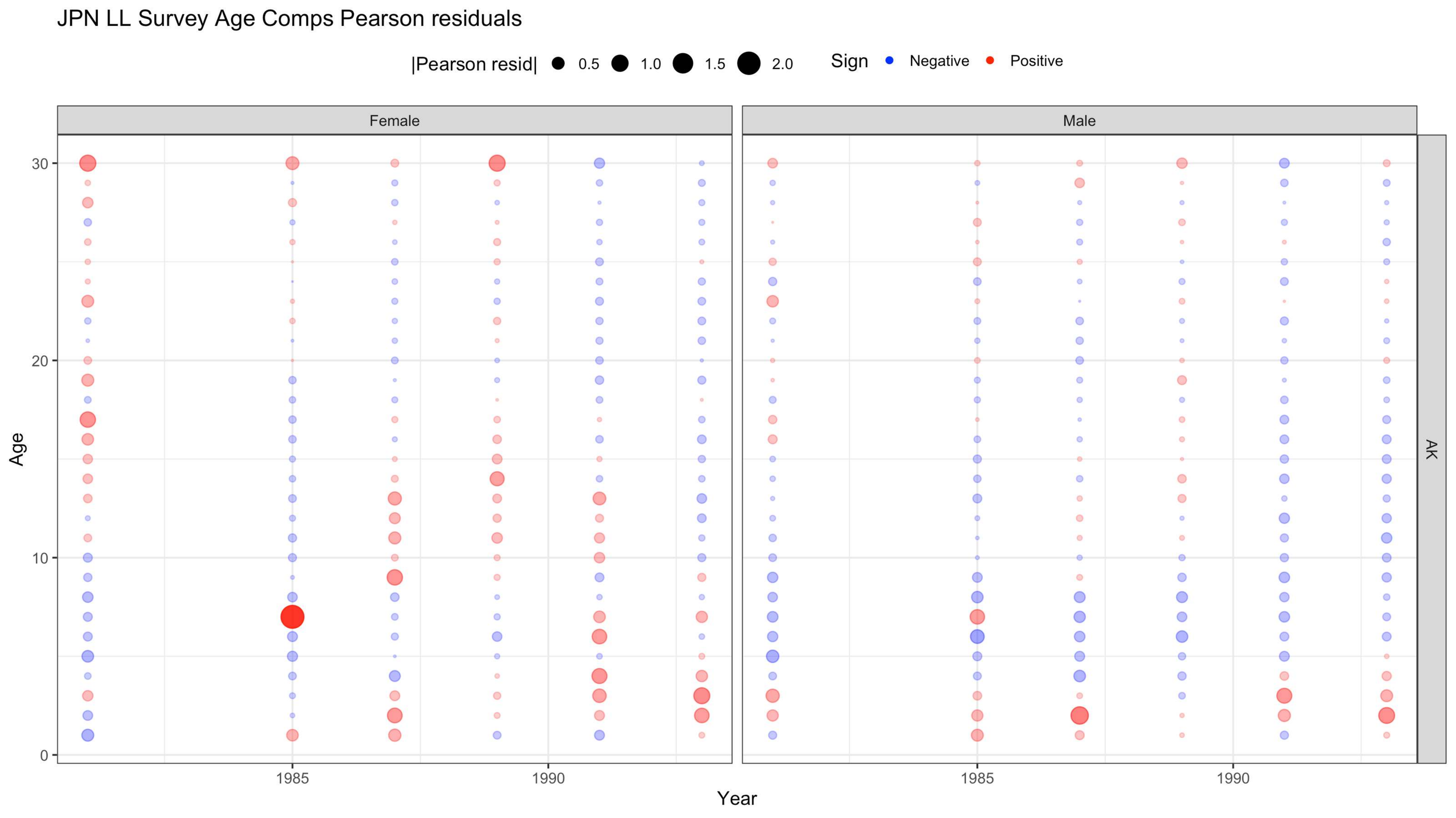


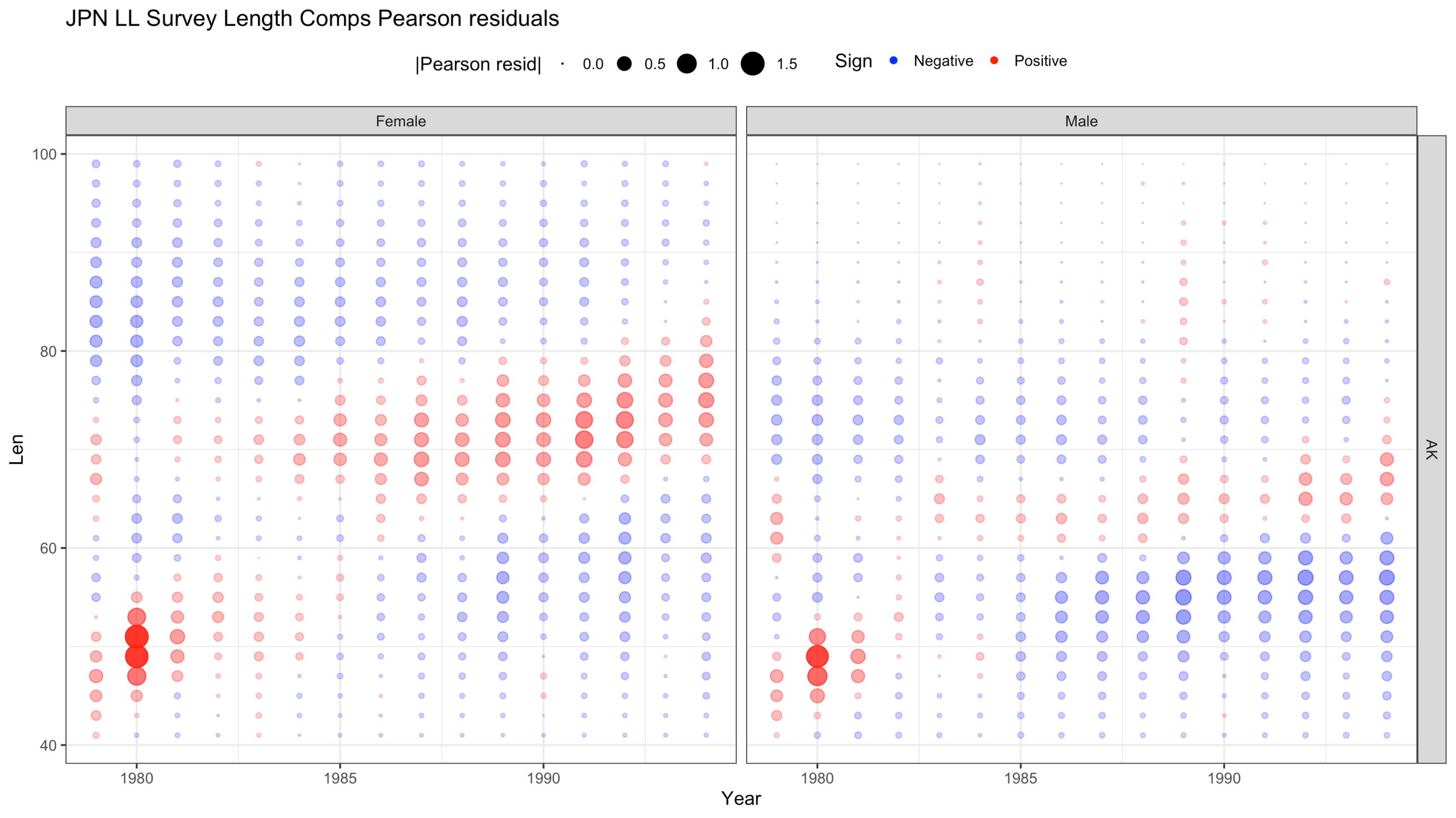

JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
Proportion
0.100
0.075
0.050
0.025
0.000
0
10
20
30
Age

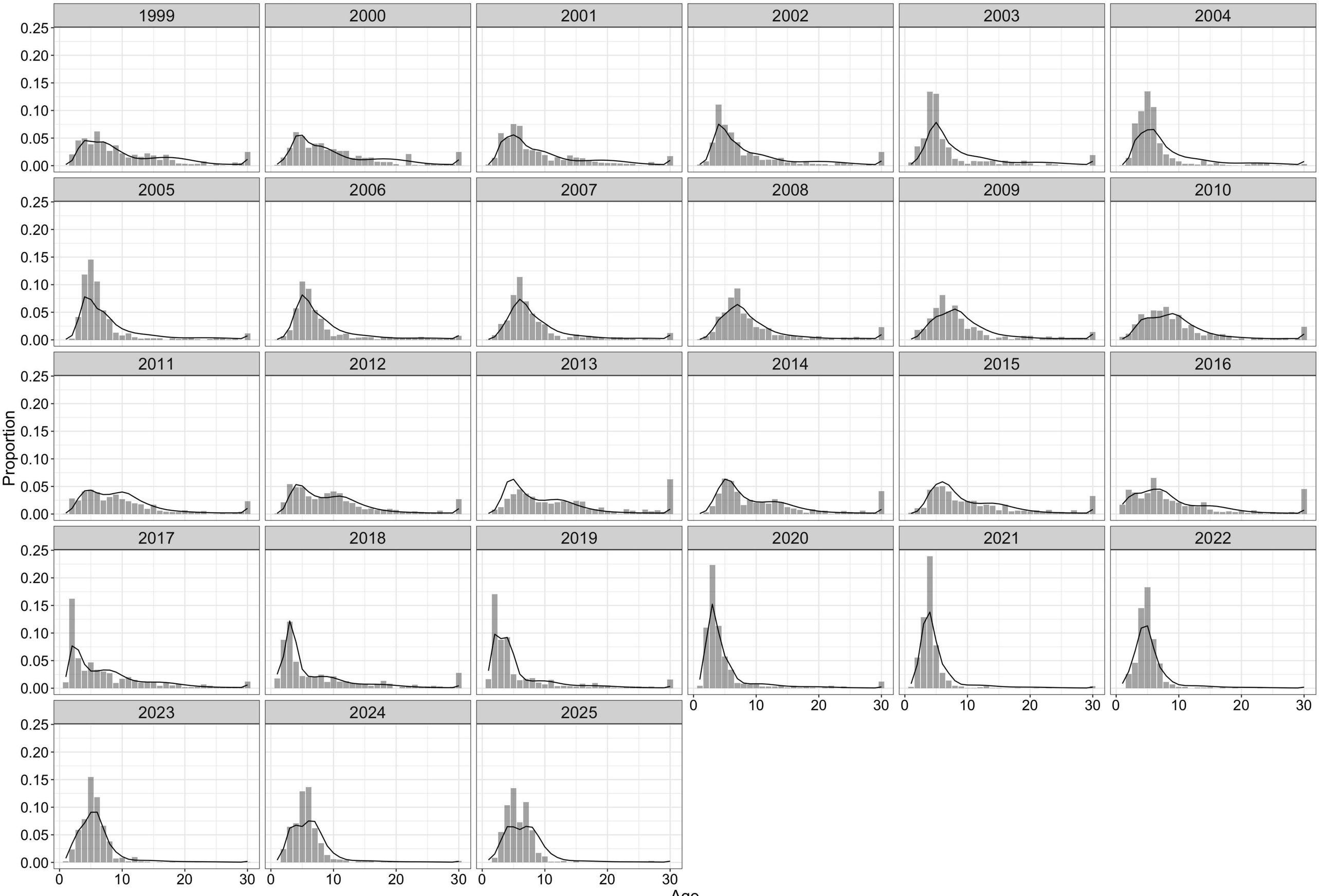
JPN LL Survey Age Comps OSA AK



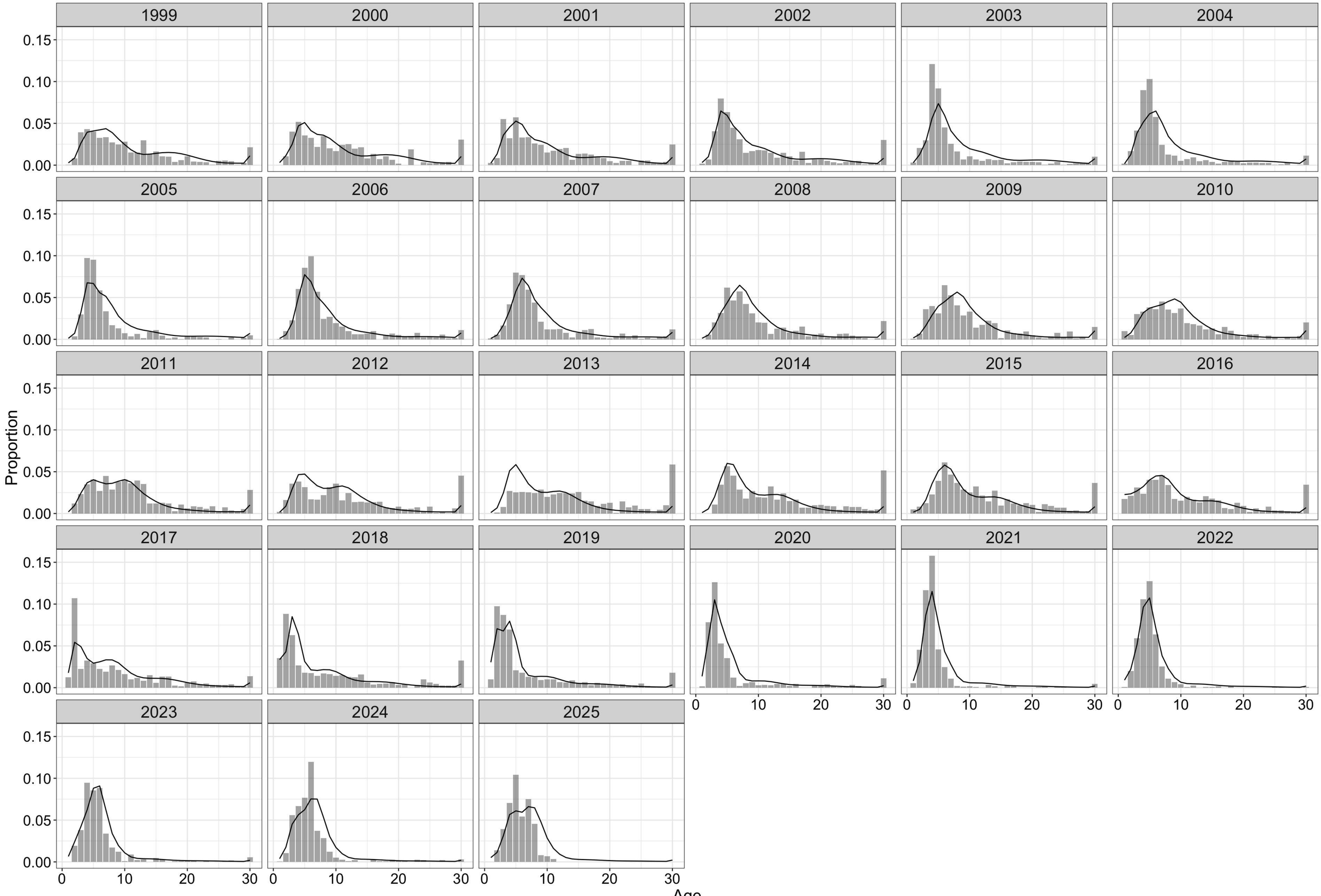

JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year


JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

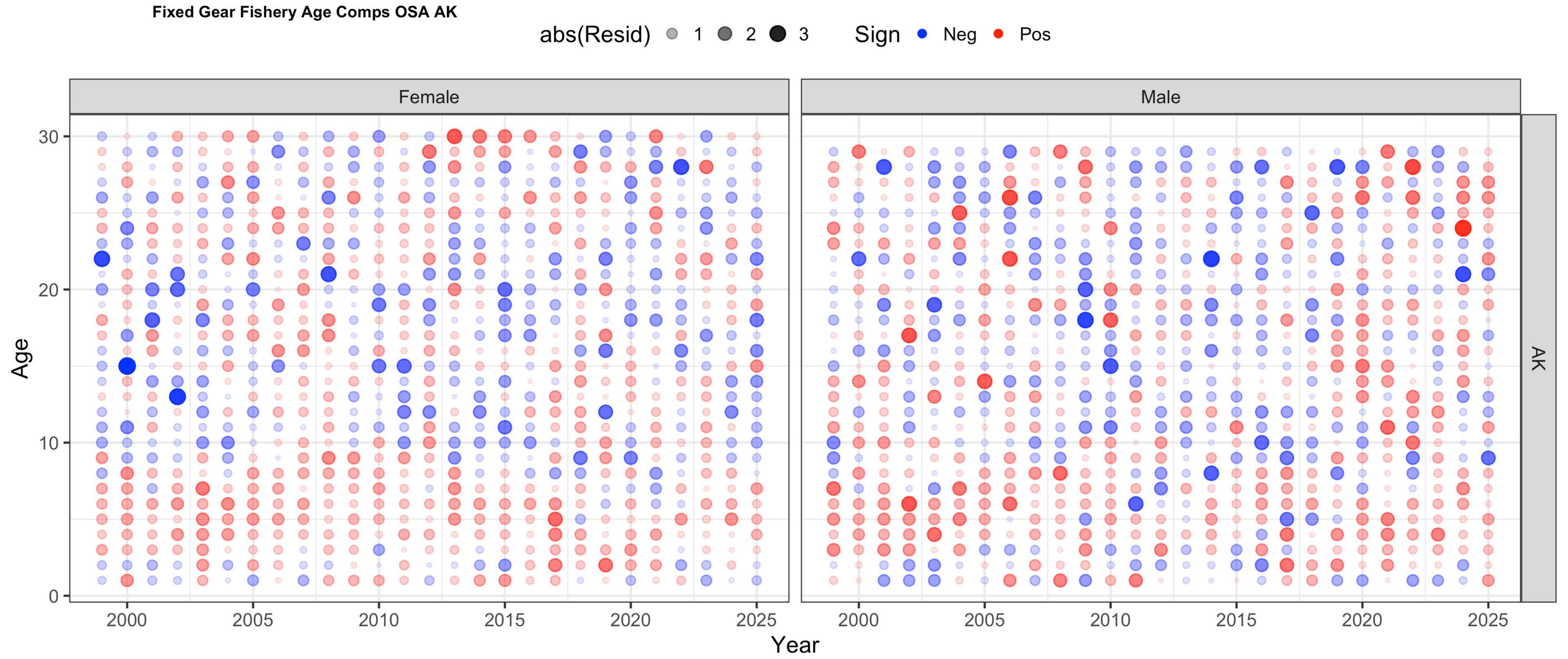
Fixed Gear Fishery Age Comps Female AK

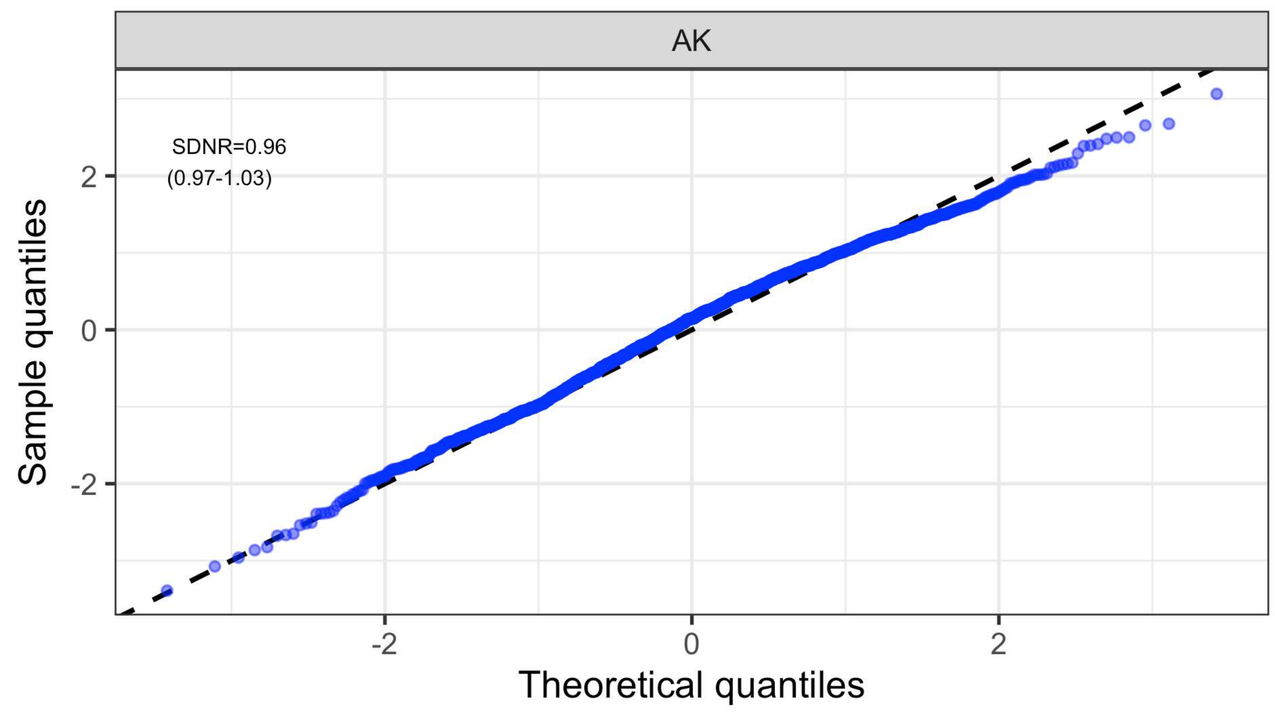


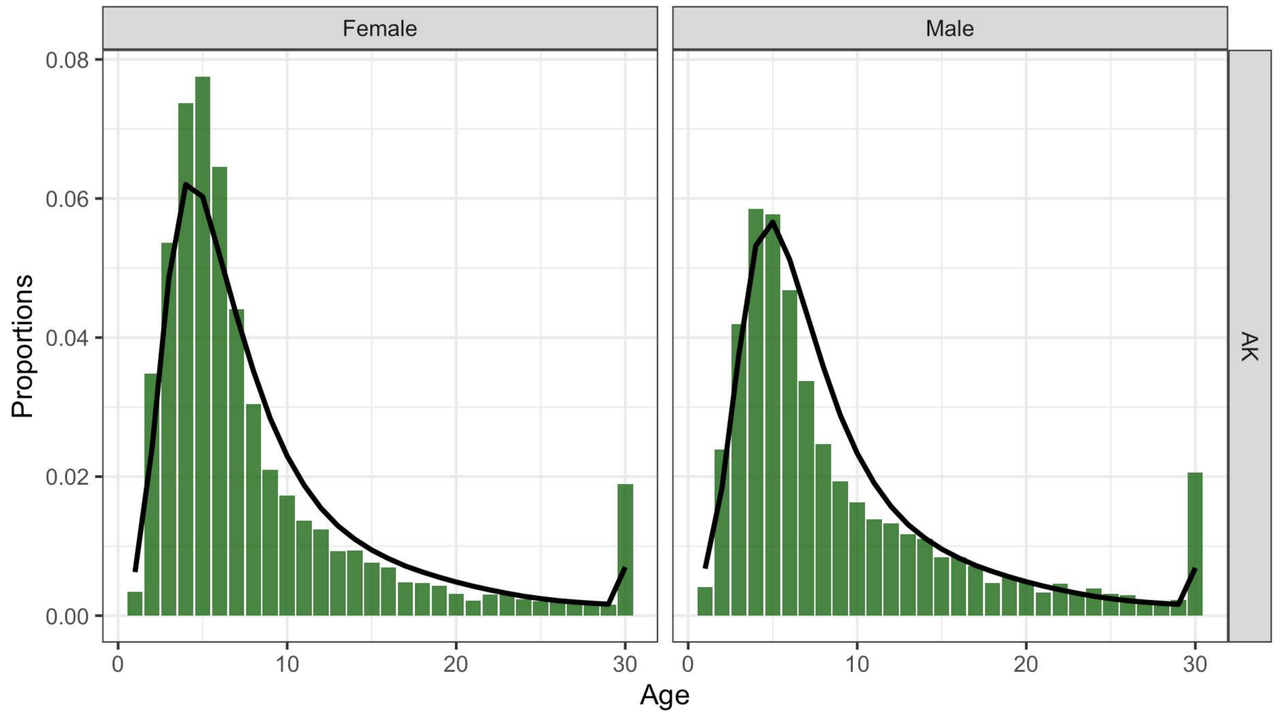
Fixed Gear Fishery Age Comps Male AK

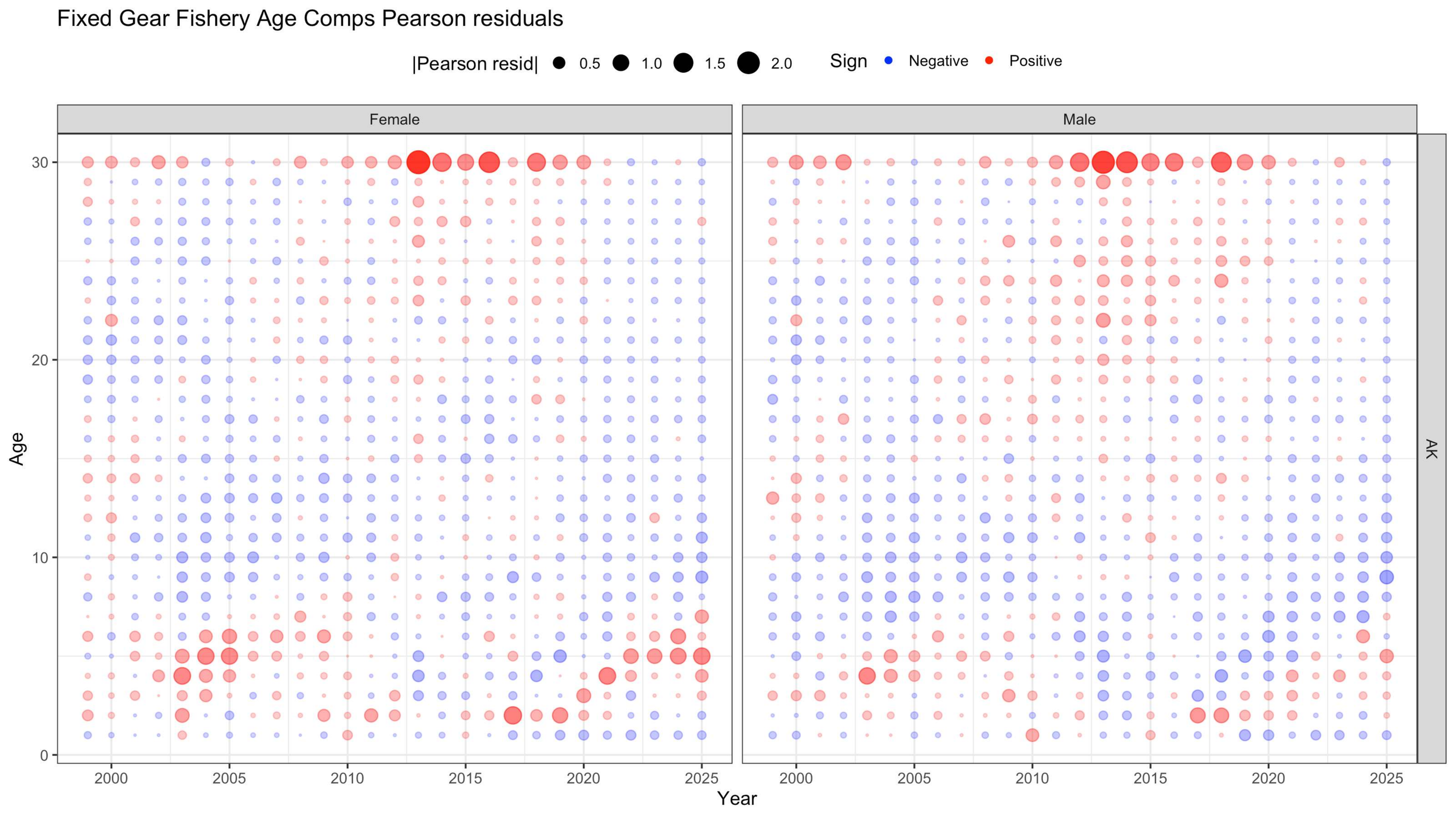


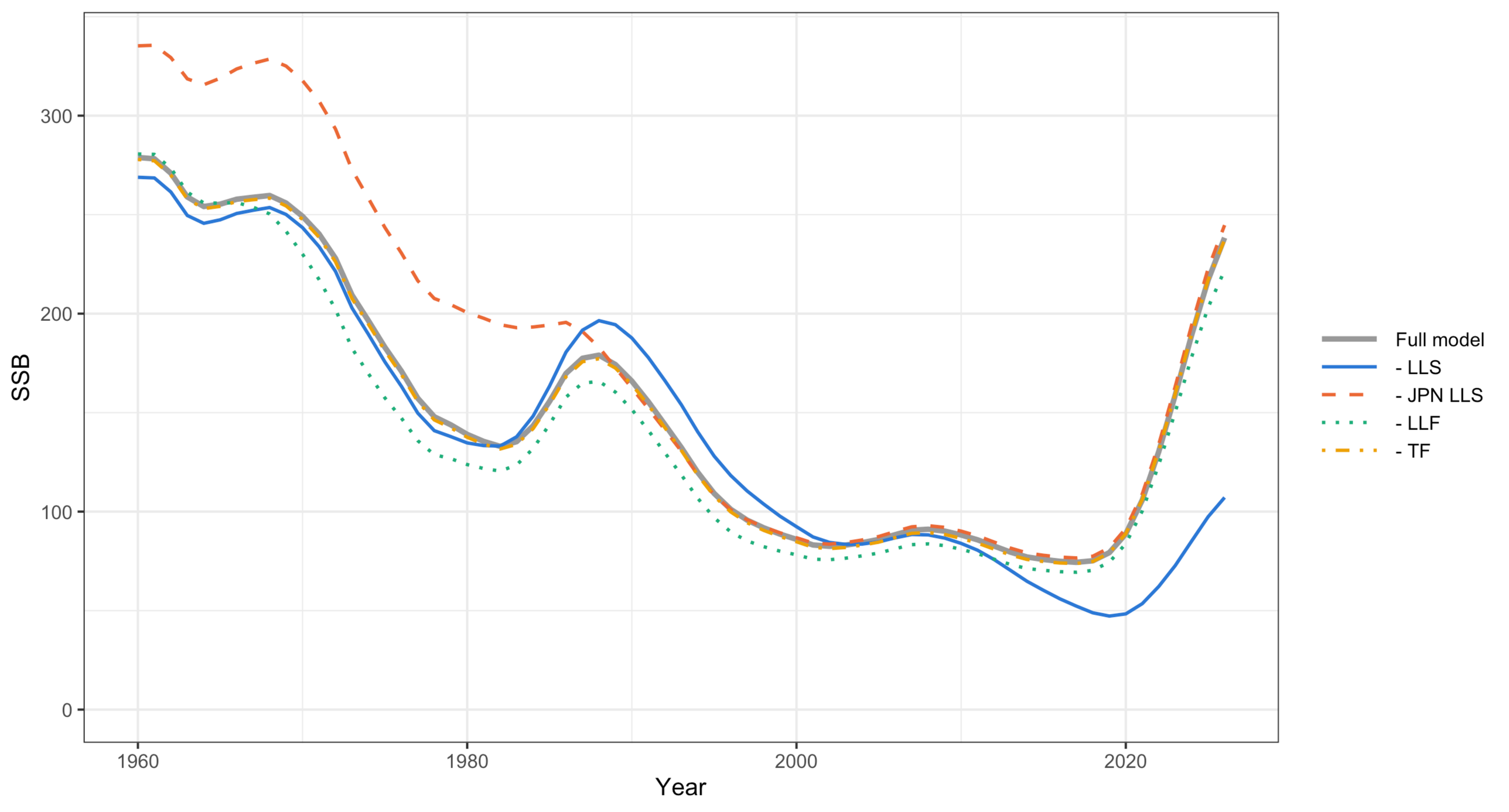
Fixed Gear Fishery Age Comps OSA AK

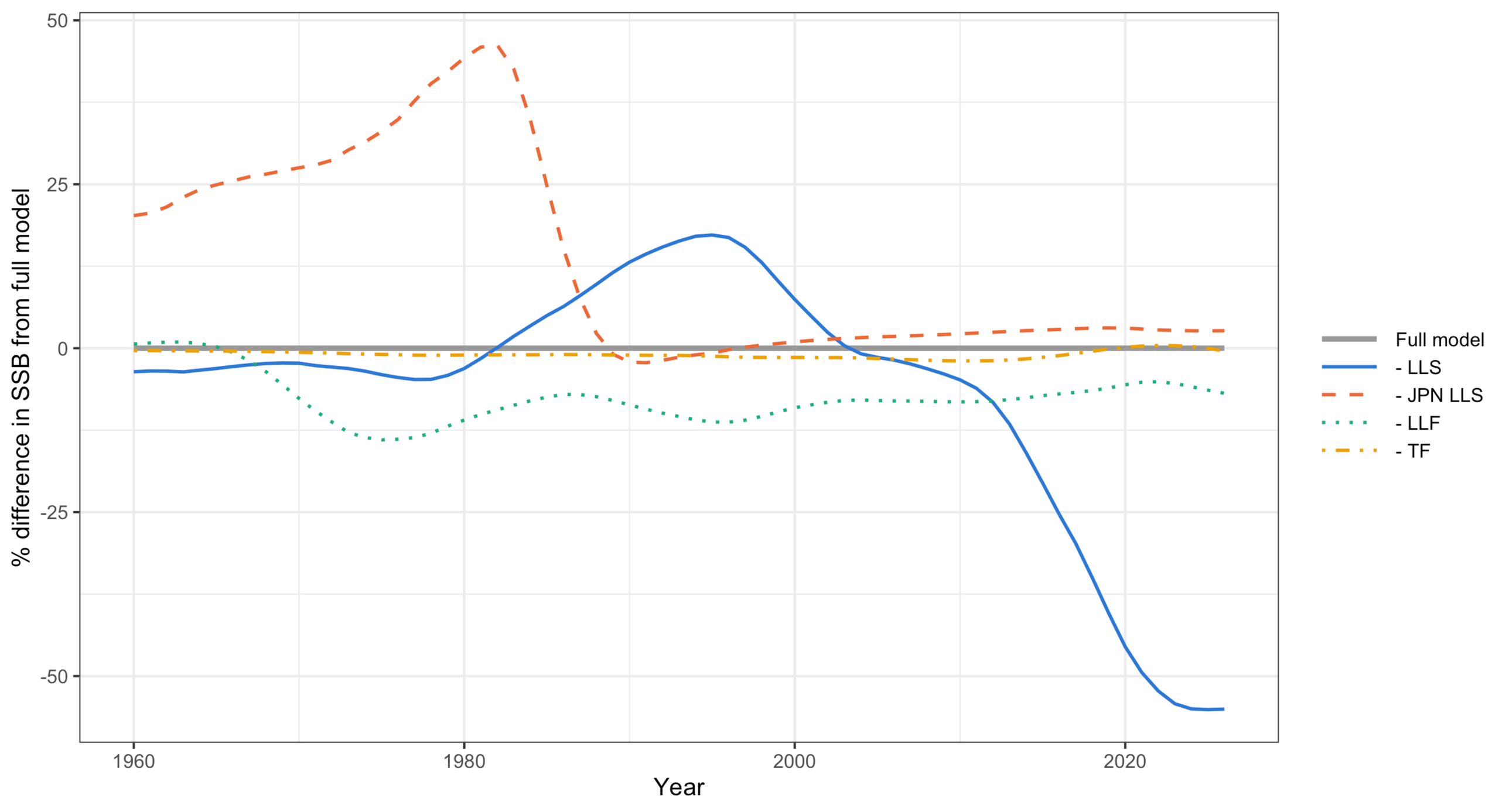


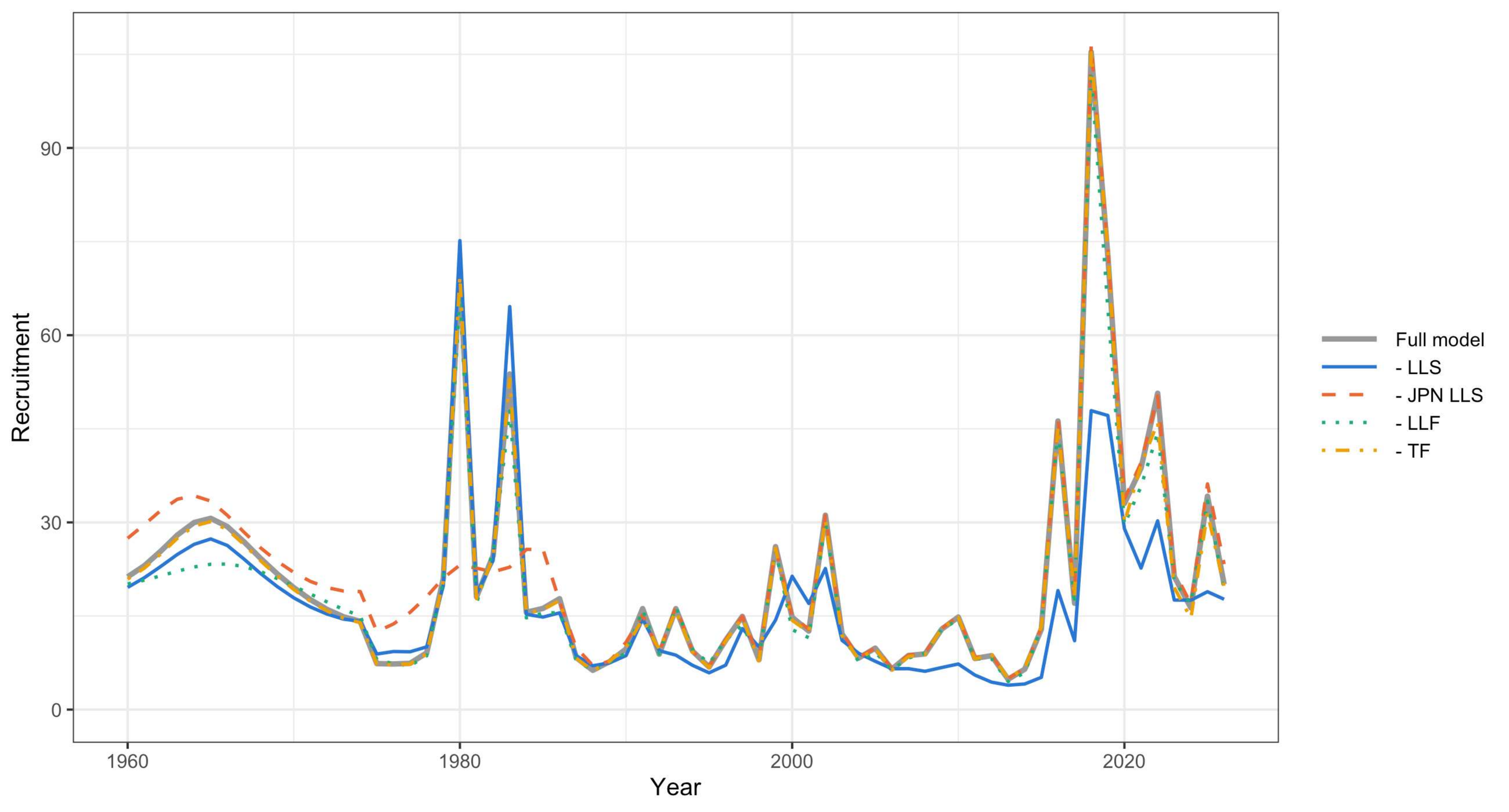


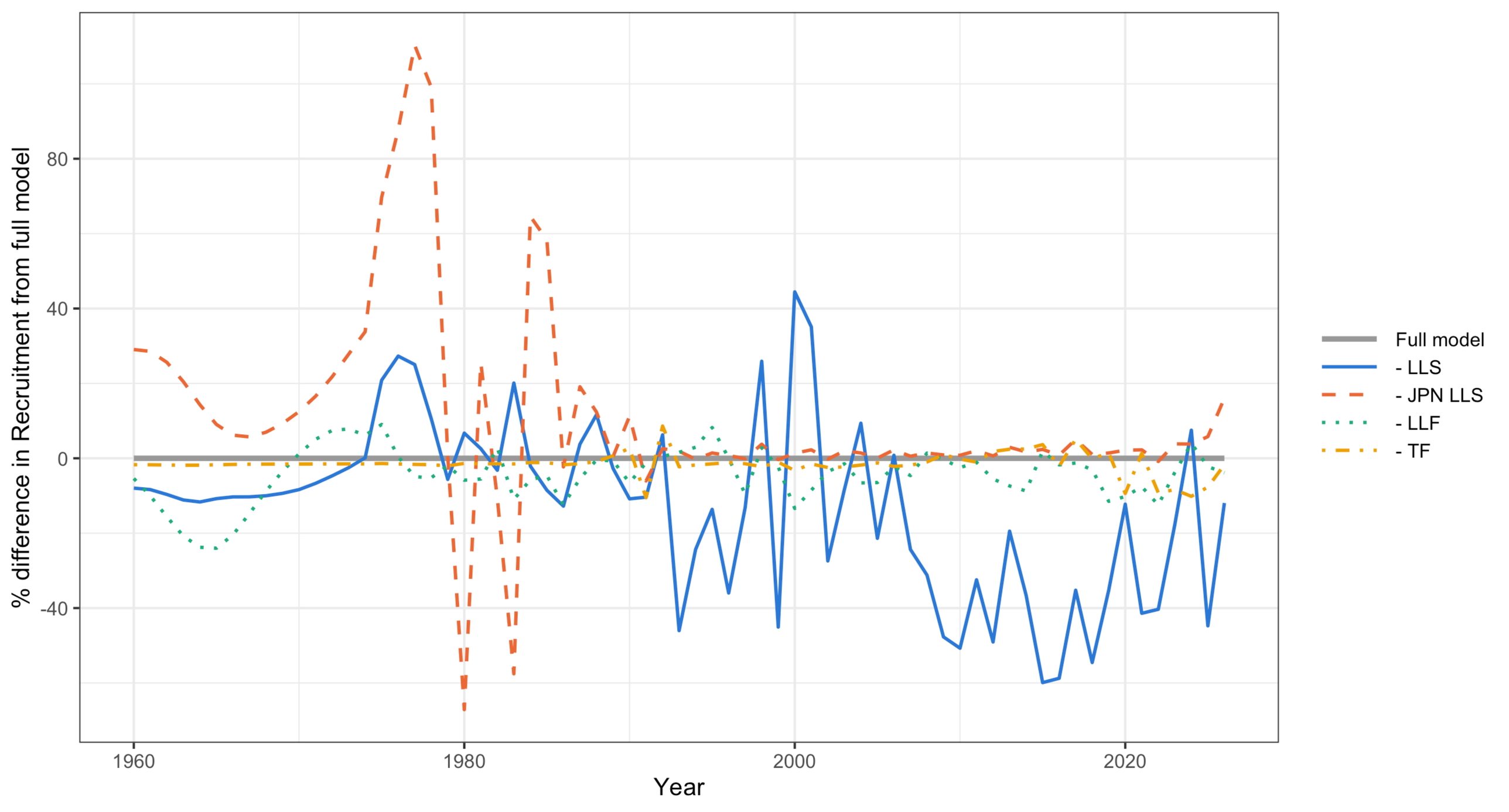



Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025


SSB
300
200
100
0
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF

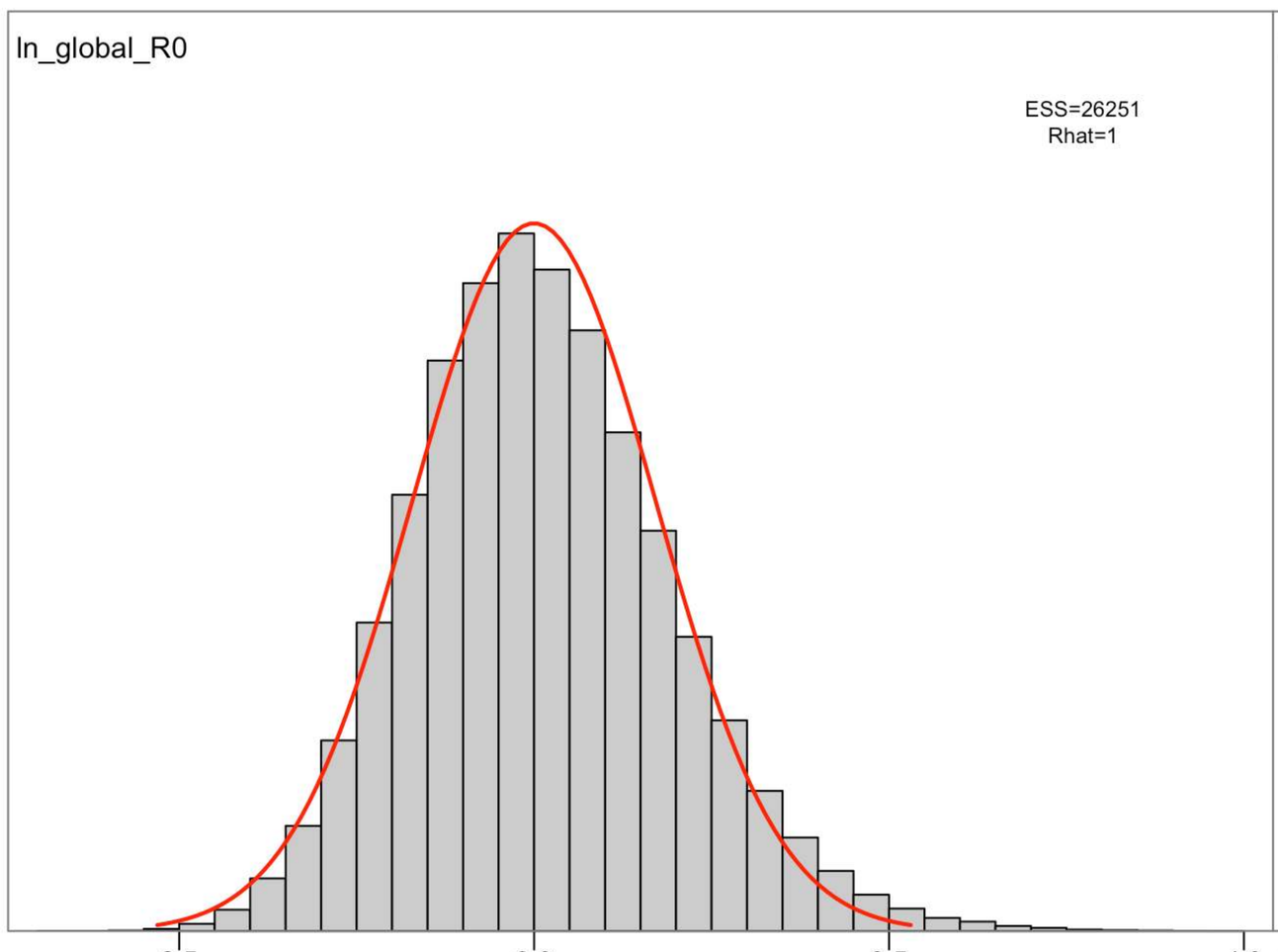

50
25
0
-25
-50
% difference in SSB from full model
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF


Recruitment
90
60
30
0
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF


% difference in Recruitment from full model
80
40
0
-40
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF

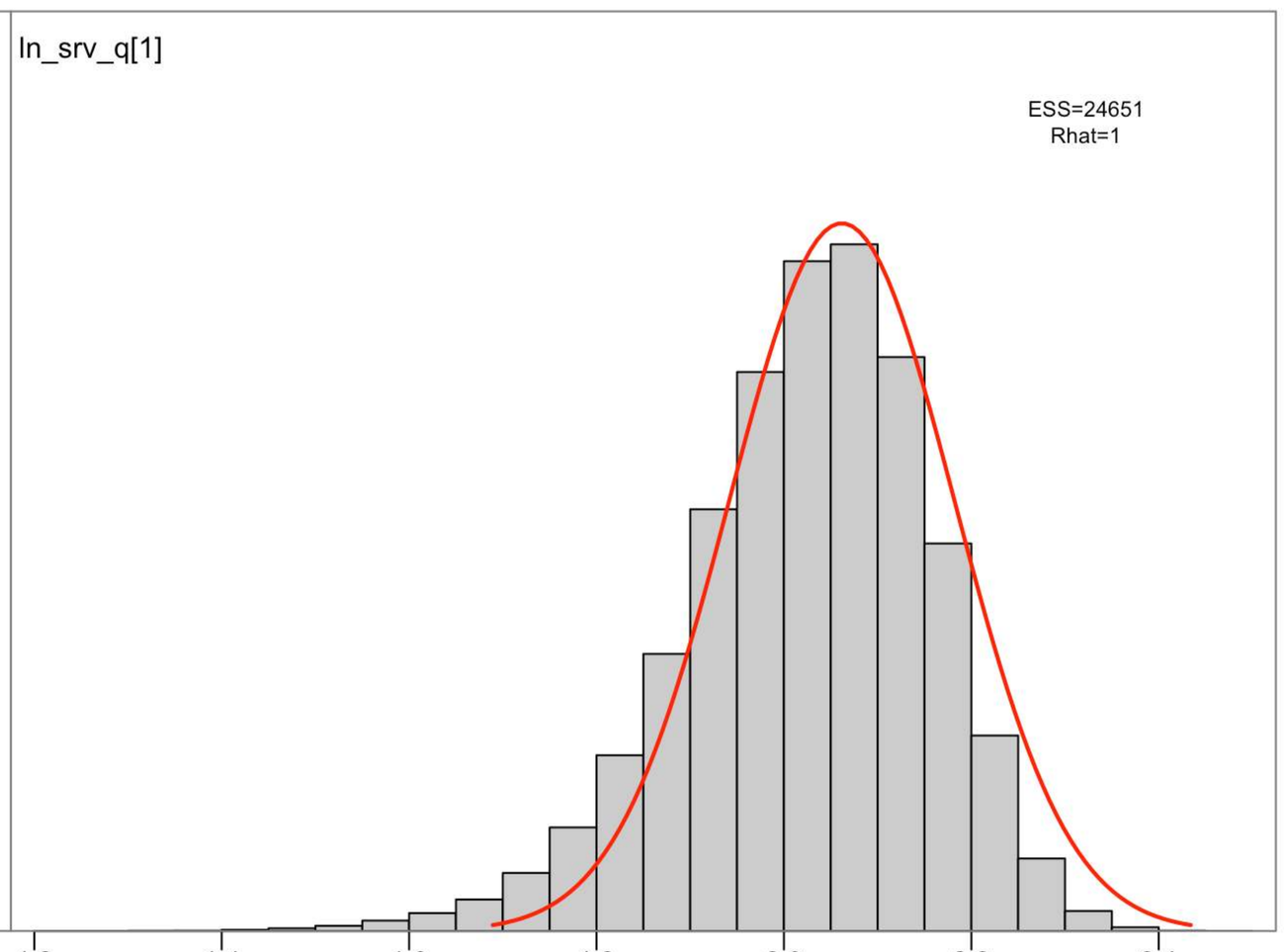
ln_global_R0

ESS=26251
Rhat=1



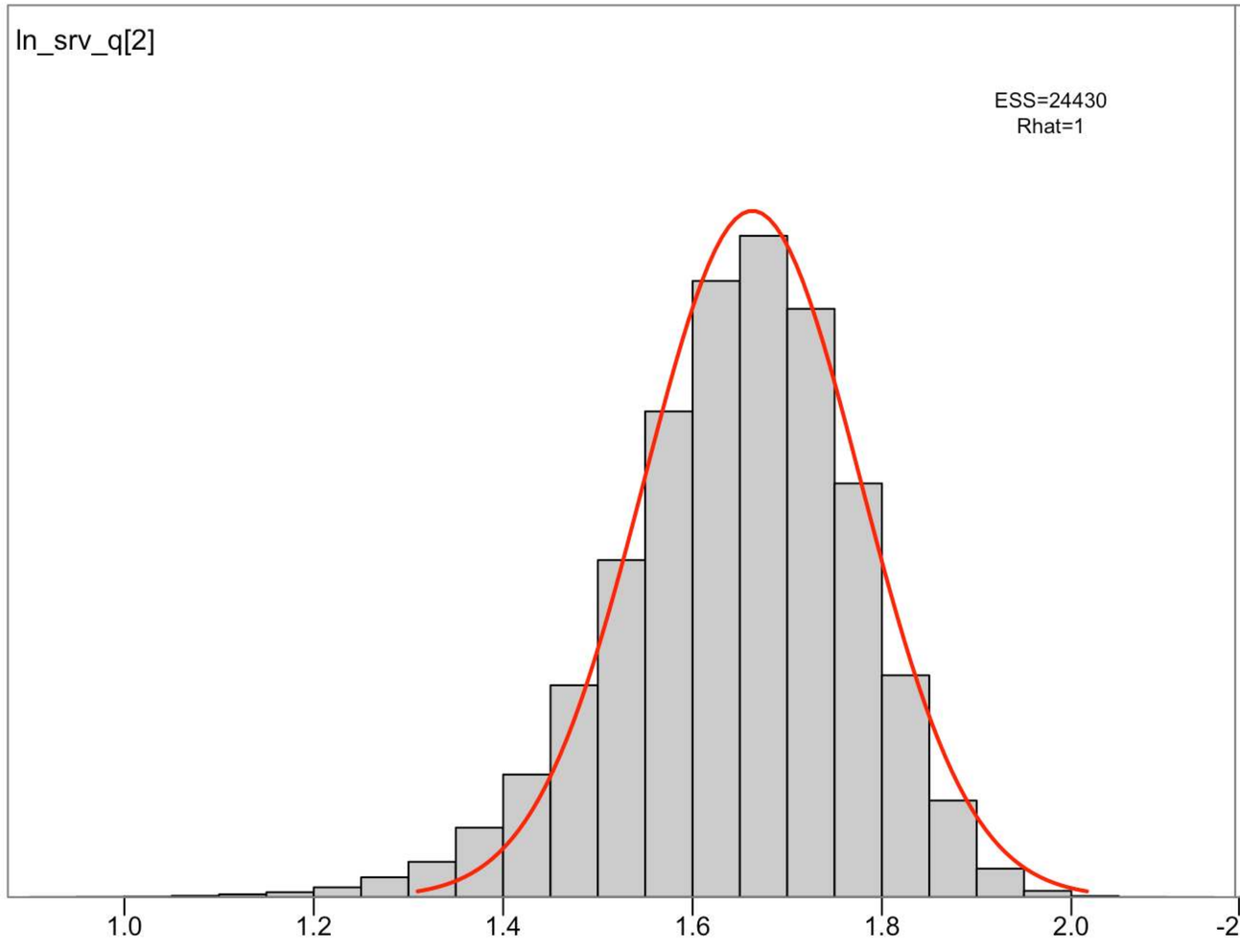
ln_srv_q[1]

ESS=24651
Rhat=1



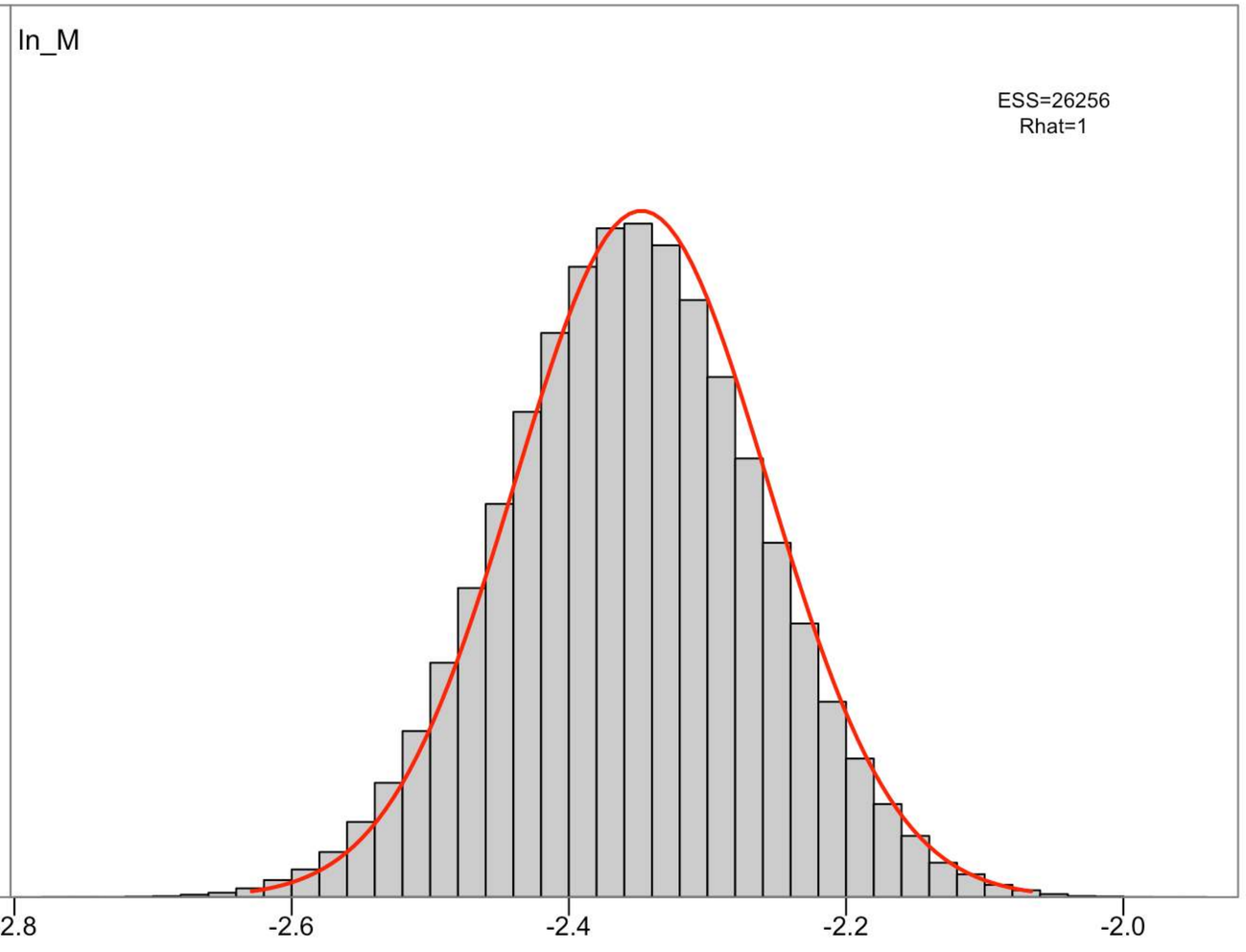
ln_srv_q[2]

ESS=24430
Rhat=1



ln_M

ESS=26256
Rhat=1



ln_global_R0

ln_srv_q[1]

ln_srv_q[2]

ln_M

ESS=26251 Rhat=1.00

-0.86

-0.83

0.9

ESS=24651 Rhat=1.00

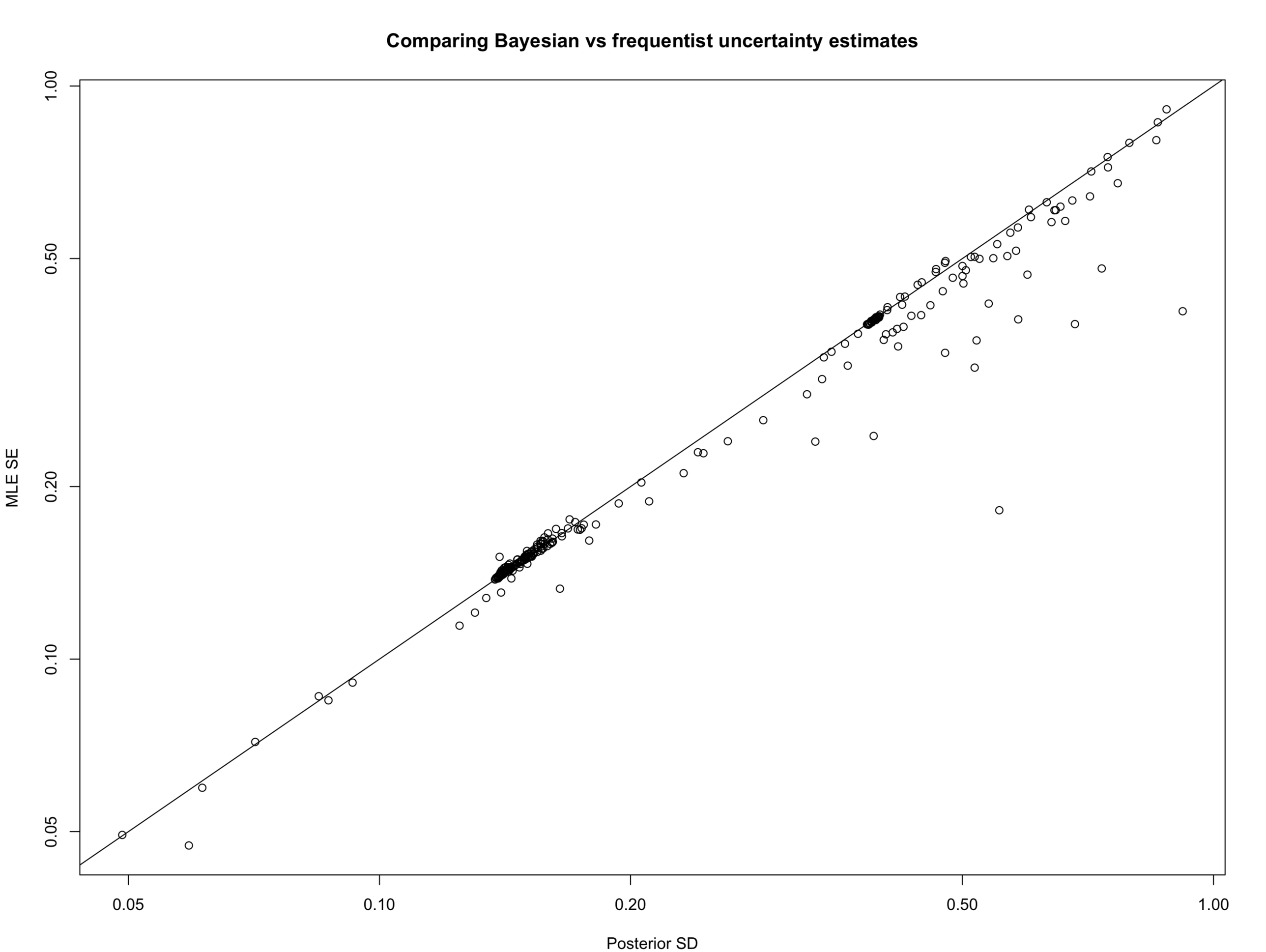
0.9

-0.87

ESS=24430 Rhat=1.00

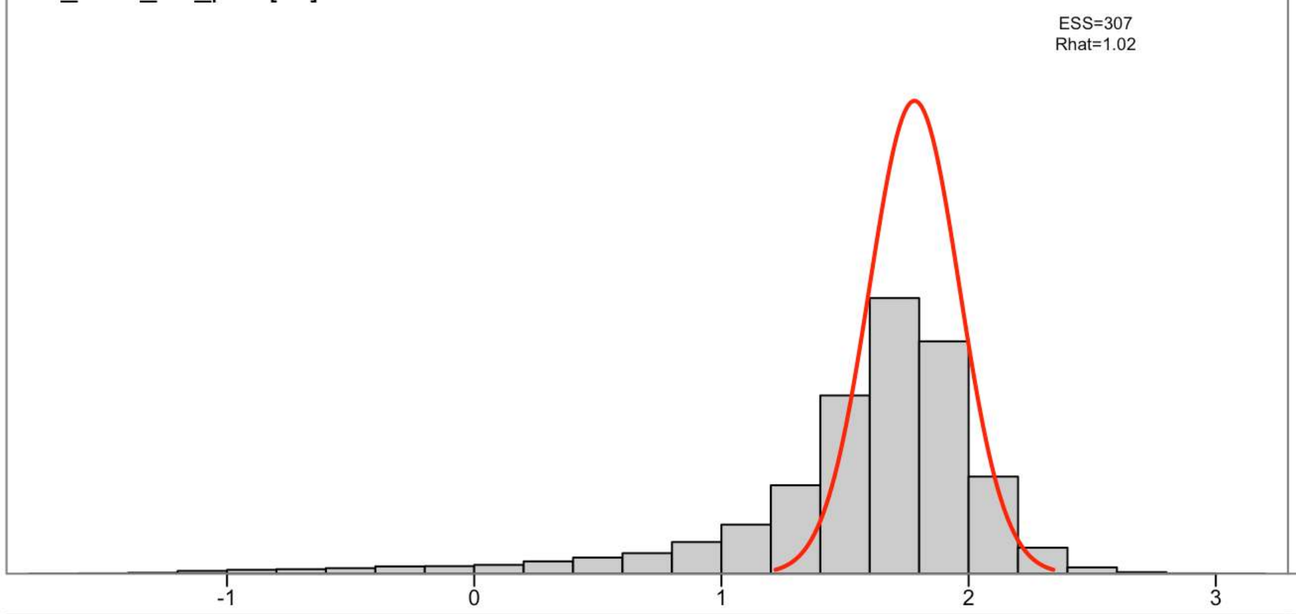
-0.83

ESS=26256 Rhat=1.00

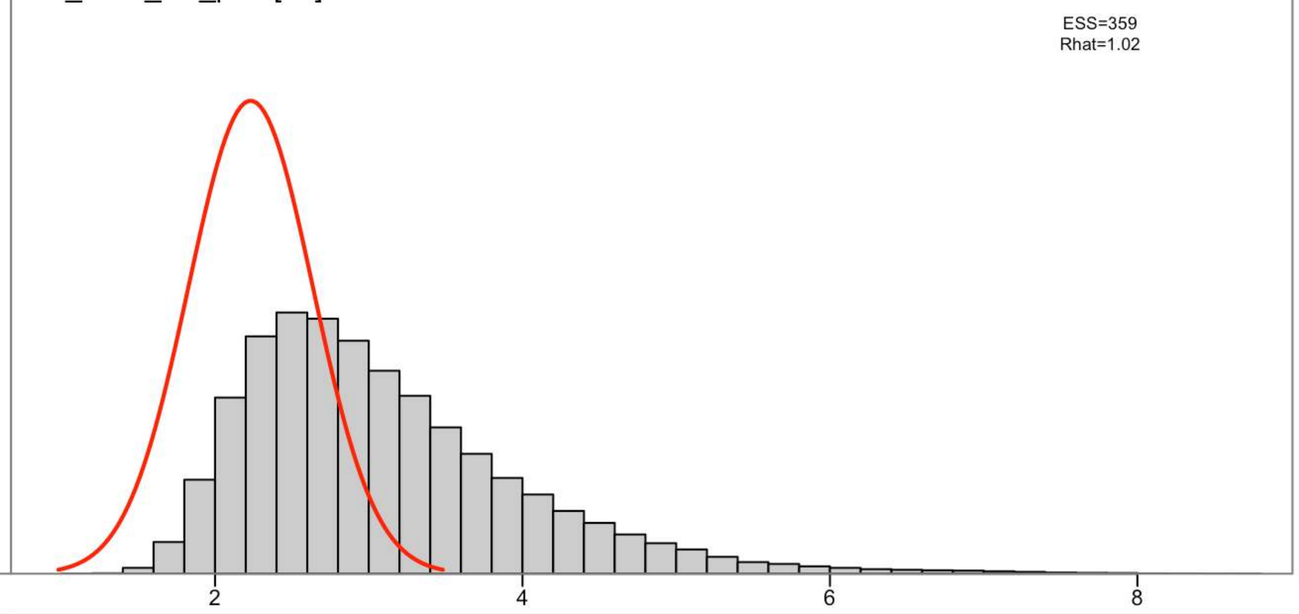

Comparing Bayesian vs frequentist uncertainty estimates
MLE SE
1.00
0.50
0.20
0.10
0.05
0.05
0.10
0.20
0.50
1.00
Posterior SD

accept_stat
1.00
0.75
0.50
0.25
0.00
divergent
1.00
0.75
0.50
0.25
0.00
energy
10000
7500
5000
2500
value
log_stepsize
2
0
-2
-4
n_leapfrog
1000
750
500
250
0
0
10000
20000
30000
40000
iteration
chain
1
2
3
4

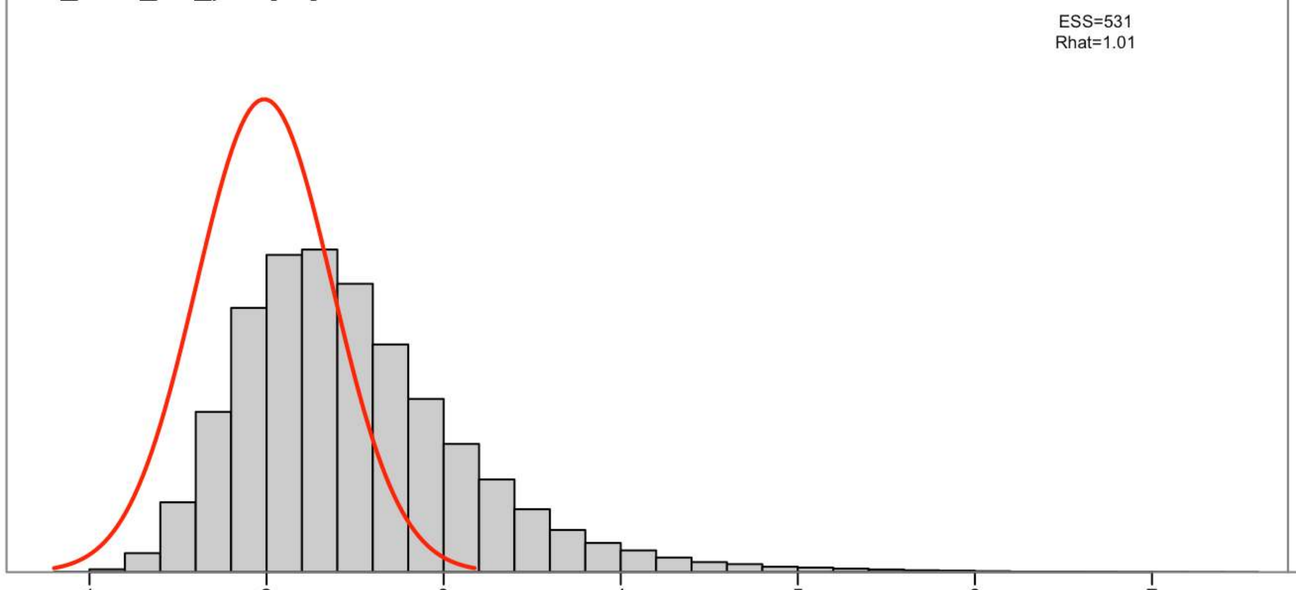
fish_fixed_sel_pars[12]



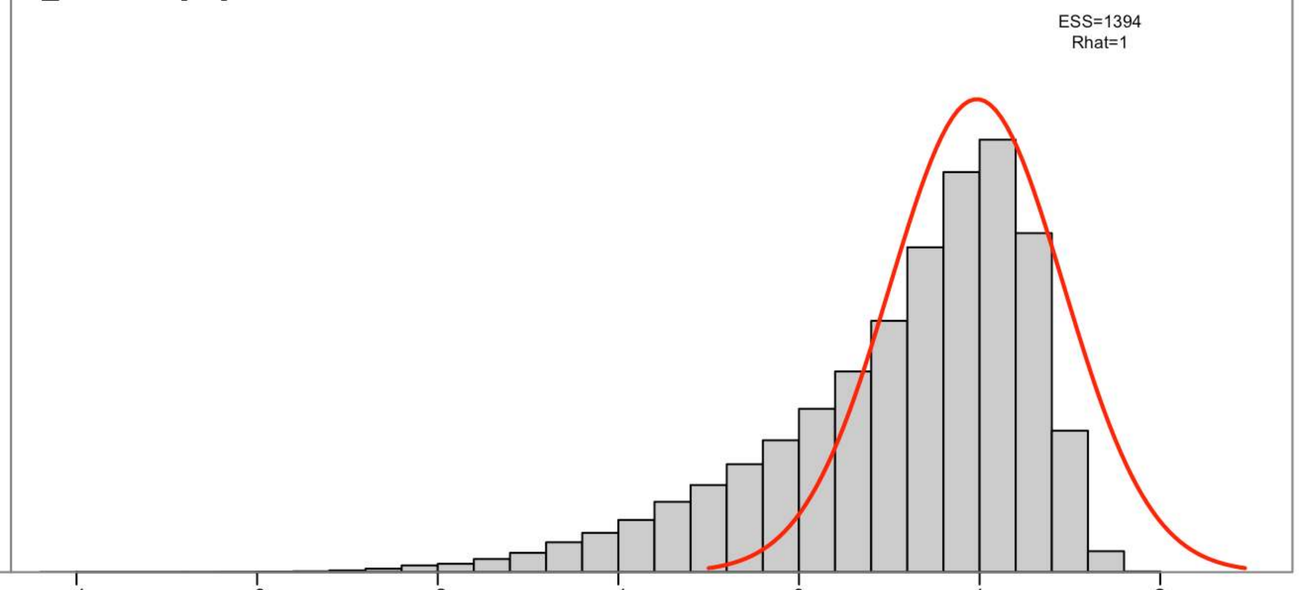
fish_fixed_sel_pars[14]



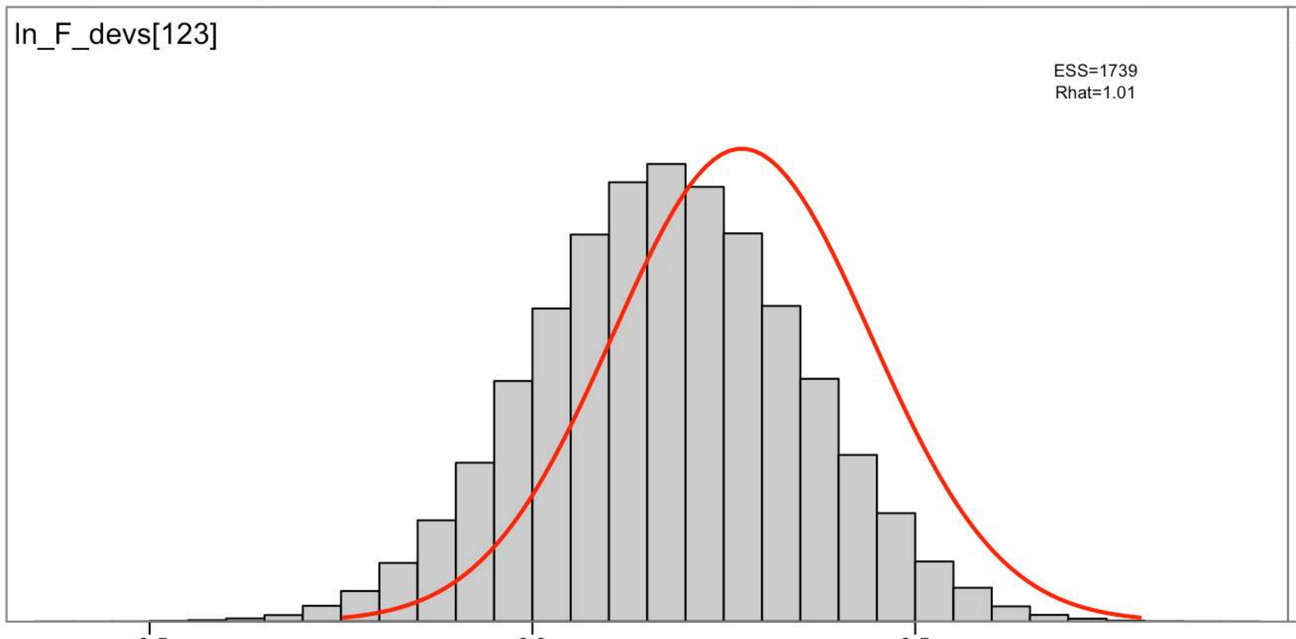
fish_fixed_sel_pars[13]



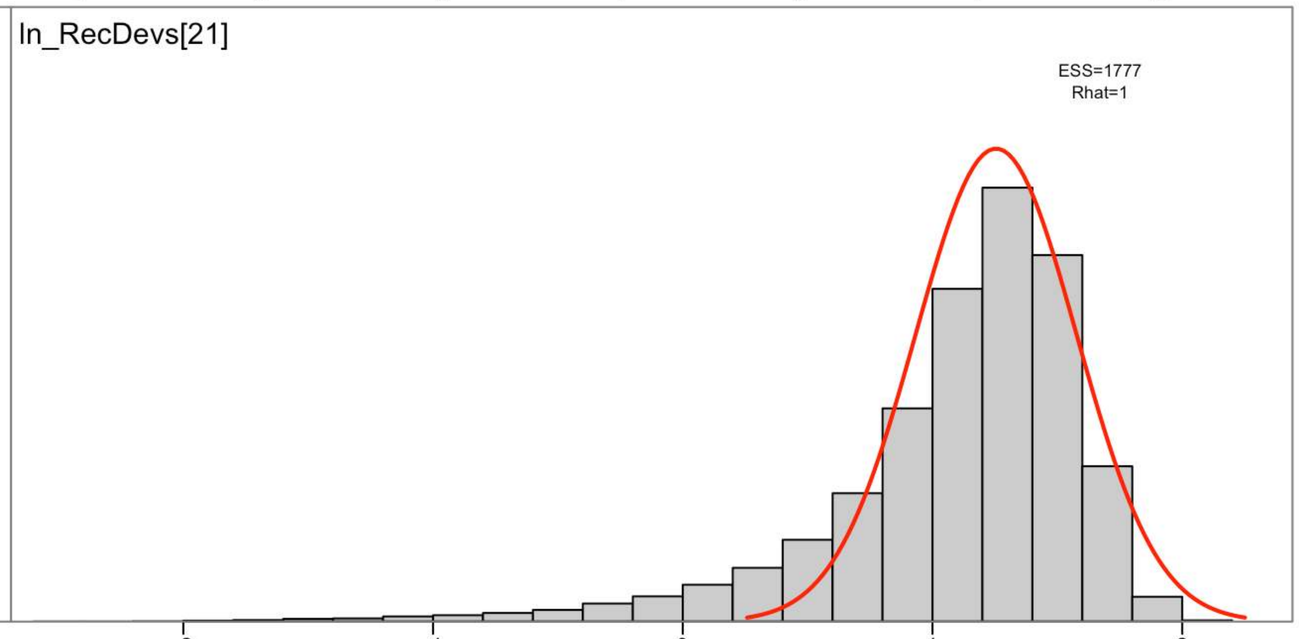
ln_RecDevs[24]



ln_F_devs[123]

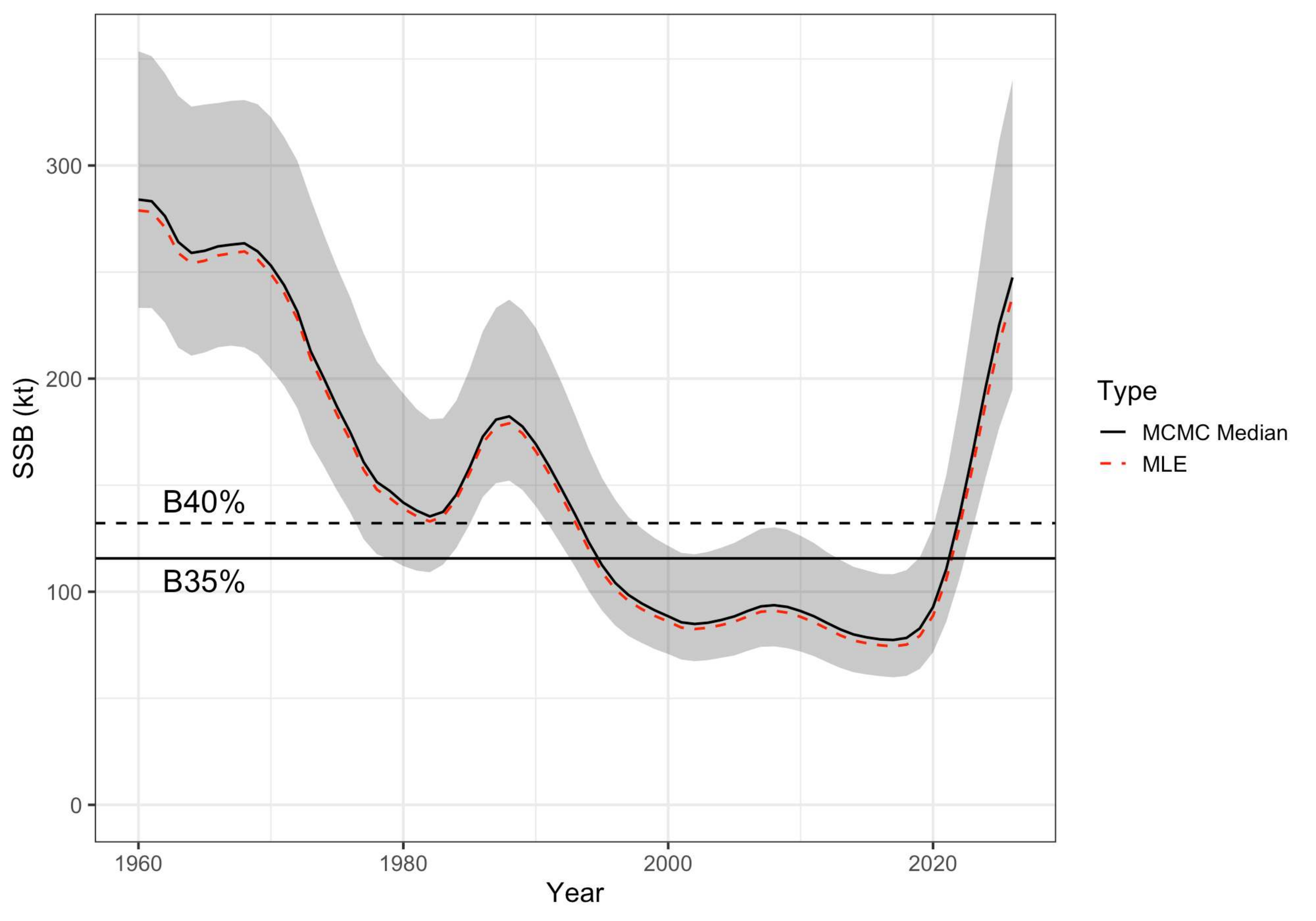


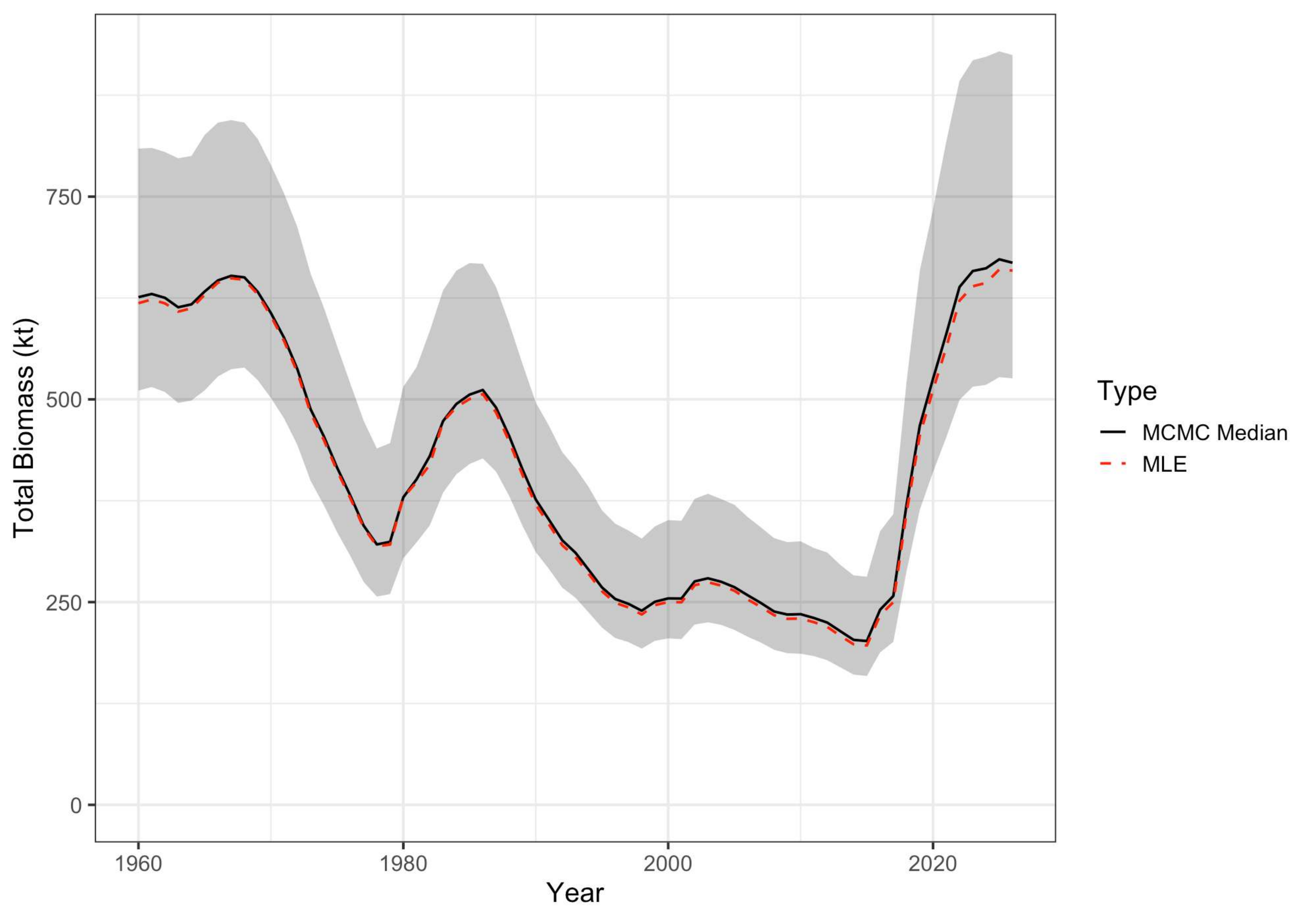
ln_RecDevs[21]

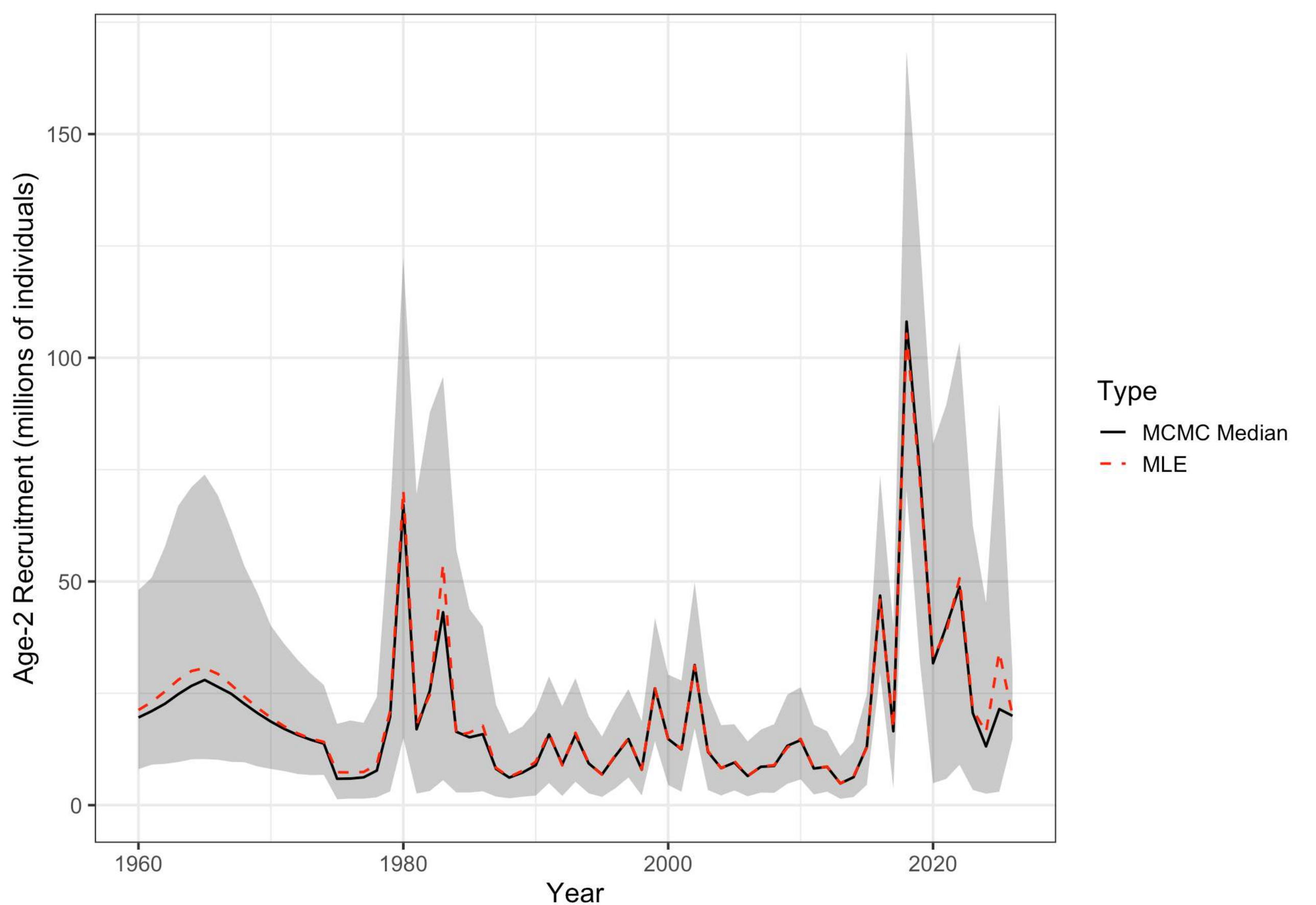


fish_fixed_sel_pars[12]
fish_fixed_sel_pars[14]
fish_fixed_sel_pars[13]
ln_RecDevs[24]
ln_F_devs[123]
ln_RecDevs[21]

ESS=307 Rhat=1.02
-0.42
-0.19
0.42
ESS=359 Rhat=1.02
0.4
-0.27
ESS=531 Rhat=1.01
ESS=1394 Rhat=1.00
ESS=1739 Rhat=1.01
ESS=1777 Rhat=1.00


SSB (kt)
300
200
100
0
B40%
B35%
1960
1980
2000
2020
Year
Type
MCMC Median
MLE


750
500
250
0
Total Biomass (kt)
1960
1980
2000
2020
Year
Type
MCMC Median
MLE


Age-2 Recruitment (millions of individuals)
150
100
50
0
1960
1980
2000
2020
Year
Type
MCMC Median
MLE

Type: Age, Catch, Index, jnLL, Length, Penalty, Prior

25.12

Likelihood

300
200
100
0
-100
-200
-300

Catch nLL
Fishery Age nLL
Fishery Index nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL

Likelihood Component

Numbers-at-Age

Sex — Female ···· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

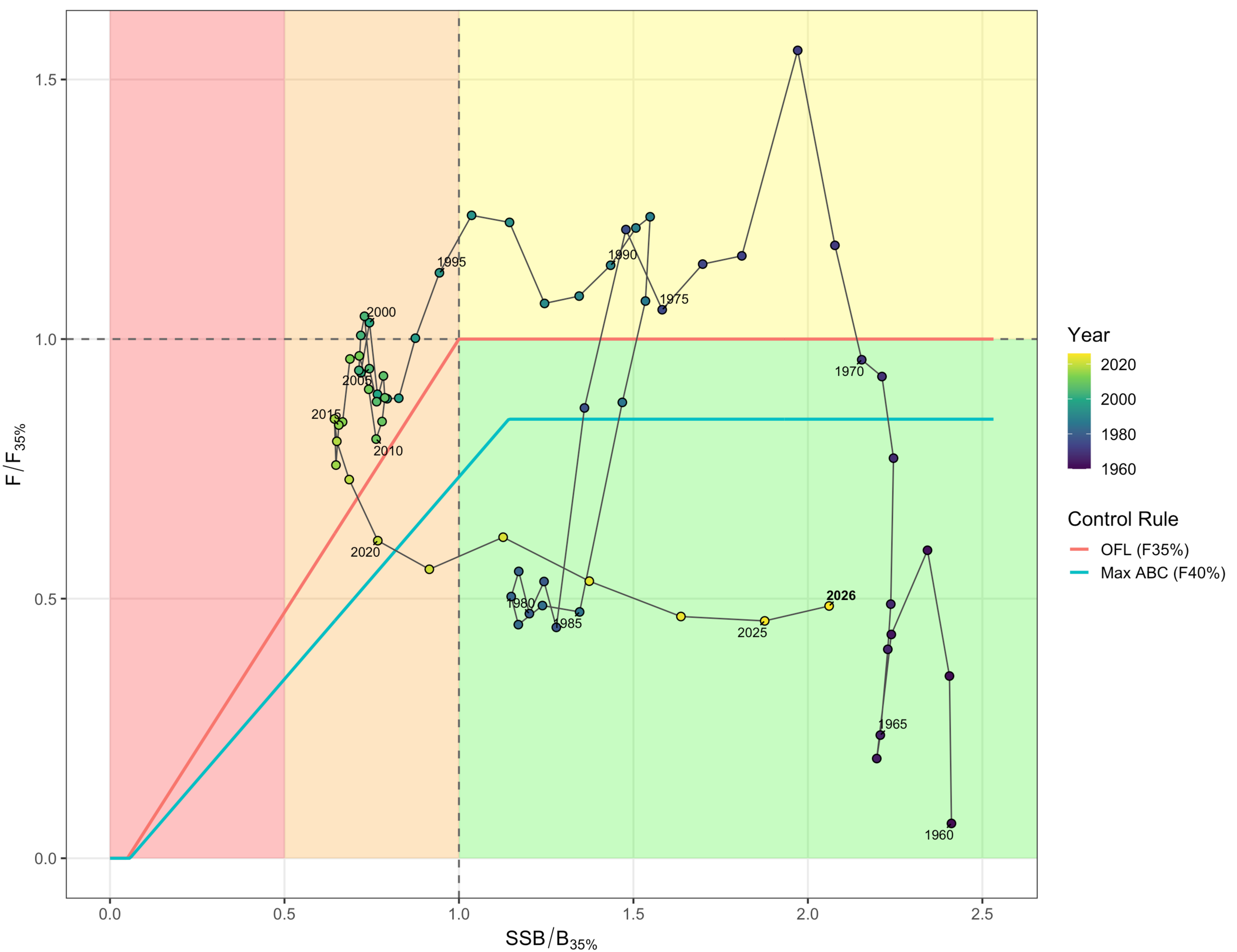
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

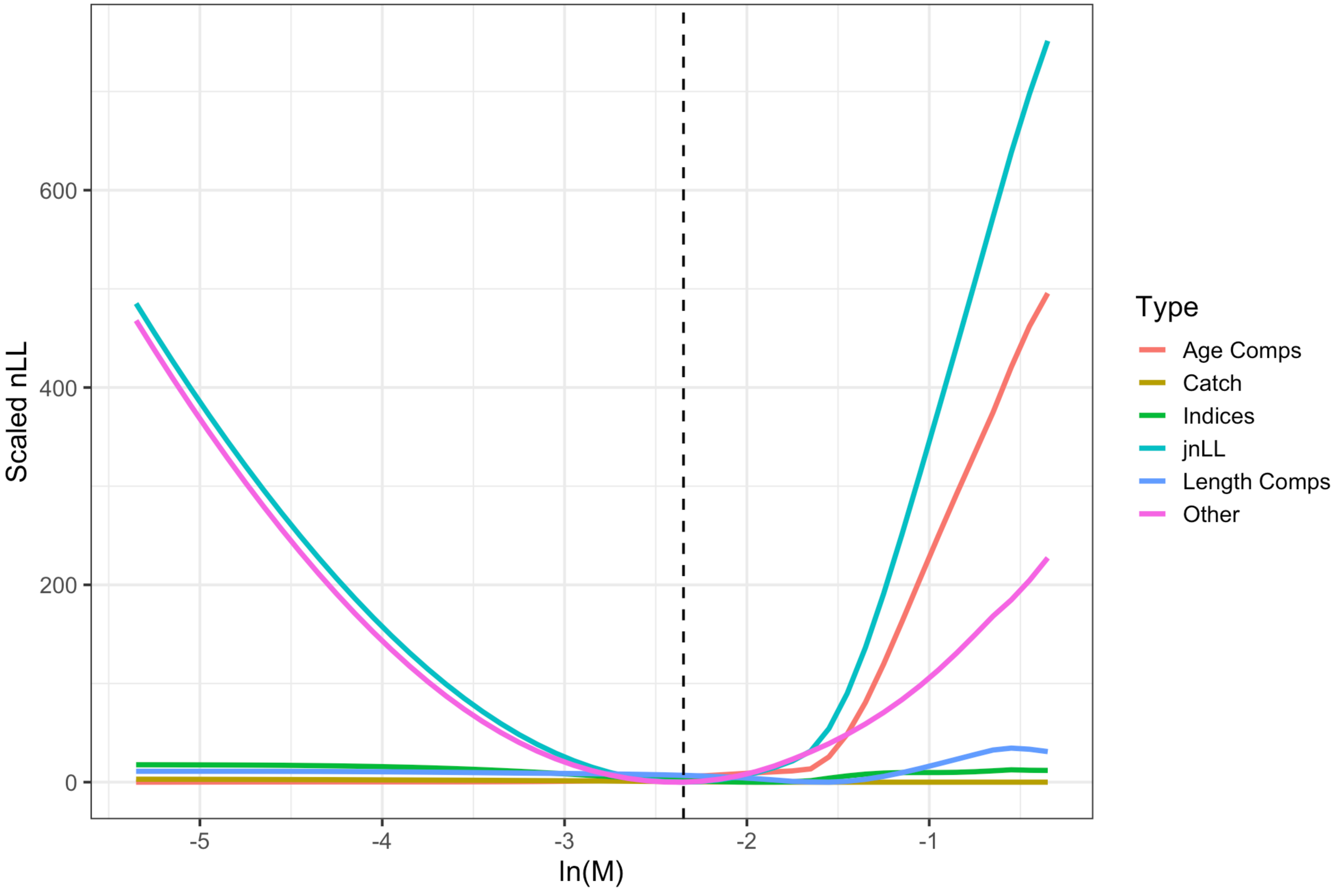
Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

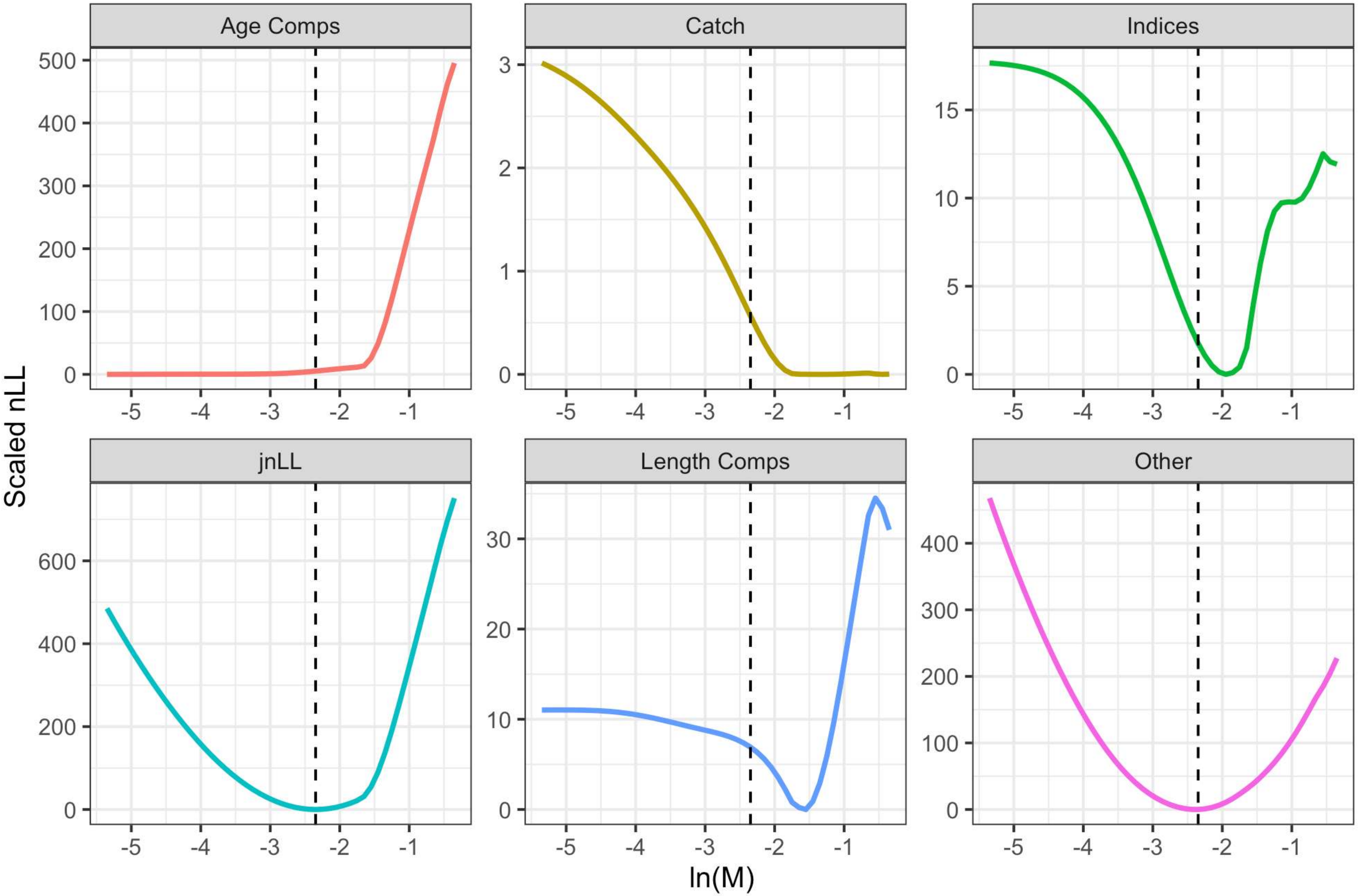
Year


F/F35%
SSB/B35%
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
2.0
2.5
1995
1990
1975
2000
2005
2015
2010
2020
1970
1980
1985
2025
2026
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Natural mortality profile



Natural mortality profile



Recruitment (R0) profile

Scaled nLL

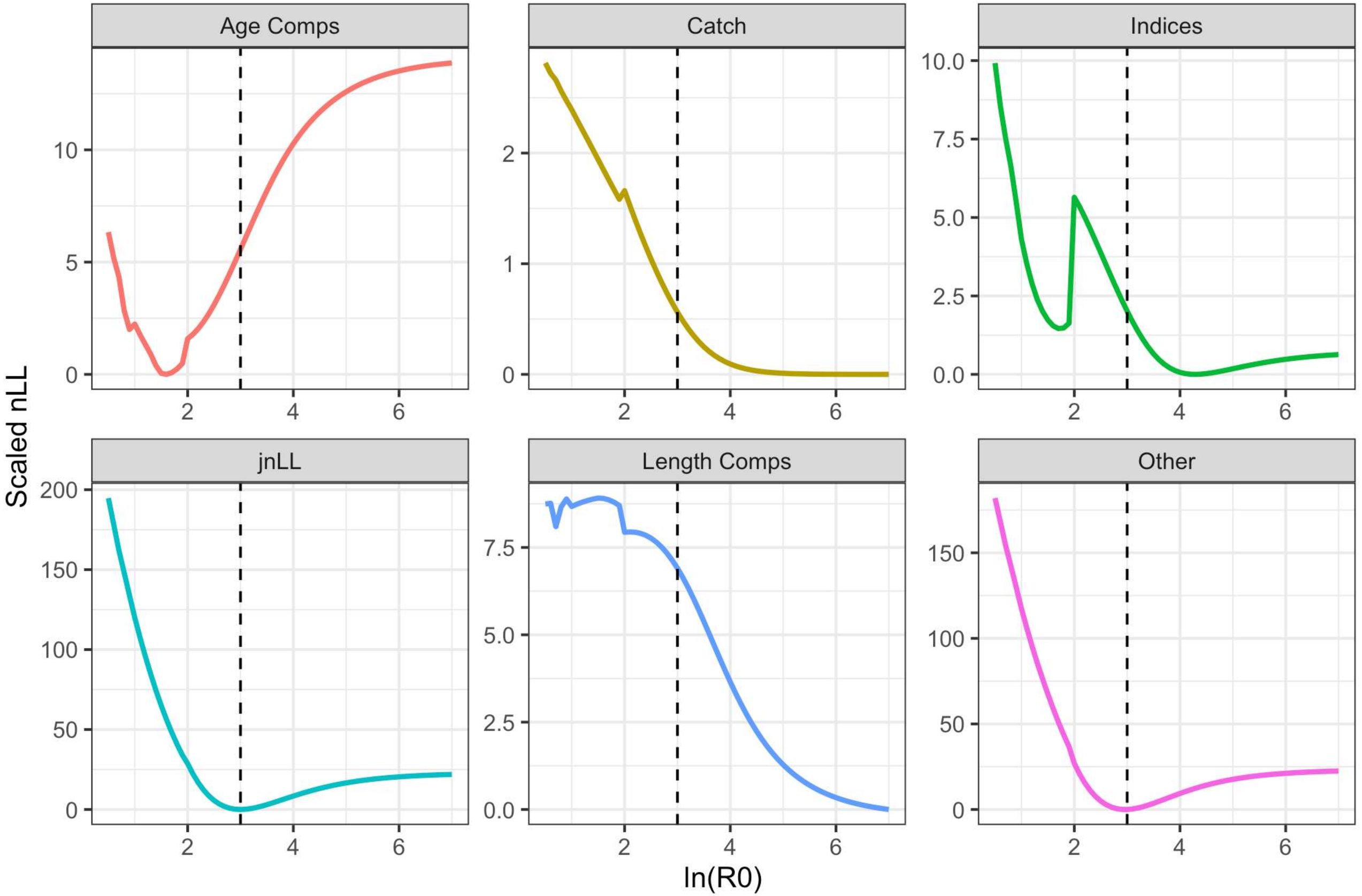
200
150
100
50
0

2
4
6

ln(R0)

Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

Recruitment (R0) profile



Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.3769

Mohns Rho: -0.0183

0.25
0.00
-0.25
-0.50
-0.75

Relative Difference from Terminal Year

AK

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

90
60
30
0

200
100
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

90

60

30

0

1

2

3

4

5

6

7

Years since cohort was last estimated

Cohort

2016

2017

2018

2019

2020

2021

2022

2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.377

SSB

Mohns Rho: -0.0308

AK

Relative Difference from Terminal Year

2
1
0

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

90
60
30
0

200
100
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

90

60

30

0

1

2

3

4

5

6

7

Years since cohort was last estimated

Cohort

2016

2017

2018

2019

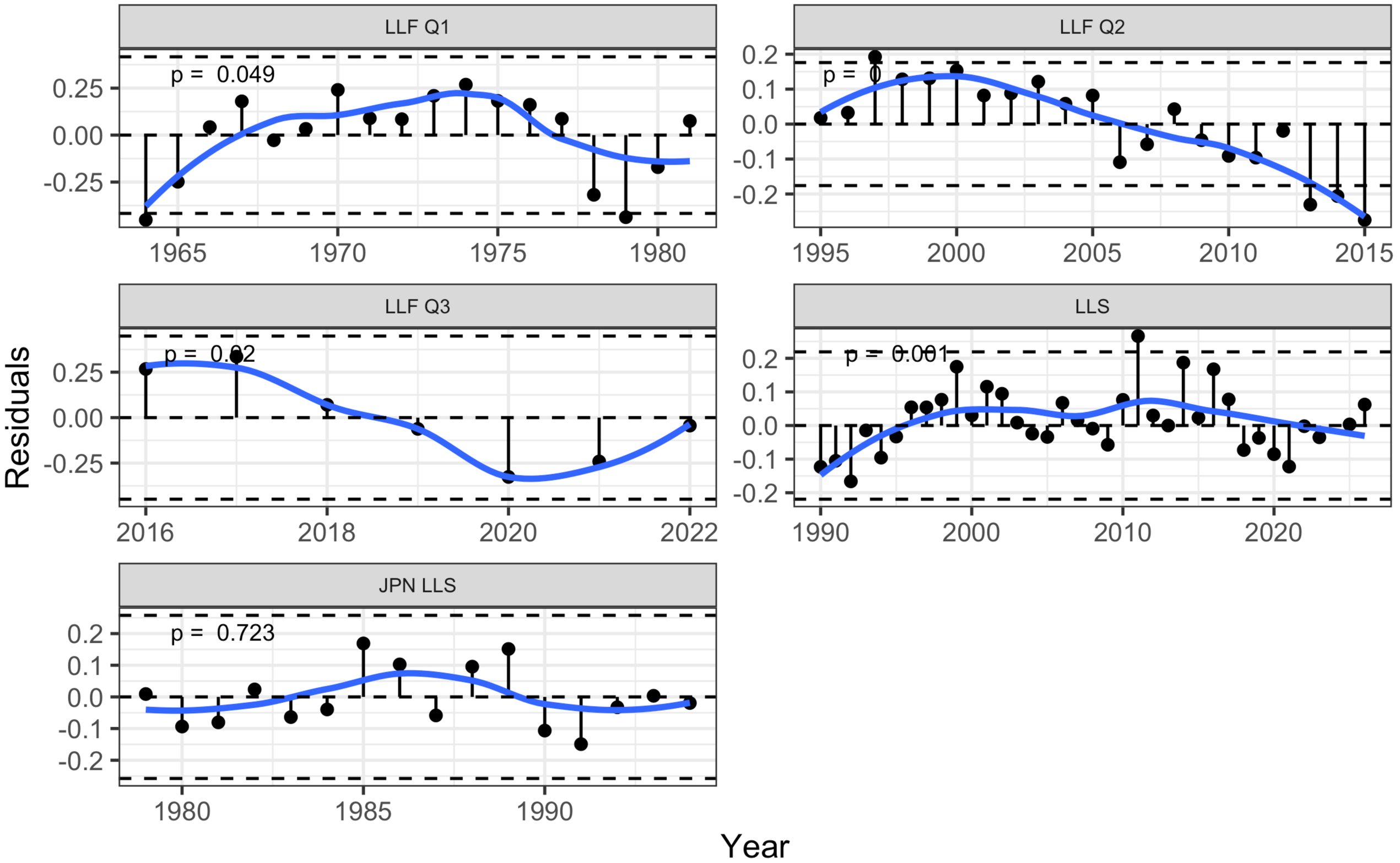
2020

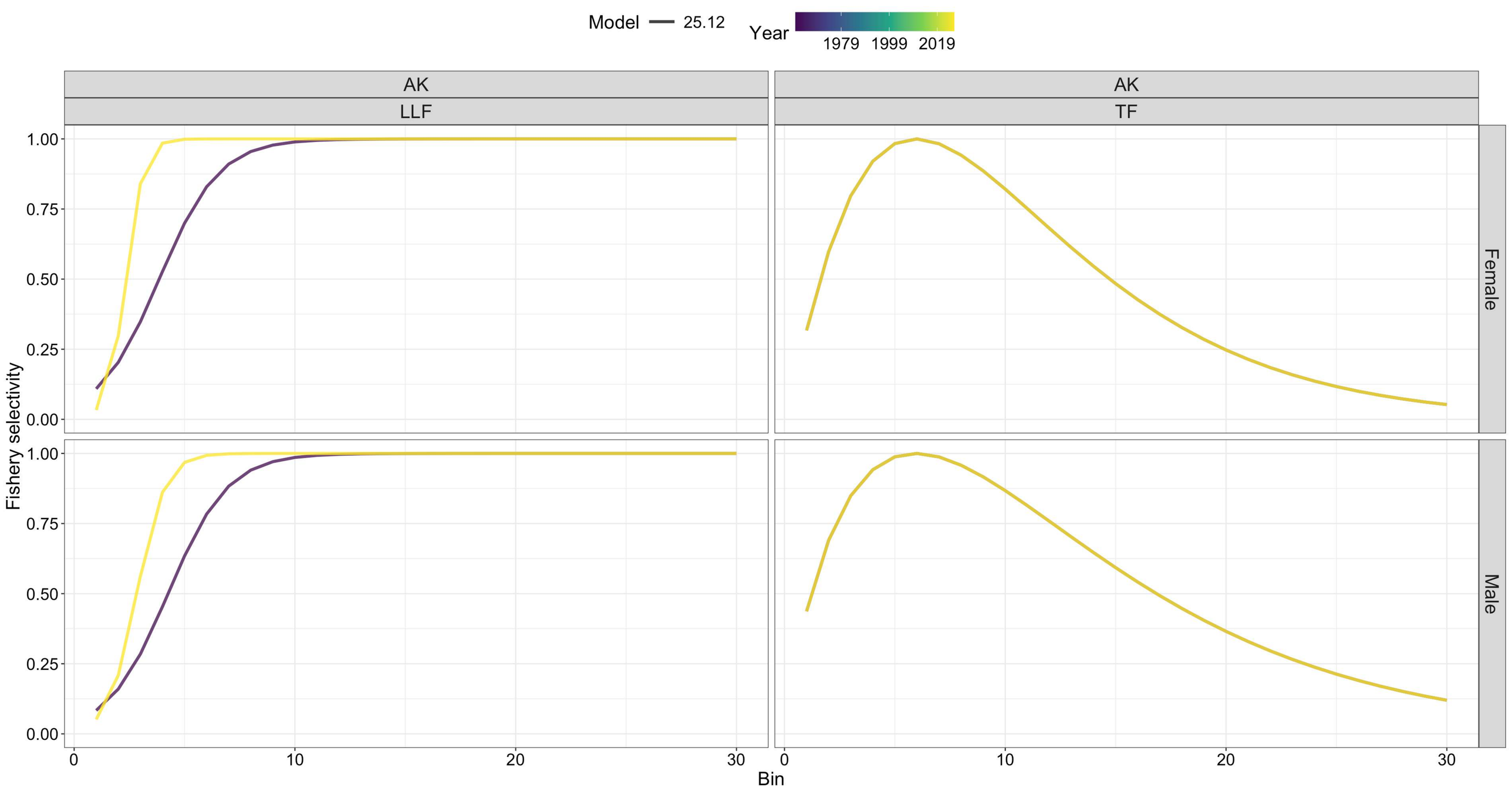
2021

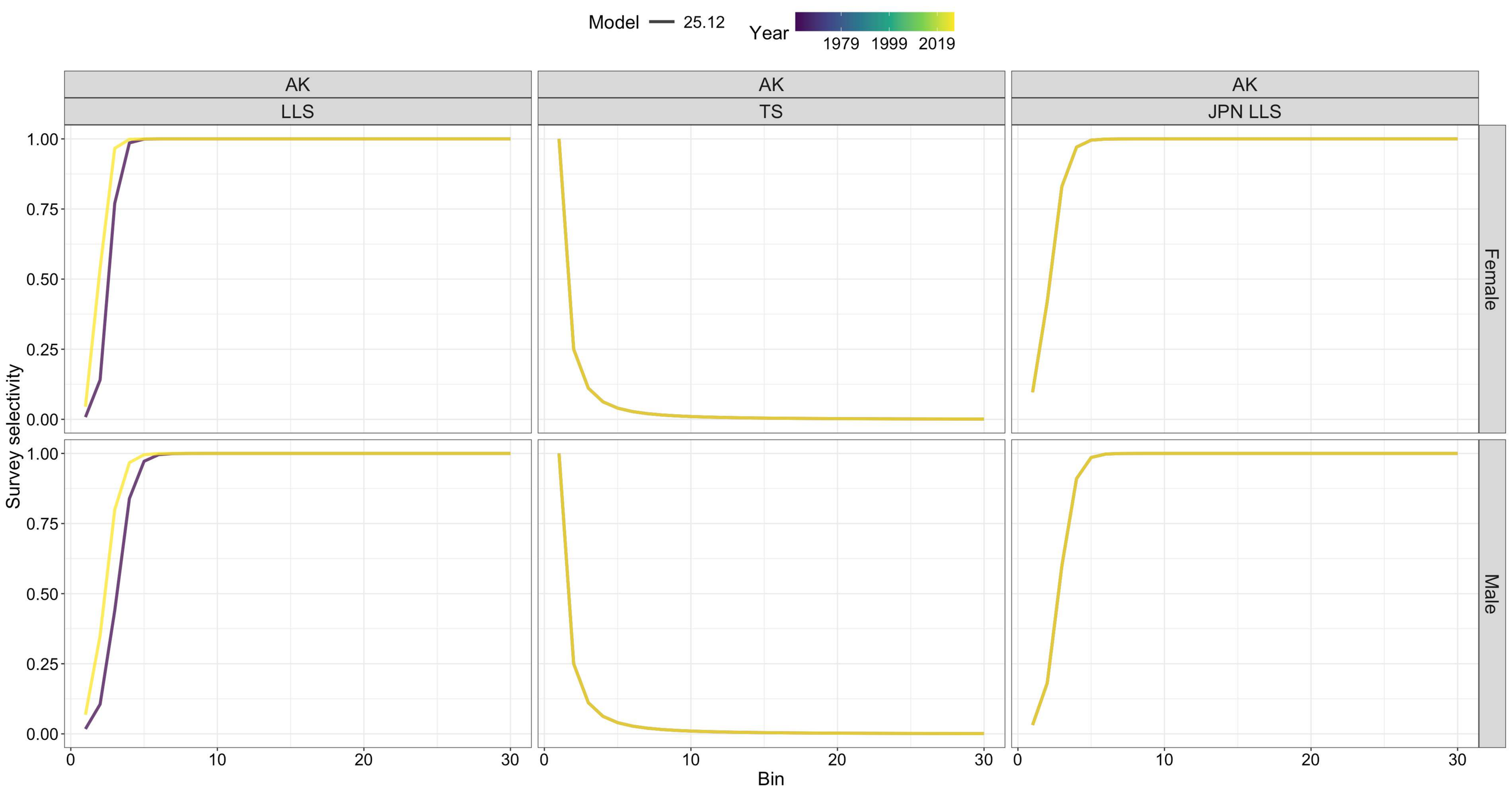
2022

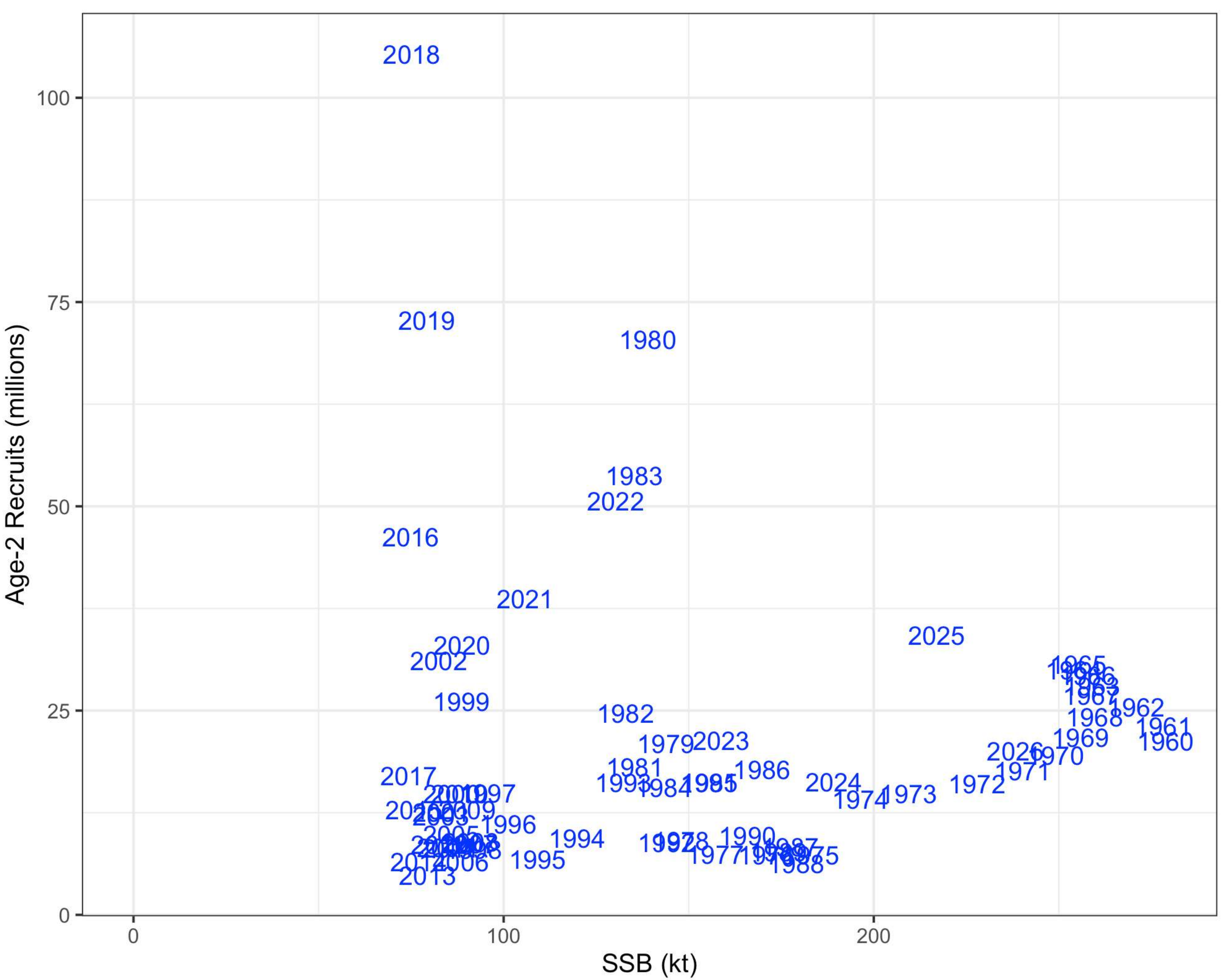
2023

Index Residuals Runs Test




Model 25.12
Year
1979 1999 2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model 25.12
Year
1979 1999 2019
AK
LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Age-2 Recruits (millions)
SSB (kt)
0
25
50
75
100
0
100
200
2018
2019
1980
1983
2022
2016
2021
2020
2002
1999
2025
1982
1979
2023
1981
1986
2017
1997
1996
1994
1995
2013
1965
1962
1968
1961
1969
1960
1970
1971
2024
1974
1973
1972
1990
1988

Population and Catch Trends

Recruitment (millions of fish)

100
75
50
25
0

300
200
100
0

SSB or Catch (kt)

1960
1980
2000
2020

Year

Catch
Recruitment
SSB

Model 25.12

AK

Dynamic SSB0

Recruitment

SSB

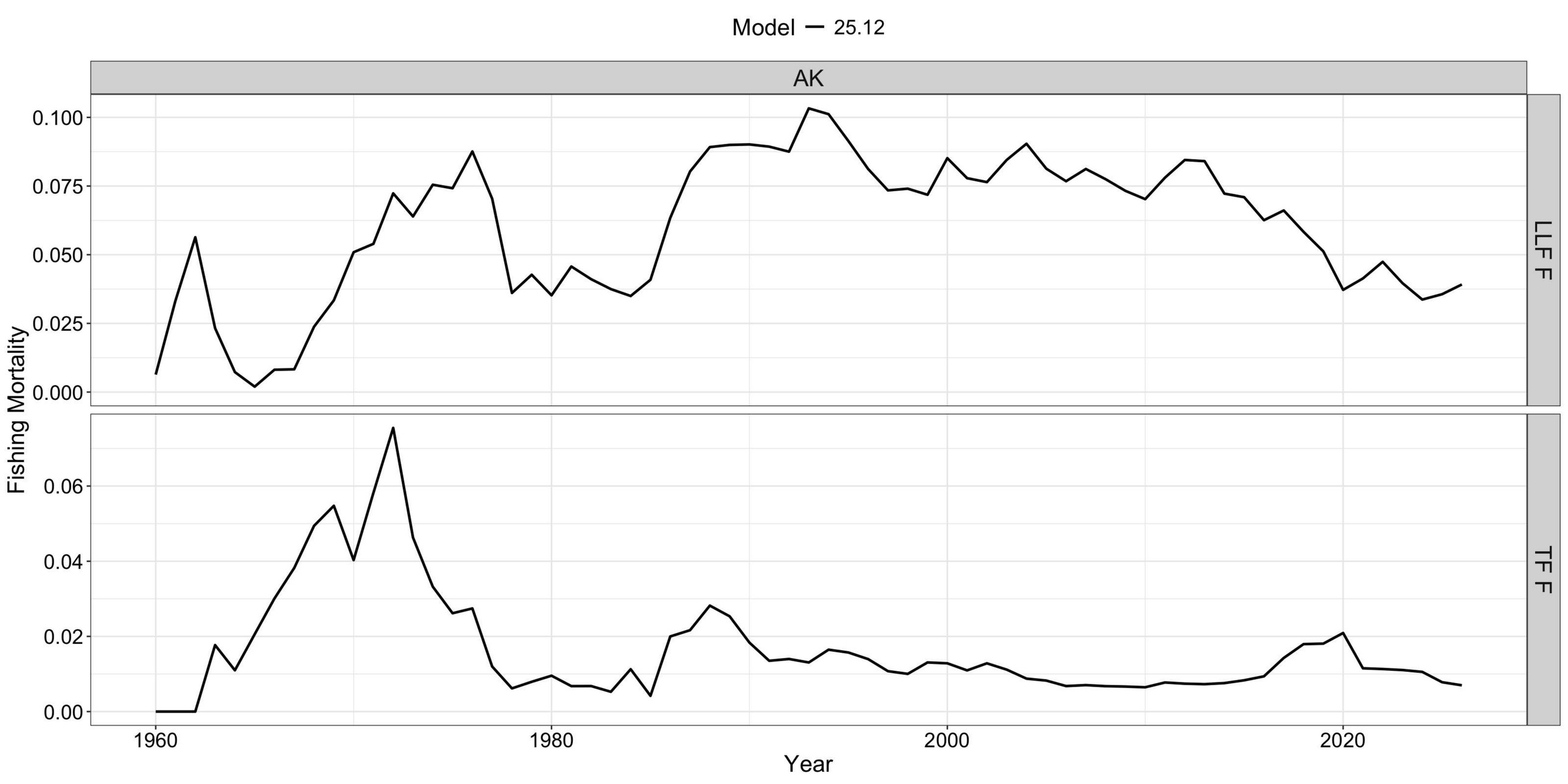
Total Biom

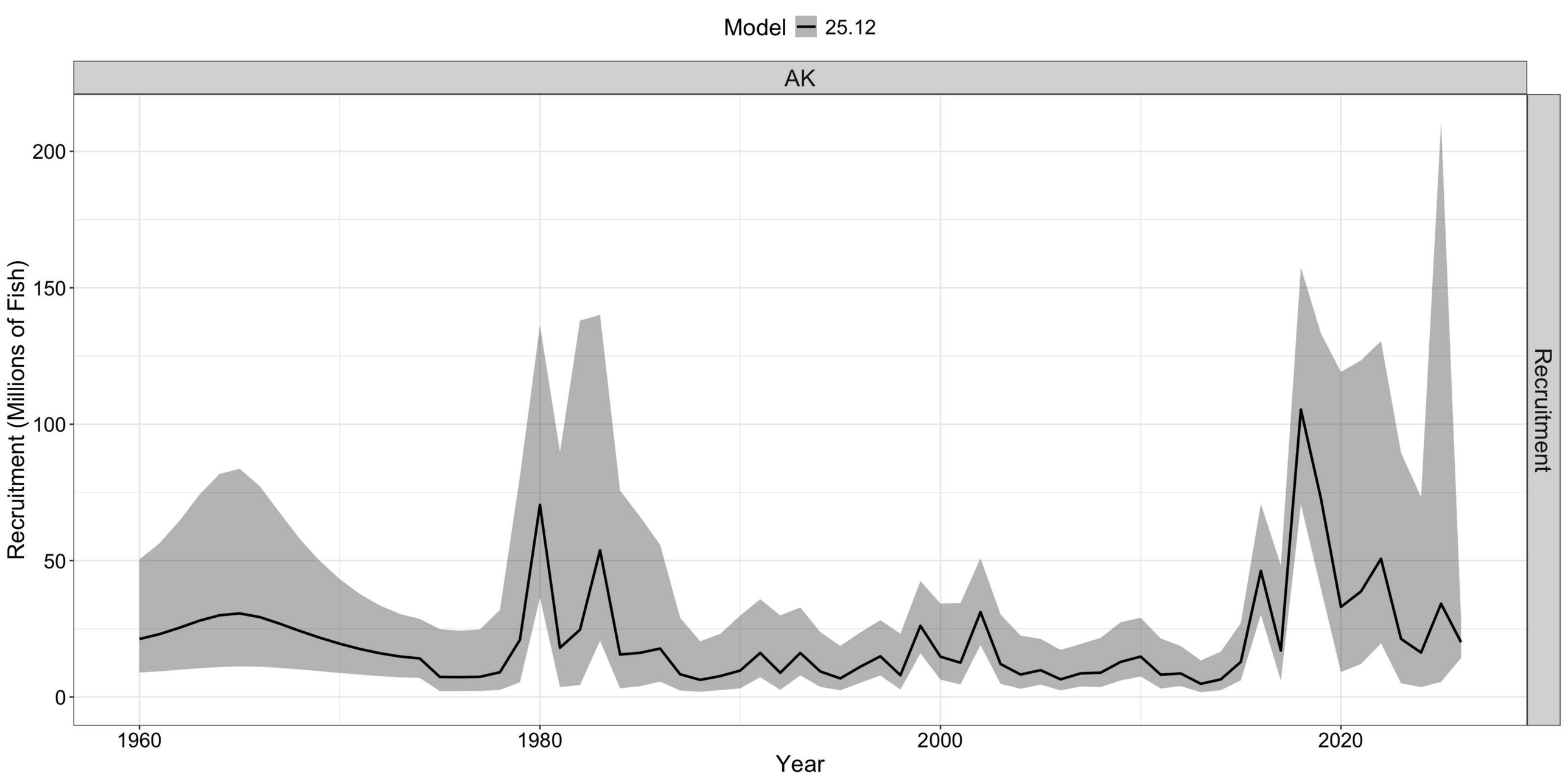
LLF F

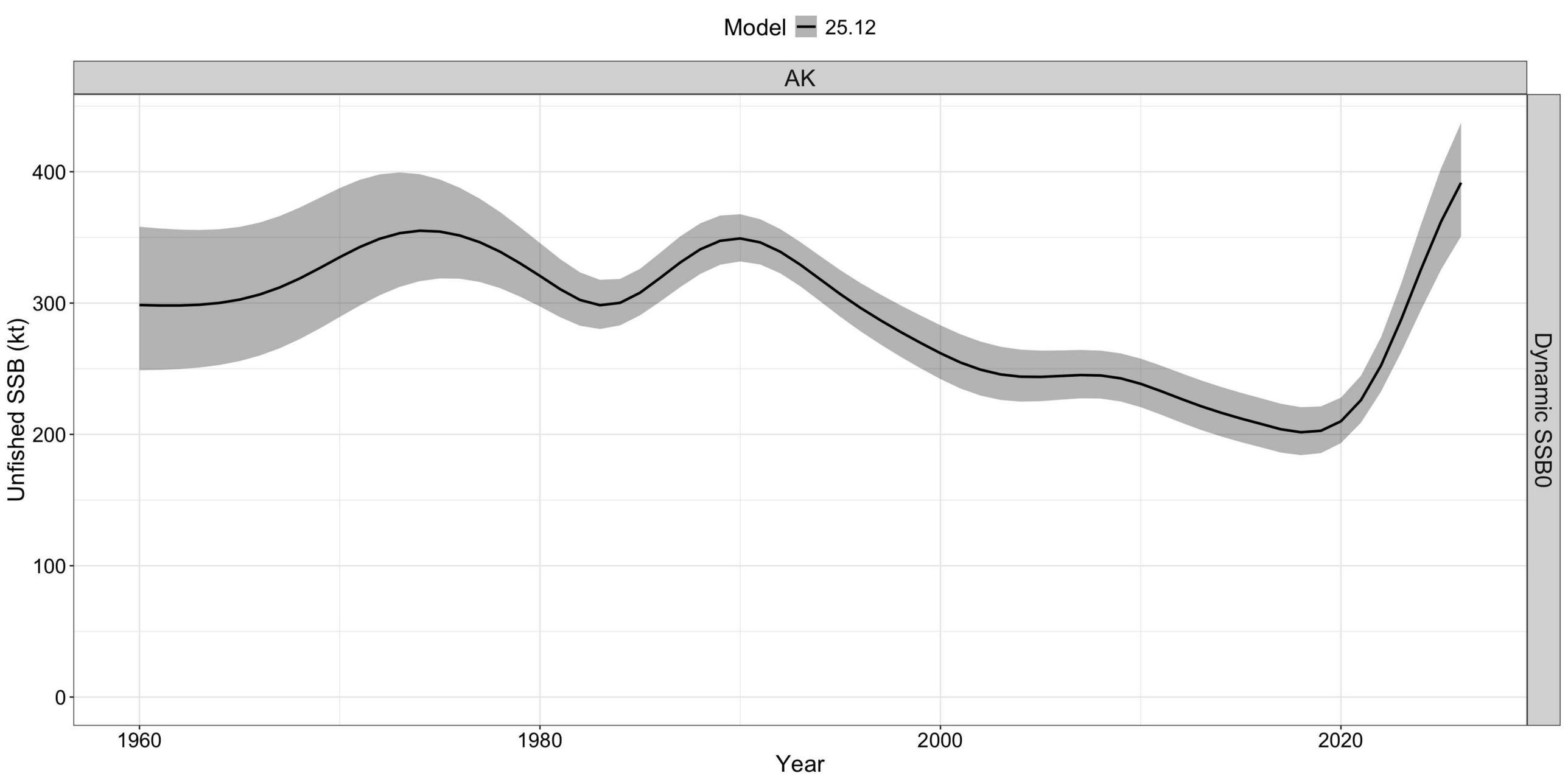
TF F

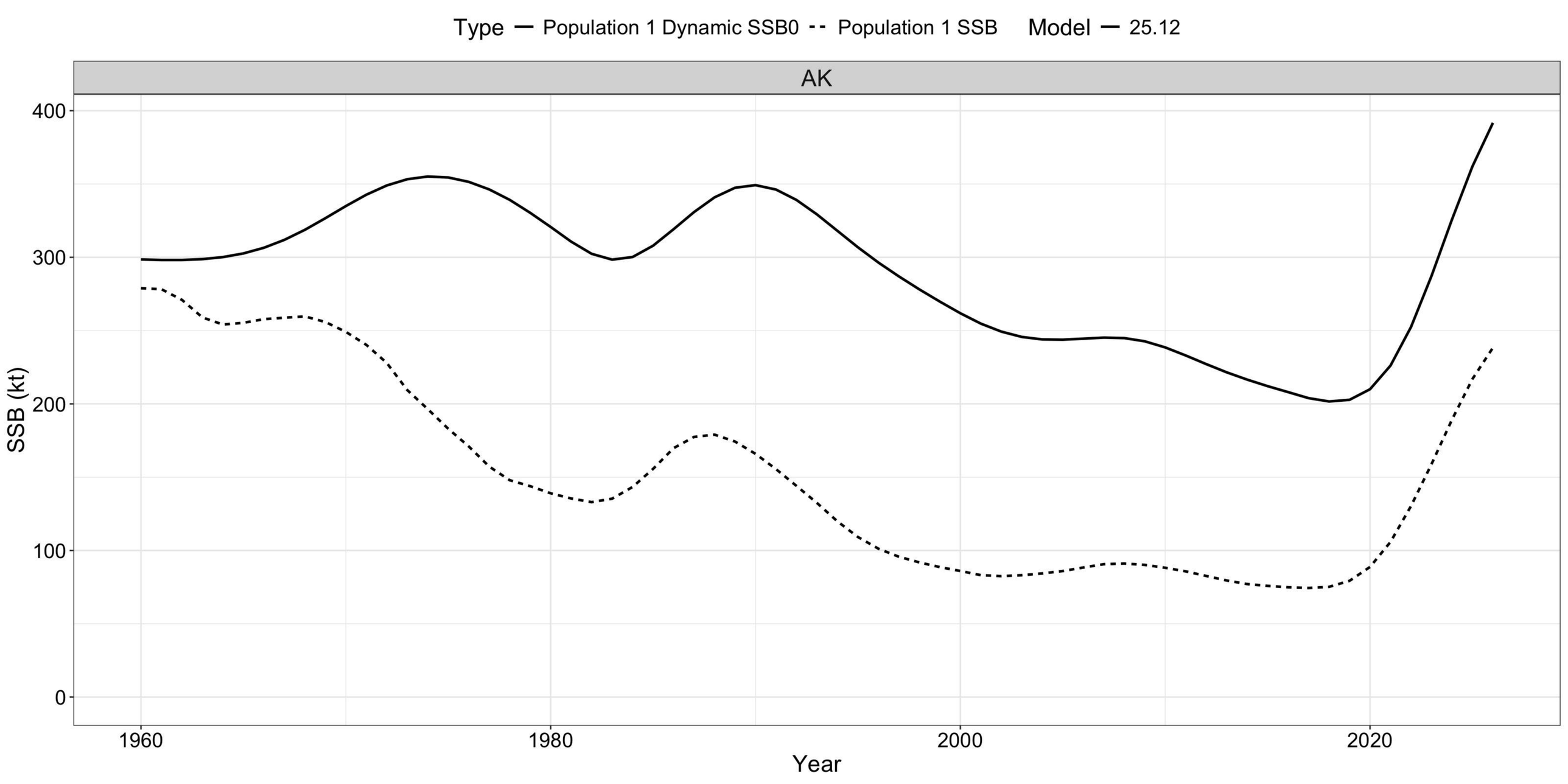
Value

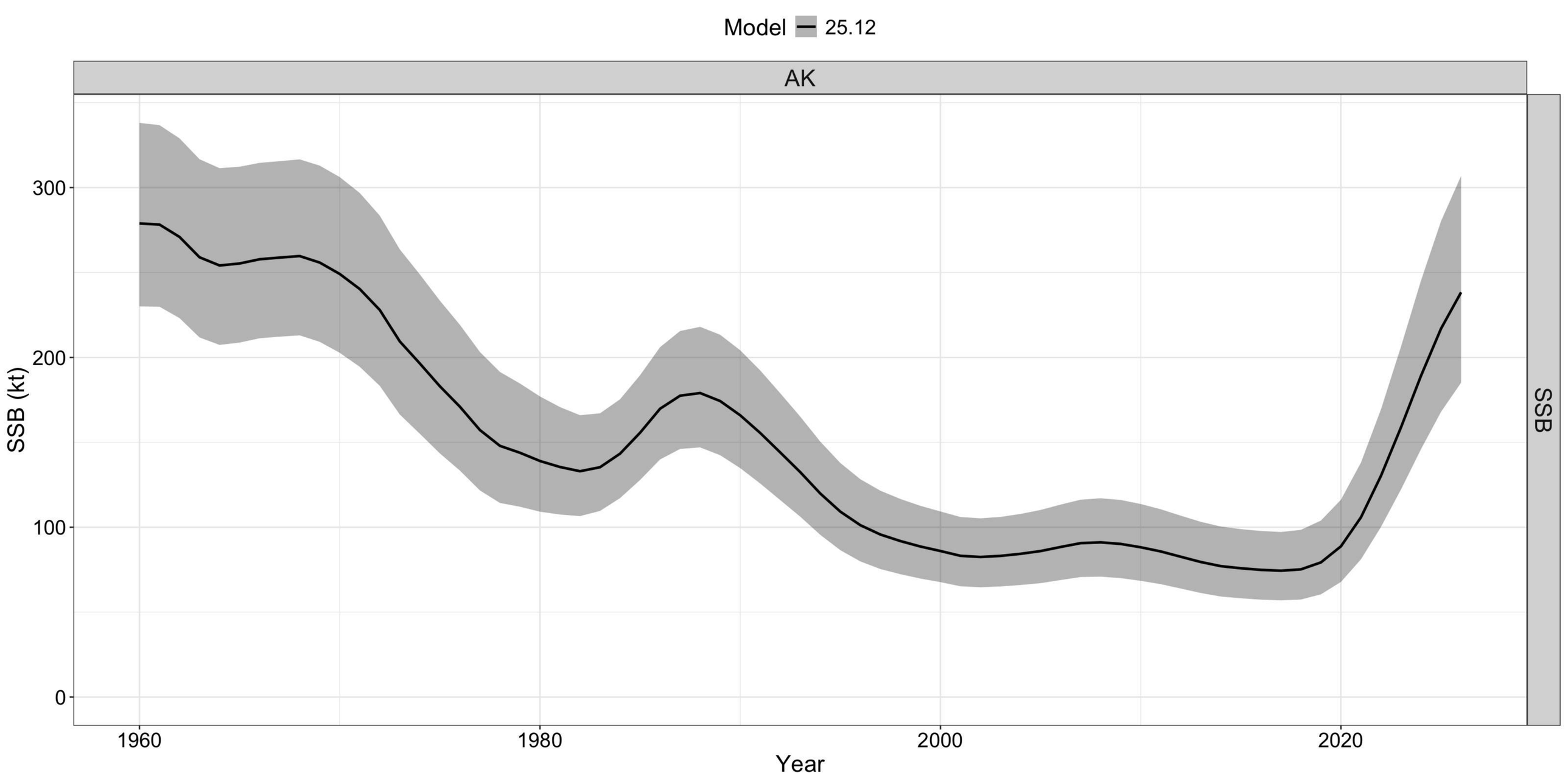
Year

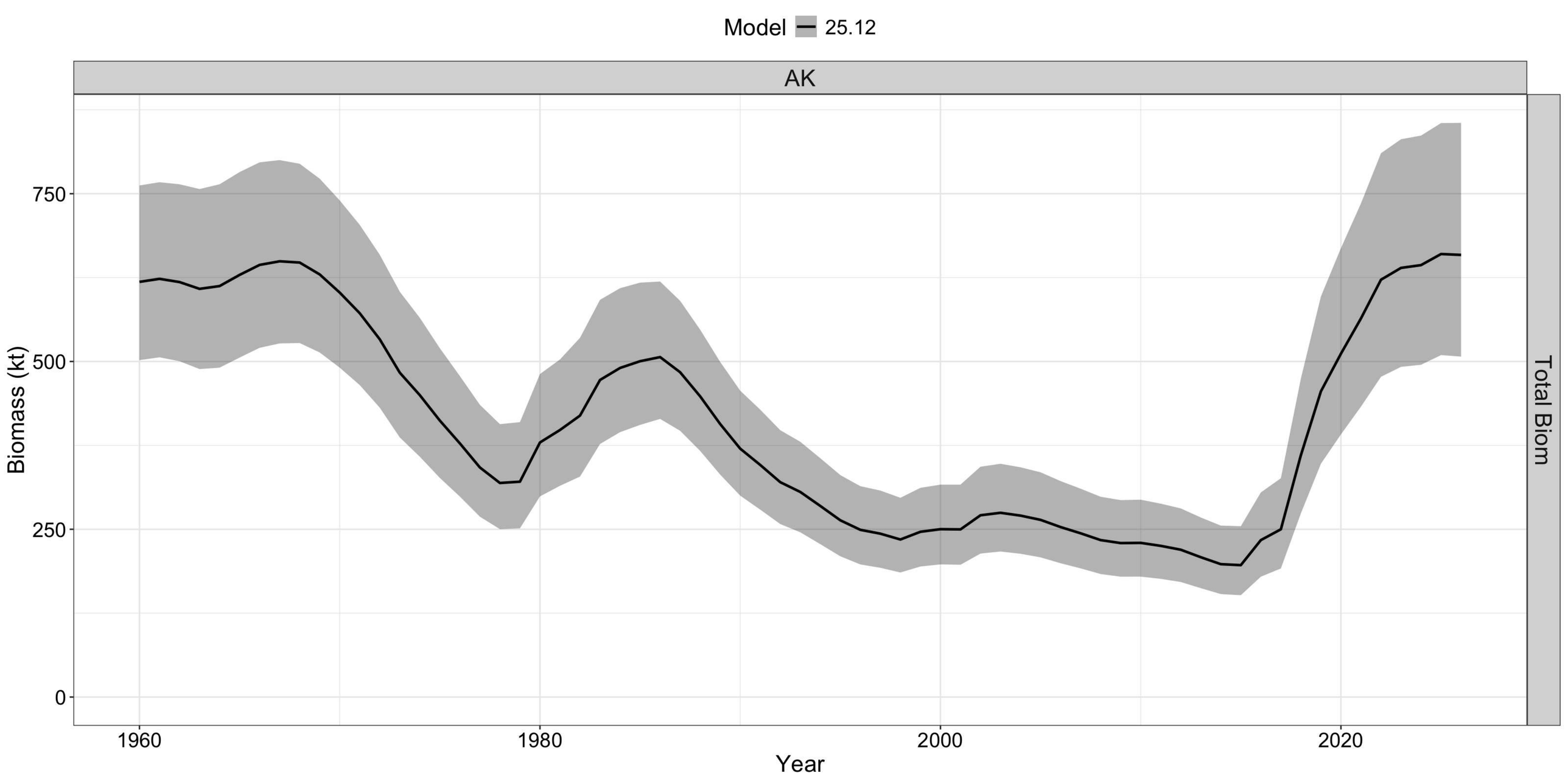

Model — 25.12
AK
LLF F
TF F
Fishing Mortality
0.100
0.075
0.050
0.025
0.000
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year


Model 25.12
AK
Recruitment (Millions of Fish)
200
150
100
50
0
1960
1980
2000
2020
Year
Recruitment

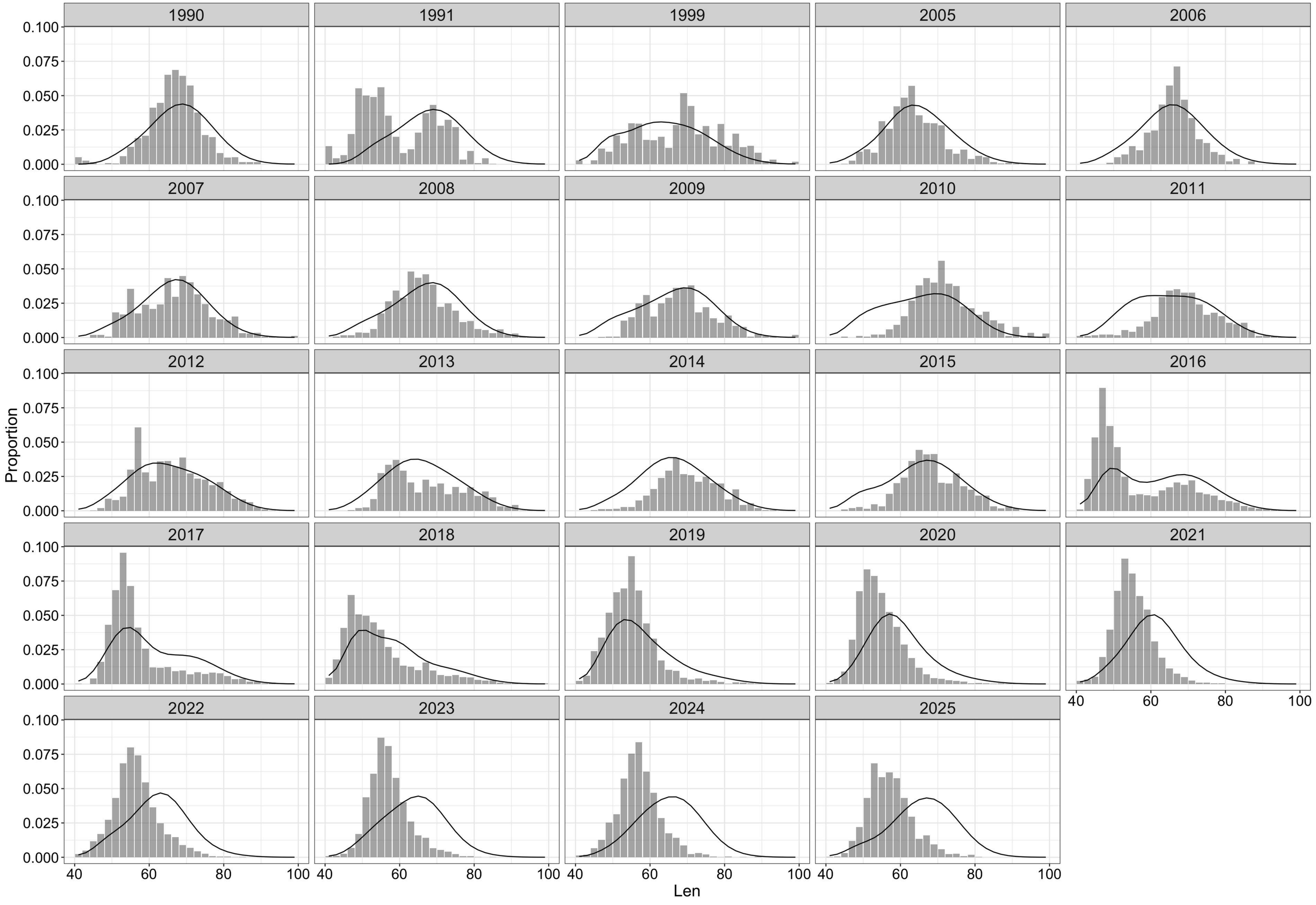

Model 25.12
AK
Unfished SSB (kt)
400
300
200
100
0
1960
1980
2000
2020
Year
Dynamic SSB0


Type — Population 1 Dynamic SSB0 -- Population 1 SSB Model — 25.12
AK
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year

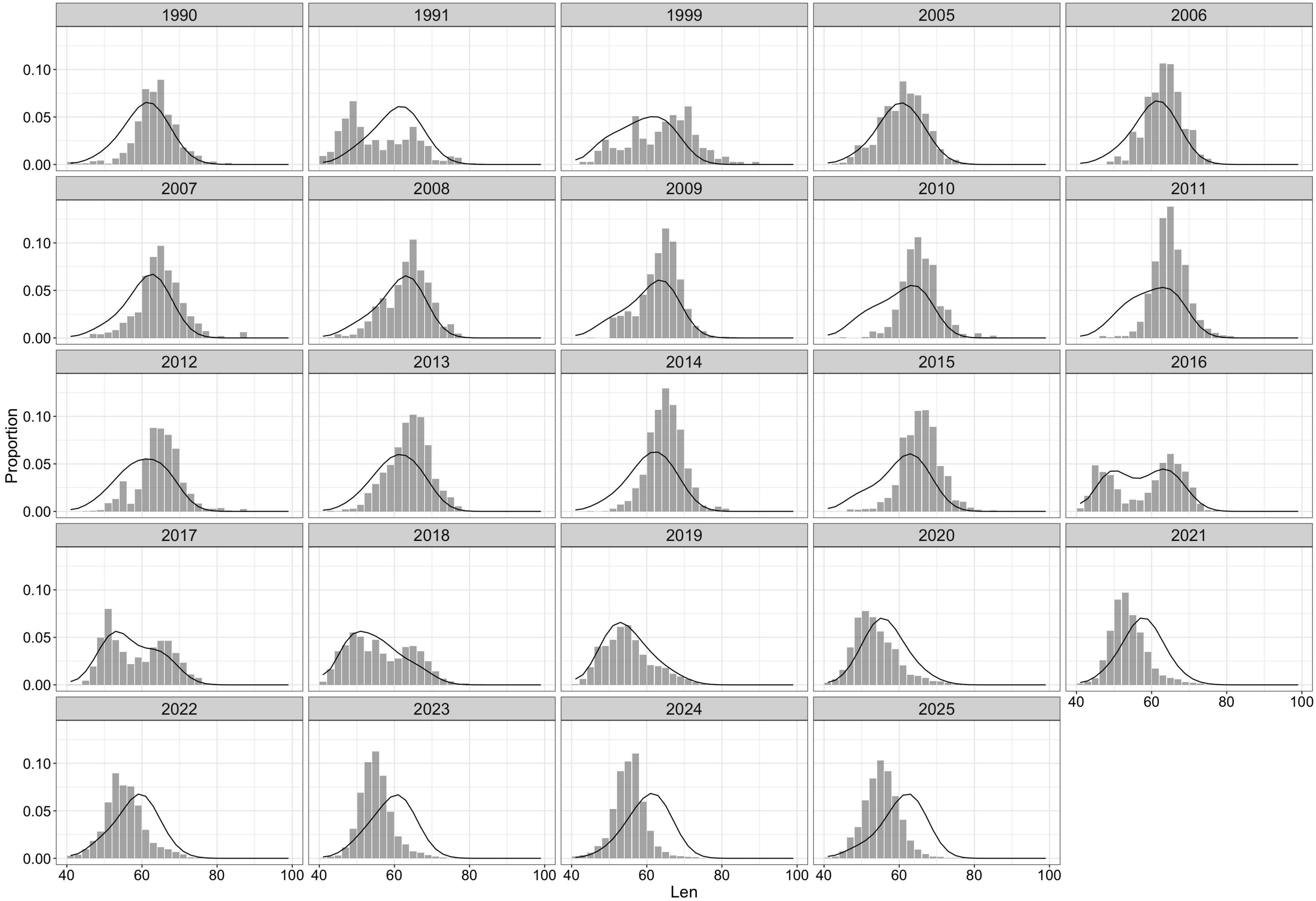

Model 25.12
AK
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year
SSB


Model 25.12
AK
Total Biom
Biomass (kt)
750
500
250
0
1960
1980
2000
2020
Year

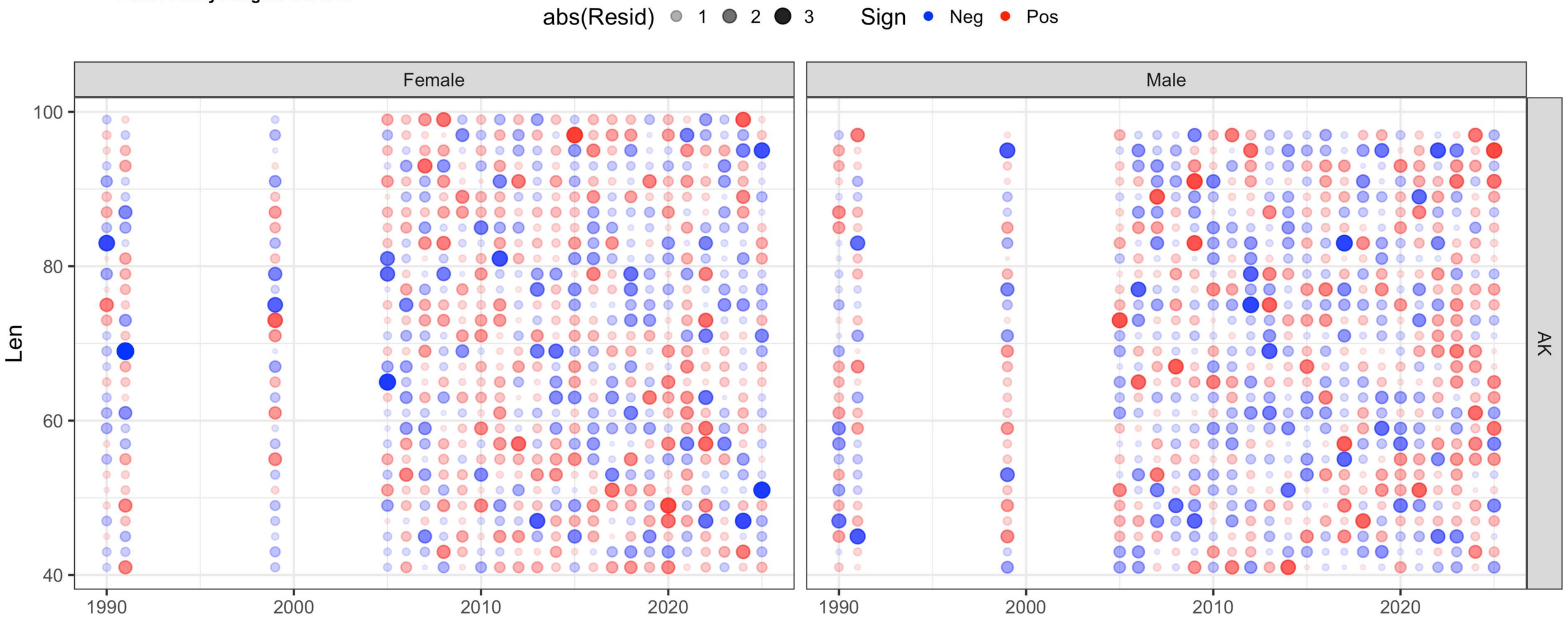
Trawl Fishery Lengths Female AK

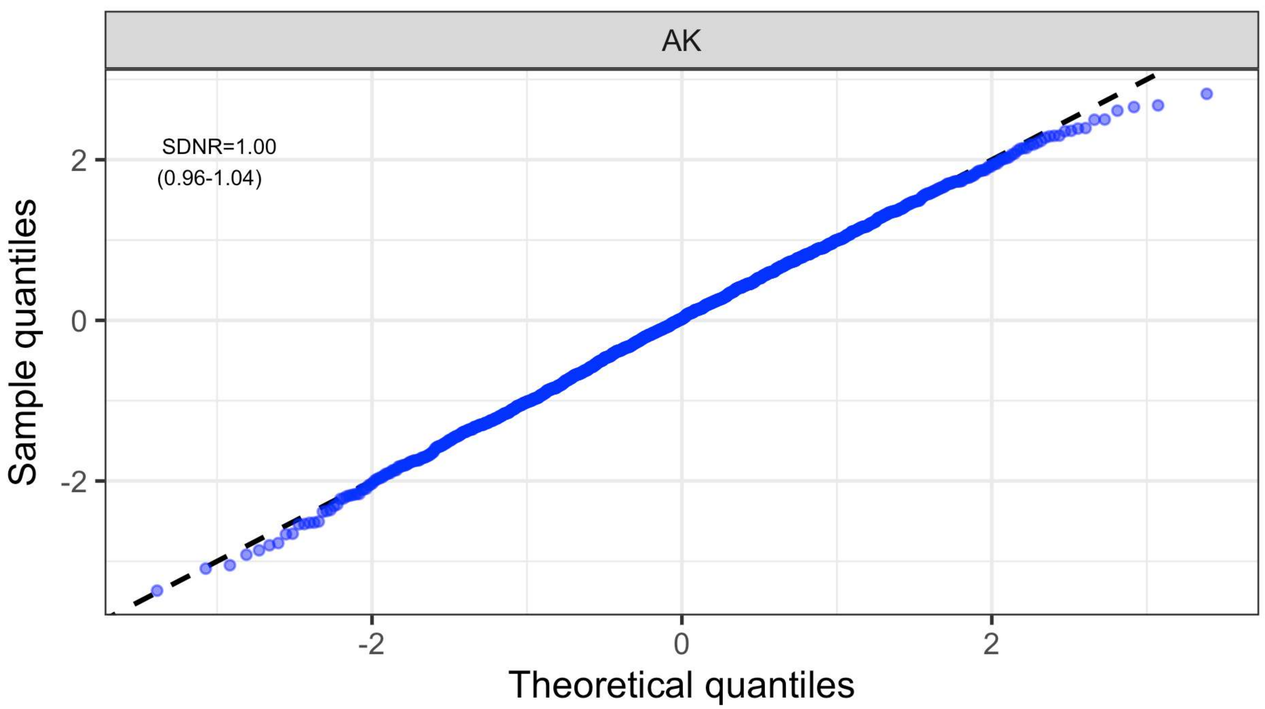


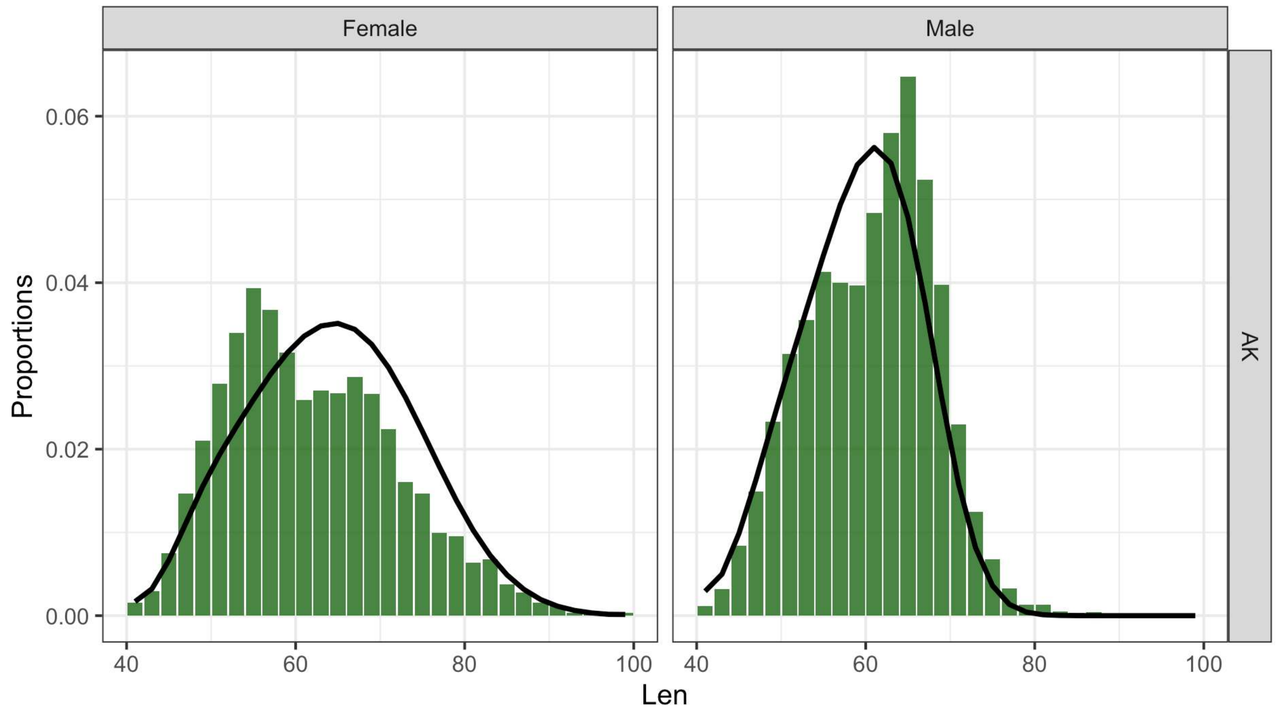
Trawl Fishery Lengths Male AK



Trawl Fishery Lengths OSA AK

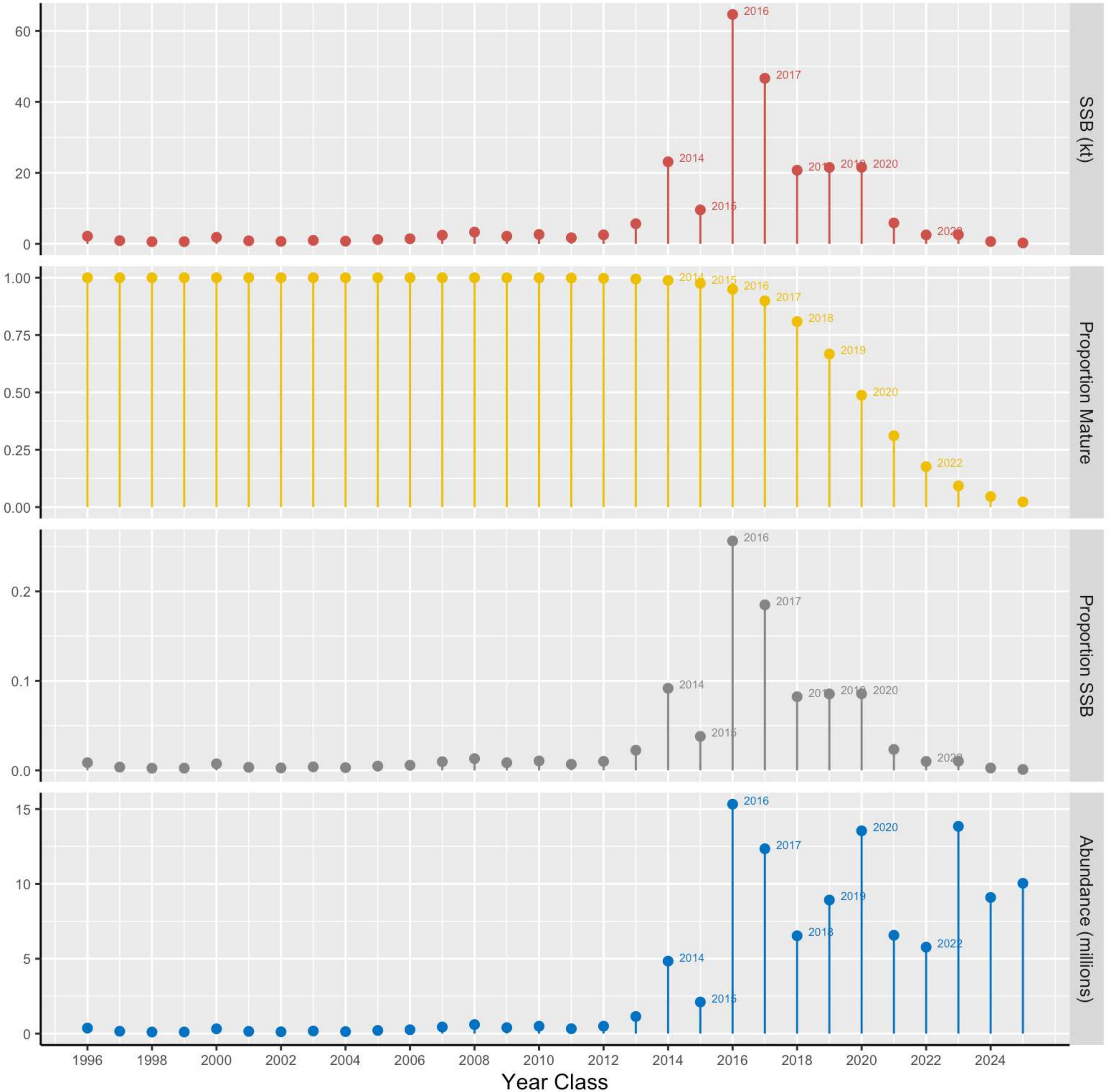






Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

Contribution to 2027 SSB by Year Class



JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

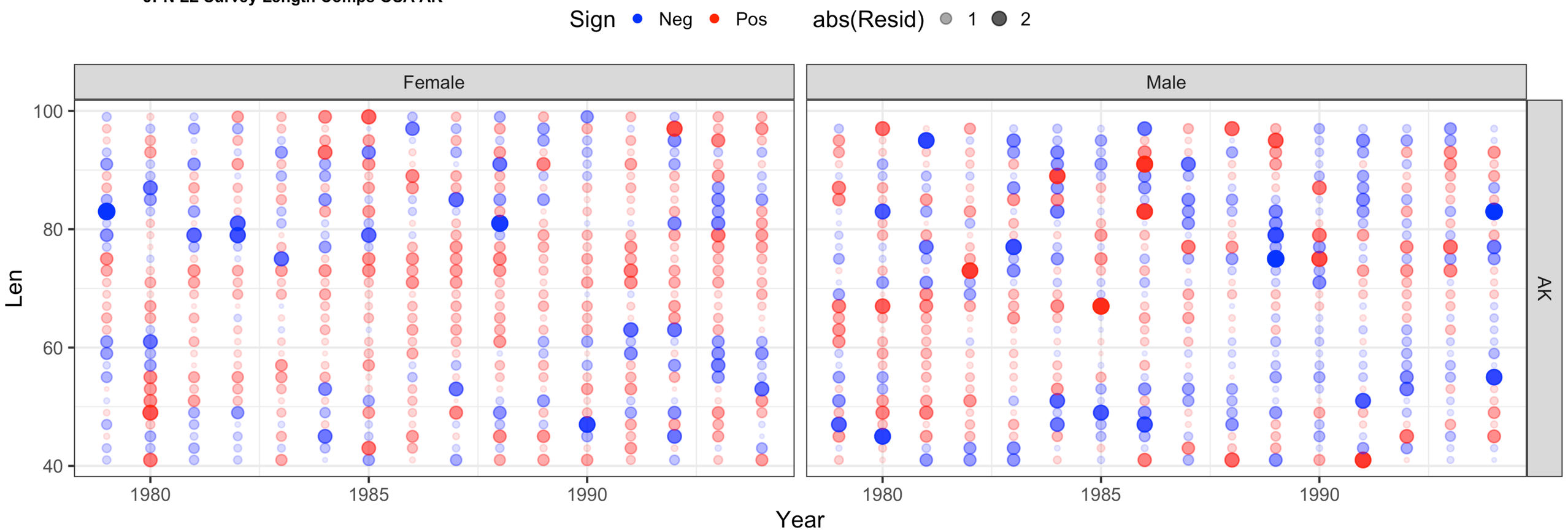
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

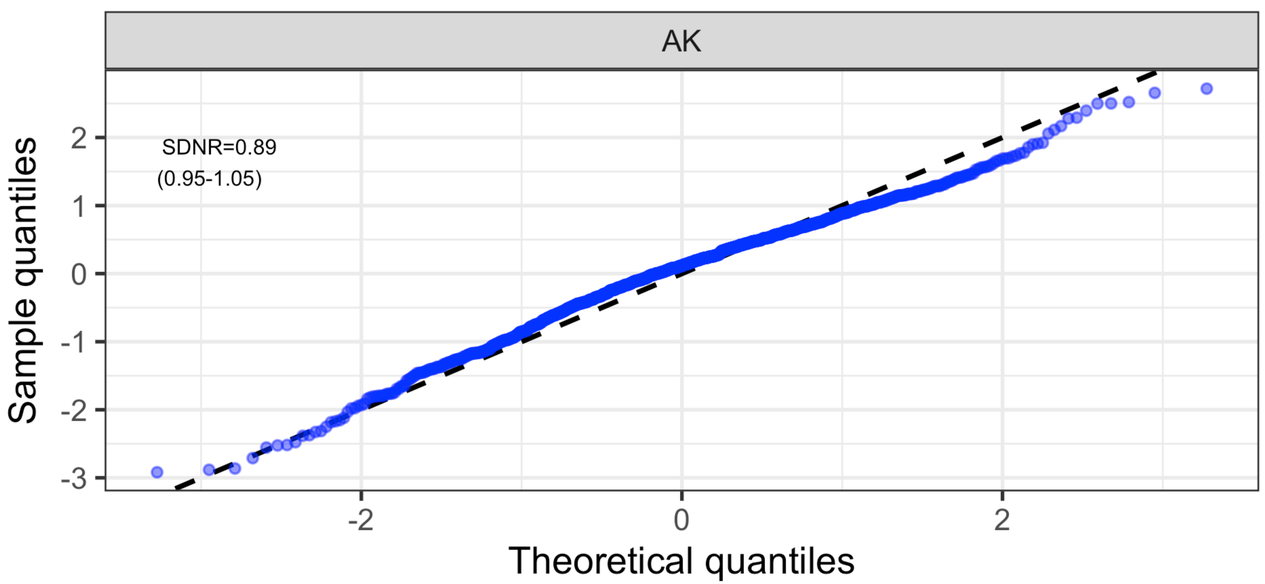
JPN LL Survey Length Comps Male AK

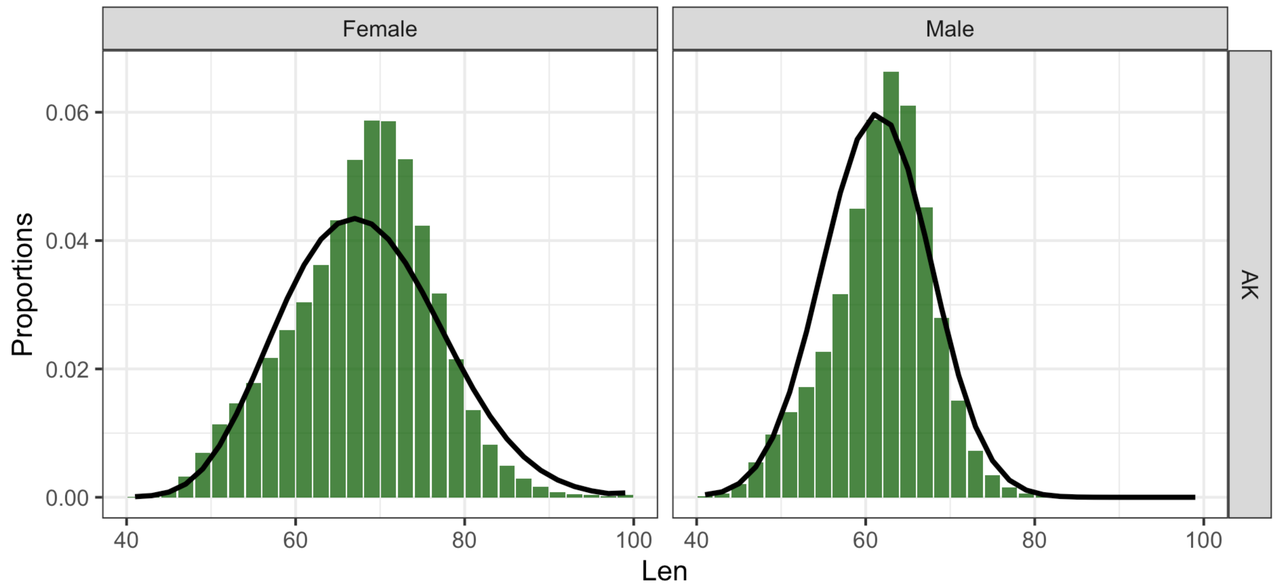
1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK

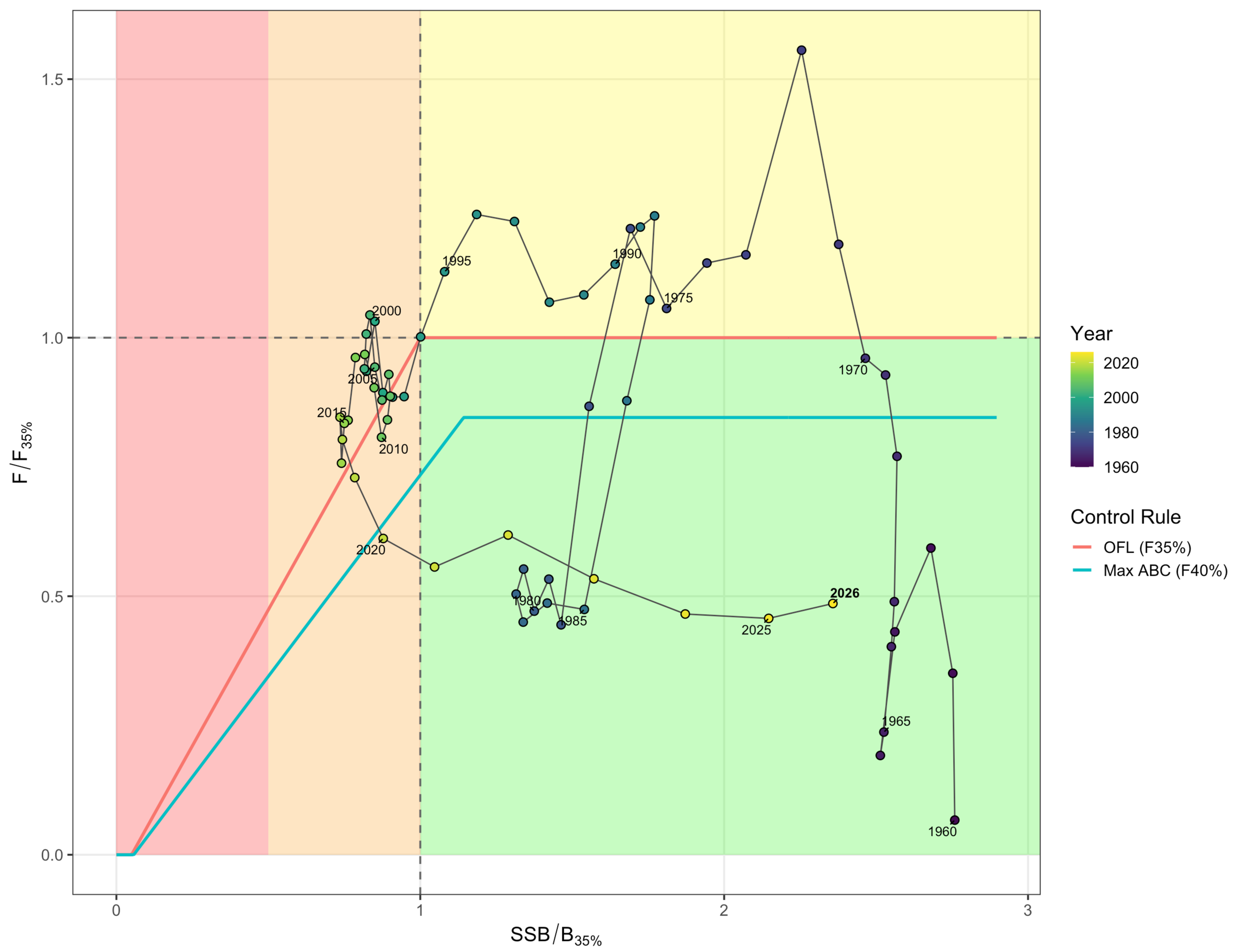






Source	SE	Data_Wt	ISS	ESS
LL Survey Index	0.09	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.09	1.00	NA	NA
Fixed Gear Fishery CPUE	0.20	1.00	NA	NA
LL Survey Ages	NA	3.56	20	71.30
JPN LL Survey Lengths	NA	0.94	20	18.73
JPN LL Survey Ages	NA	0.65	20	12.94
Fixed Gear Fishery Ages	NA	0.79	20	15.78
Trawl Fishery Lengths	NA	0.25	20	4.99

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
25.12	1	238.3376	391.6886	0.04615667	43.06031	132.1734	0.08034	1.80322	0.60849	0.57453


1.5
1.0
0.5
0.0
F/F35%
0
1
2
3
SSB/B35%
1995
1990
1975
2000
2005
2015
2010
1970
2020
1980
1985
2025
2026
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.096	0.096
M Male (estimated)	0.114	0.114	0.096	0.096
Average Recruitment (millions)	25.9	25.9	21.1	21.1
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	651327	642525
Projected Female Spawning Biomass (t)	219,714	241,217	252158	260404
B100% (t)	302,672	302,672	330434	330434
B40% (t)	121,069	121,069	132173	132173
B35% (t)	105,935	105,935	115652	115652
FOFL	0.102	0.102	0.095	0.095
max FABC	0.087	0.087	0.08	0.08
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	50570	49907
OFLw (t)	58,532	57,797	50439	49776
max ABC (t)	50,283	49,651	43057	42492
ABC (t)	50,283	49,651	43057	42492
ABCw (t)	50,111	49,482	42926	42361
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.15

0.10

0.05

0.00

1960

1980

2000

2020

2040

Year

S1: FABC (F40)
S1a: Author's F (FABC Ratio) Year 1, FABC After
S2: Author's F (FABC Ratio) Years 1+2
S3: 5-year Average F
S4: Half max FABC (1/2 * F40)
S5: No Fishing
S6: Overfished (FOFL)
S6a: Author's F (FABC Ratio) Year 1, FOFL After
S7: Approaching Overfishing (FABC -> FOFL)
Fishing Mortality
0.000
0.025
0.050
0.075
2010
2020
2030
2040
Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

250

500

750

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

250

500

750

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0

20

40

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

1960
1980
2000
2020
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0 10 20 30 40 50

2010 2020 2030 2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
25
50
75
100

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

25

50

75

100

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

2010

2020

2030

2040

Year

Year	Apportion_Type	BS	AI	WGOA	CGOA	WY	SEO
2026	Carryover_No_wAI	0.30	0.20	0.11	0.21	0.06	0.12
2026	REMA_LLS_No_wAI	0.28	0.23	0.10	0.21	0.06	0.12
2026	Carryover	0.27	0.26	0.10	0.20	0.06	0.11
2026	REMA_LLS	0.26	0.30	0.09	0.19	0.05	0.11
2026	GOA_Fix	0.30	0.24	0.09	0.20	0.05	0.12

AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2014
2016
2018
2020
2022
2024
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI

AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2021
2022
2023
2024
2025
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	13037	11421	22550	4687	9622	2652	5589	47008
2027	Carryover_No_wAI	12878	8585	21463	4722	9014	2576	5151	42926
2027	REMA_LLS_No_wAI	12019	9873	21034	4293	9014	2576	5151	42926
2027	Carryover	11590	11160	20175	4293	8585	2576	4722	42926
2027	REMA_LLS	11161	12878	18887	3863	8156	2146	4722	42926
2027	GOA_Fix	12878	10302	19746	3863	8585	2146	5152	42926
2028	Carryover_No_wAI	12708	8472	21181	4660	8896	2542	5083	42361
2028	REMA_LLS_No_wAI	11861	9743	20757	4236	8896	2542	5083	42361
2028	Carryover	11437	11014	19910	4236	8472	2542	4660	42361
2028	REMA_LLS	11014	12708	18639	3812	8049	2118	4660	42361
2028	GOA_Fix	12708	10167	19486	3812	8472	2118	5084	42361

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	0	0	0	0	0	0	0	0
2027	Carryover_No_wAI	-1	-25	-5	1	-6	-3	-8	-9
2027	REMA_LLS_No_wAI	-8	-14	-7	-8	-6	-3	-8	-9
2027	Carryover	-11	-2	-11	-8	-11	-3	-16	-9
2027	REMA_LLS	-14	13	-16	-18	-15	-19	-16	-9
2027	GOA_Fix	-1	-10	-12	-18	-11	-19	-8	-9
2028	Carryover_No_wAI	-3	-26	-6	-1	-8	-4	-9	-10
2028	REMA_LLS_No_wAI	-9	-15	-8	-10	-8	-4	-9	-10
2028	Carryover	-12	-4	-12	-10	-12	-4	-17	-10
2028	REMA_LLS	-16	11	-17	-19	-16	-20	-17	-10
2028	GOA_Fix	-3	-11	-14	-19	-12	-20	-9	-10

Year	Region	OFLw	ABCw	TAC	Catch	Year2	Region	OFLw	ABCw	Year3	Region	OFLw	ABCw
2026	BS	NA	13037	8996	3339	2027	BS	NA	12878	2028	BS	NA	12708
2026	AI	NA	11421	7440	632	2027	AI	NA	8585	2028	AI	NA	8472
2026	GOA	NA	22550	22550	18191	2027	GOA	NA	21463	2028	GOA	NA	21181
2026	WGOA	NA	4687	4687	3458	2027	WGOA	NA	4722	2028	WGOA	NA	4660
2026	CGOA	NA	9622	9622	7597	2027	CGOA	NA	9014	2028	CGOA	NA	8896
2026	WY	NA	2652	2652	2271	2027	WY	NA	2576	2028	WY	NA	2542
2026	SEO	NA	5589	5589	4865	2027	SEO	NA	5151	2028	SEO	NA	5083
2026	Total	57797	47008	38986	22161	2027	Total	50439	42926	2028	Total	49776	42361

Region	Year	Biomass (4+)	OFLw	ABCw	TAC	Catch
GOA	2025	305000	NA	24038	22836	18387
GOA	2026	299000	NA	23737	22550	18191
GOA	2027	279615	NA	21463	NA	NA
GOA	2028	274950	NA	21181	NA	NA
BS	2025	175000	NA	13898	8496	3909
BS	2026	172000	NA	13723	8996	3339
BS	2027	153111	NA	12878	NA	NA
BS	2028	150557	NA	12708	NA	NA
AI	2025	153000	NA	12175	7940	948
AI	2026	151000	NA	12022	7440	632
AI	2027	168562	NA	8585	NA	NA
AI	2028	165750	NA	8472	NA	NA